Beyond Recipients: Sibling Spillovers of College Financial Aid*

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Abstract

This paper studies the spillover effects of college financial aid on younger siblings. We exploit the eligibility rules of *Ser Pilo Paga* (SPP), a large-scale merit- and need-based scholarship program in Colombia, to estimate how having a college-going older sibling affects younger siblings' educational, labor market, and criminal outcomes. When an older sibling qualifies for SPP, younger siblings are more likely to enroll in higher education and in higher-quality institutions, experience temporary declines in formal employment consistent with increased educational investment, and are 71 percent less likely to be convicted of a crime. The probability that younger siblings themselves qualify for an SPP scholarship rises by 42 percent. Younger siblings of students who narrowly missed the eligibility threshold, and whose families therefore received no financial transfer from the program, also show significant gains in enrollment and reductions in criminal convictions. This pattern is consistent with aspirational and motivational spillovers that operate independently of household resource relief. Spillover effects on enrollment and test scores emerge only when the older sibling performs well in college, are largest in first-generation college families and among siblings close in age, and are present in standardized test scores as early as age eight or nine. The evidence points to aspirations, motivation, and revised beliefs about the returns to education as the primary channels through which college access for one child reshapes the trajectories of others in the same household.

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1 Introduction

Access to higher education remains stratified by family income. In Latin America, students from the top income quintile are roughly 45 percentage points more likely to attend college than those from the bottom, and gaps of similar magnitude persist across most developing economies. Governments have responded by expanding financial aid programs for low-income students. Most evaluations of these programs measure effects on direct recipients. But educational decisions are rarely made in isolation. They unfold within families, where siblings observe one another's trajectories, share information about opportunities, and revise what they believe is achievable. A scholarship that changes one child's educational path may reshape others in the same household, not only because it transfers money to them, but because it changes what they see as possible. If these spillovers are large, the true return to financial aid is systematically understated by analyses that focus only on recipients.

This paper studies the spillover effects of college financial aid on younger siblings. We examine *Ser Pilo Paga* (SPP), a large-scale merit-based financial aid program introduced in Colombia in 2014. The program provided full scholarships, covering tuition and a living stipend, to high-achieving students from low-income households who scored in roughly the top decile of the national secondary school exit exam, *Saber 11*, and met an income threshold based on Colombia's official poverty index, *Sisben*. Eligible students could enroll only in one of the country's 33 accredited high-quality universities, 12 public and 21 private. Because the vast majority of these institutions are located in major cities, many recipients had to relocate. The program ran from 2015 to 2018, awarding approximately 40,000 scholarships in total.

To causally identify spillover effects, we exploit SPP's stringent eligibility rules as a source of quasi-experimental variation within a difference-in-differences design. The program granted financial aid to students from income-eligible households who scored above a cutoff on the Saber 11 exam. Our sample consists of households with at least two children in which the older sibling took the Saber 11 between 2009 and 2017. We define three groups based on the older sibling's position relative to the cutoff. Eligible households (T1) have an older sibling who scores above the SPP cutoff and meets the income threshold, making them eligible to receive the scholarship and enroll in an accredited high-quality university. Almost-eligible households (T2) have an older sibling who scores just below the cutoff while also meeting the income threshold. These siblings narrowly missed the scholarship eligibility and receive no financial transfer from the program. The control group consists of income-eligible households whose older sibling scored in the 70th–80th percentile of the test distribution, placing them far from the cutoff. Timing of exposure is determined by the cohort in which the older sibling takes the exam, allowing us to compare younger siblings' outcomes before and after the introduction of SPP. The identifying assumption is that, absent the program, the outcomes of younger siblings in all three groups would have followed parallel trends.

This design is constructed to separately identify two types of spillover shocks to the household. The first is an informational and aspirational shock, captured by the T2 comparison: observing an older sibling come close to qualifying may lead younger siblings to revise upward their beliefs about what is achievable for someone from their background, raising their own effort and educational investment. Because T2 older siblings receive no scholarship, any effects on their younger siblings cannot be attributed to financial transfers, providing a clean test of whether resource relief is necessary for spillovers to materialize. The second is a combined informational, aspirational, and resource shock, captured by T1, where the older sibling both qualifies and enrolls, exposing the household to all channels simultaneously. As a placebo, we examine siblings from ineligible families, who could not have participated in the program regardless of exam performance.

Our analysis proceeds in three steps. We first characterize the household-level shock that younger siblings are exposed to. SPP eligibility raises the probability that a younger sibling has a college-going older sibling by 19.2 percentage points, a 48 percent increase above baseline, with enrollment shifting into high-quality private universities outside the home municipality. Their formal employment and earnings fall during the college years and recover substantially by year six, consistent with the short-run cost and long-run premium of selective college attendance (Hoekstra, 2009; Zimmerman, 2014; Londoño-Vélez et al., 2025). Among almost-eligible households, the probability of having an older sibling enrolled in a higher education institution also increases by 2.4 percentage points, concentrated in high-quality public universities within the home municipality. The two groups expose younger siblings to qualitatively different household shocks, which we use to identify the spillover effects.

We then turn to the main analysis: the effects on younger siblings who are not directly treated by the program. Among younger siblings in eligible households, the probability of postsecondary enrollment rises by 2.9 percentage points, a 9 percent increase above a baseline rate of 31.4 percent, with gains concentrated in high-quality private universities. Younger siblings are also 2.5 percentage points more likely to qualify for the SPP scholarship themselves, shift toward STEM fields, and are more likely to enroll outside their home municipality, mirroring the pattern documented for older siblings. Formal employment and earnings fall in the short run and show a recovery by year six, consistent with younger siblings investing more in education at the cost of delayed labor market entry. The probability of being convicted of a crime falls by 1.7 per 1,000, a 71 percent reduction relative to a baseline rate of 2.4 per 1,000, with reductions concentrated in economically motivated and drug-related offenses and no significant change in violent crime. Younger siblings in almost-eligible households show a similar pattern of institutional upgrading: enrollment gains are concentrated in high-quality public universities, mirroring the shift documented for their older siblings. Criminal convictions fall by 0.9 per 1,000, a 38 percent reduction relative to baseline. Because almost-eligible older siblings did not qualify for the scholarship and generated no direct financial transfer to the household, these effects are difficult to attribute to resource relief and are more consistent with the change in the older sibling's educational trajectory itself.

Finally, we explore two mechanisms underlying the sibling spillover effects. Three pieces of evidence support a role model and aspiration interpretation of the spillovers. First, younger siblings show improvements in academic performance at ages when information about college admissions or financial aid is not yet relevant to their own decisions. Using national standardized exam data, we find significant gains in math and Spanish scores among younger siblings as early as age eight or nine, years before they face any college-related decision. Improvements in school effort, including punctuality, emerge at age eleven. The presence of effects this early is consistent with a general shift in motivation and engagement with school rather than a response to specific information about educational opportunities.

Second, what moves younger siblings' outcomes is not the older sibling's eligibility but their visible success in college. Restricting to families whose older sibling enrolled in one of Colombia's leading private universities, we interact the treatment indicator with whether the older sibling performed well academically in their first semester. Spillover effects on younger siblings' test scores and enrollment are large and significant only in families where the older sibling succeeds. Eligibility alone generates no significant effect. Because all older siblings in this subsample attend the same institution under comparable financial and academic conditions, the results isolate variation in the older sibling's performance rather than in resources or institution type.

Third, the pattern of heterogeneous effects across family types is consistent with aspiration and role model effects. Enrollment spillovers are concentrated among younger siblings with an age gap of four years or less from the older sibling, where the aspiration signal is most immediate, and absent for those with a larger age gap. Effects are also concentrated in families where neither parent has postsecondary education, where the older sibling is the first in the family to access higher education and whose trajectory provides the most novel signal about what is attainable for someone from their background. Crime reductions are driven by pairs where the older sibling is male, regardless of the younger sibling's gender, consistent with a gender-specific role model effect on criminal behavior.

The most natural alternative explanation is that the scholarship relieves household financial constraints, freeing resources that benefit younger siblings. We find little support for this interpretation. Almost-eligible younger siblings, whose older siblings did not qualify for the scholarship and received no financial transfer, show significant enrollment gains and reductions in criminal convictions. Families where the older sibling was employed before college, those facing the largest resource disruption when the older sibling transitions to full-time study, show the largest enrollment spillover effects, not the smallest. We cannot fully separate informational from aspirational effects, but the evidence is more consistent with younger siblings responding to what they observe about their older sibling's trajectory than with households redistributing financial resources.

This paper makes four contributions to the literature.

The first contribution is to show that the effects of college financial aid extend beyond direct recipients to their younger siblings, and to trace these spillovers across educational, labor market, and criminal outcomes. Prior work on sibling spillovers in higher education focuses almost exclusively on educational choices: whether younger siblings apply to and enroll in the same institution as their older sibling (Altmejd et al., 2021), enroll in the same major (Joensen and Nielsen, 2018; Dahl et al., 2024), or attend college at all (Barrios-Fernández, 2022). We trace sibling spillovers from a college scholarship to labor market earnings and criminal convictions, margins that contribute to welfare analysis of college financial aid policy. Our findings show that these downstream effects are large. Younger siblings delay labor market entry in the short run, consistent with increased educational investment, and are 71 percent less likely to be convicted of a crime, with reductions concentrated in economically motivated and drug-related offenses. The crime reductions are consistent with higher opportunity costs of criminal activity among younger siblings who invest more in education.

We also study how and when spillovers occur. We use administrative records from one of Colombia's leading private universities and find that spillovers arise only when the older sibling performs well in college, showing that the older sibling's experience and success play a role in shaping the effect. Linking early standardized tests in third, fifth, and ninth grade, we further show that younger siblings' gains emerge years before the college decision stage, consistent with a general shift in motivation and aspirations rather than a response to specific information about college or financial aid. Second, unlike prior work on sibling spillovers in higher education, where the older sibling would have attended college and the relevant margin is which institution to choose, we study a policy that shifts the extensive margin of higher education, moving scholarship recipients from not attending college to attending selective universities. This counterfactual matters: for families at the extensive margin, the older sibling's college attendance is new, making the signal to younger siblings about what is attainable more novel. Third, we find that the program's sibling spillover effect is not confined to directly eligible families. Younger siblings of students on the margin of eligibility also show gains in college enrollment and reductions in criminal convictions, suggesting that the policy's reach and its spillover potential extend beyond its intended beneficiaries.

This paper also contributes to the literature on how parents allocate resources among children within the household. Existing studies focus on early childhood, when parental investments are more flexible and children's investment returns are still uncertain. Identifying these reallocations is empirically challenging, as endogeneity often limits within-family comparisons (Barrera-Osorio et al., 2023). An open question in this literature is whether households continue to adjust resources as children grow older, since most evidence to date comes from early-life interventions. Adhvaryu and Nyshadham (2016) exploit variation in exposure to a large-scale iodine supplementation program in Tanzania and find that siblings of treated children are more likely to receive necessary vaccinations. They interpret these findings through a model in which parents are averse to inequality among their children. Carneiro et al. (2023) study sibling spillovers from a prenatal intervention in Nigeria that combined information and cash transfers and show that the program improved health and nutrition outcomes for non-treated siblings, consistent with parents equalizing investments across children. In contrast, Barrera-Osorio et al. (2023) analyze a conditional cash transfer in Bogotá and find negative spillovers on siblings. Yet the intervention itself had limited effects on the directly treated child, making those negative spillovers difficult to interpret as evidence that parents reallocate resources away from other children.

We extend this literature beyond early childhood to adolescence and young adulthood, when children are more autonomous and the opportunity costs of education are higher. We examine how families respond when an older child gains access to higher education and show that younger siblings benefit across multiple ages, around 9, 11, 15, and 17, corresponding to third, fifth, ninth, and eleventh grades. These gains persist into adulthood, with long-run effects such as reductions in criminal behavior. At later ages, parents likely have clearer information about their children's abilities and prospects, which could in principle lead them to reinforce rather than compensate investments. Yet our findings show no evidence of crowding out, the evidence is not consistent with parents reallocating resources away from younger siblings, even when the treated child is older and the intervention occurs later in life.

Third, this paper contributes to the literature in the economics of crime that studies the determinants of criminal behavior (Chalfin and McCrary, 2017; Doleac, 2023). Previous research shows that keeping youth in school reduces youth offending (Jacob and Lefgren, 2003; Luallen, 2006; Anderson, 2014; Berthelon and Kruger, 2011), that stronger labor-market conditions lower crime and recidivism (Yang, 2017; Schnepel, 2018; Britto et al., 2022), and that reduced access to mental healthcare increases incarceration (Doleac, 2023). While most of this work emphasizes enforcement or labor-market incentives, this paper argues that criminal involvement is also a function of perceived opportunity and aspirations. We show that expanding access to college for one member of the household can influence the criminal behavior of others who are not direct beneficiaries. When an older sibling gains entry to higher education, younger siblings, who face the same economic environment but a new reference point, become significantly less likely to engage in crime. The results suggest that exposure to opportunity and to a credible role model within the household can reduce criminal behavior by reshaping aspirations and perceived returns to legal activities. In this sense, we highlight a complementary mechanism to deterrence: expanding access to education changes how individuals perceive their prospects, lowering the relative appeal of crime.

More broadly, this paper contributes to understanding how educational opportunities for one family member affect the beliefs and behavior of others in the same household. Prior work documents that exposure to role models outside the family, including female political leaders (Beaman et al., 2012) and cinematic figures (Riley, 2024), raises educational aspirations and attainment. We provide evidence that a sibling who visibly succeeds in higher education generates similar effects from within the household, and we assemble multiple converging tests that point toward this channel rather than household resource redistribution. Enrollment spillovers are largest among younger siblings close in age to the older sibling and in families where no parent has postsecondary education. Crime reductions are concentrated in same-gender pairs. Academic performance improves among younger siblings as early as age eight or nine, before any college-related information could plausibly influence their decisions. And within a subsample where older siblings attends the same institution and we can identify college GPA, spillover effects emerge only when the older sibling performs well academically, showing that what matters is not whether the older sibling received financial aid, but whether they succeeded visibly.

The paper proceeds as follows. Section 2 describes the Colombian higher education system and the design of the SPP program. Section 3 describes the data sources and sample construction. Section 4 presents descriptive evidence on sibling correlations and motivating evidence on spillovers around the eligibility cutoff. Section 5 presents the empirical strategy. Section 6 reports the first-stage effects on older siblings and the spillover effects on younger siblings across educational, labor market, and criminal outcomes. Section 7 examines the mechanisms underlying the spillovers. Section 8 presents robustness checks. Section 9 concludes.

2 Institutional Background and Program Details

2.1 Context: Higher Education in Colombia

By 2013, only 4 percent, 11 percent, and 34 percent of individuals in poor, vulnerable, and middle-class Colombian households, respectively, had attained tertiary education. Among heads of poor households, only 36 percent had completed secondary education or higher (World Bank Group, 2015). These figures underscore the deep inequality in educational attainment across income groups and highlight the persistent barriers to accessing post-secondary education in the country. Even when access is achieved, the quality and value-added of higher education often remain insufficient (Camacho et al., 2017).

The higher education (HE) system in Colombia is regulated by Law 30 of 1992, which governs the organization and operation of both public and private HE institutions. In the early 2000s, the Ministry of Education introduced new regulations to strengthen quality assurance. Institutions are required to undergo a certification process for each academic program to obtain a *Qualified Registry*, which is a prerequisite for operation and inclusion in the National System of Higher Education Institutions (SNIES). In addition to this mandatory certification, institutions may voluntarily apply for a *High-Quality Accreditation*, which involves an extensive self-evaluation and external peer-review process. This accreditation is regarded as a proxy for excellence in education provision (Camacho et al., 2017). Fewer than 10 percent of institutions held this high-quality accreditation during the period analyzed.

Colombia's HE system includes four main types of institutions that differ by program length and degree level: (i) *Professional Technical Institutions*, which offer one- to three-year technical programs; (ii) *Technological Institutions*, focused on applied technical education; (iii) *University Institutions*, which can offer professional degrees and postgraduate specializations; and (iv) *Universities*, which are the only institutions authorized to confer master's and doctoral degrees.[1] By 2014, there were 290 registered higher education institutions, of which

[1] Articles 58–60 of Law 30 (1992) regulate the creation of public institutions, while Act 1478 (1994)

27 percent were private. Among the 83 institutions classified as universities, only 32 (12 public and 20 private) had received high-quality accreditation as of October 1, 2014 (SNIES data). Admission rates reflected both capacity constraints and high selectivity: while private institutions admitted around 86 percent of applicants, public HEIs admitted only about 36 percent. These figures reveal both a shortage in the supply of public higher education and significant quality segmentation within the system.

Colombia's HE system is tightly linked to standardized testing administered by the Institute for the Assessment of Education (ICFES). Two exams are particularly relevant: the high school exit exam (*Saber 11*) and the higher education graduation exam (*Saber Pro*). The *Saber 11* test has been mandatory since 1980 for graduation from high school and admission into HEIs, serving a role similar to the SAT or ACT in the United States. Its near-universal take-up—over 90 percent of seniors—makes it an excellent proxy for the universe of high school graduates. The *Saber Pro* exam, introduced in 2003 and made mandatory for all higher education graduates by 2011, assesses key cognitive competencies such as quantitative reasoning, critical reading, and written communication (Busso et al., 2023). In 2013, 576,094 students took the *Saber 11* exam. Approximately 17 percent of these students subsequently enrolled in higher education, and among those who did, roughly 30 percent attended a high-quality accredited institution. The exam results play a decisive role in post-secondary admissions, with around four-fifths of HEIs using *Saber 11* scores as a key admission criterion (Londoño-Vélez et al., 2020). However, for many economically disadvantaged students, the primary barriers to accessing higher education are financial. In some private universities, tuition fees are notably high, and securing student loans or financial aid proves challenging with limited availability (Camacho et al., 2017).

Colombia experienced a remarkable expansion in educational coverage during the 2000s, mirroring regional trends. Secondary school completion increased substantially, and the number of high school graduates rose by about 30 percent between 2001 and 2011. During the same decade, college enrollment expanded by nearly 50 percent, driven largely by the creation of new academic programs. Between 1999 and 2011, the number of higher education programs nearly doubled—from 2,609 to 5,101—with most of the expansion occurring in existing institutions rather than newly established ones (Carranza and Ferreyra, 2019). While this growth broadened access, many of the new programs attracted students with lower high school test scores, suggesting that rapid expansion was accompanied by widening disparities

governs the establishment of private ones. Subsequent regulations, including Act 2566 (2003) and Law 1188 (2008), set minimum quality requirements for launching new programs. The National Training Service (SENA), created in 1957 and overseen by the Ministry of Labor, provides technical training but is not part of the HE system.

in program quality and student preparedness.

Overall, by the time the *Ser Pilo Paga* program was launched in 2014, Colombia's higher education landscape was characterized by three features: (i) unequal access across socioeconomic groups, (ii) persistent quality segmentation between accredited and non-accredited institutions, and (iii) a growing but uneven supply of higher education opportunities.

2.2 The Ser Pilo Paga Program

On October 1, 2014, Colombia launched *Ser Pilo Paga* (SPP), a large-scale merit- and need-based financial aid program targeting high-achieving students from low-income households. Fully financed with public funds, SPP enabled eligible students to enroll in any high-quality accredited university—public or private—covering full tuition and providing a monthly living stipend between one and four times the Minimum Current Legal Monthly Salary (SMMLV), depending on the municipality where the institution was located. The government planned approximately 10,000 new awards per year from 2015 to 2018, ultimately financing close to 40,000 undergraduate degrees. Owing to its scale and extensive public communication campaign, the program rapidly became one of the most visible and popular social initiatives in the country (Londoño-Vélez et al., 2020).

SPP eligibility combined a national test-score requirement with a poverty-based household classification. Students were required to satisfy four criteria: (i) sit for the national high school exit examination (*Saber 11*) in the fall of 2014, 2015, 2016, or 2017; (ii) score approximately within the top decile nationwide[2]; (iii) gain admission to a high-quality accredited higher education institution (HEI) or one undergoing accreditation renewal; and (iv) appear in the *Sisben III* poverty registry below a cutoff that varied by geographical location, with distinct thresholds for major cities, other urban areas, and rural zones.[3] Given the central role of *Saber 11* in admissions and the scarcity of affordable credit—particularly for private universities—SPP represented a substantial relaxation of financial constraints for academically qualified low-SES students (Camacho et al., 2017).

A first wave of empirical evidence documents large and immediate effects on access and college quality. Exploiting the stringent eligibility threshold for the inaugural cohort,

[2]Students taking the test in the fall of 2014 had to score in the 90th percentile, those in 2015 in the 91st, in 2016 in the 95th, and in 2017 in the 96th.

[3]Students' families had to be registered in version III of the System for the Identification of Potential Beneficiaries of Social Programs (*Sisben*) with the following maximum scores: 57.21 (out of 100) for families residing in the 14 main cities, 56.32 for those in other urban areas, and 40.75 for those in rural areas.

Londoño-Vélez et al. (2020) show that SPP eligibility increased immediate college enrollment by roughly 27–32 percentage points (about 60–85%), effectively closing the socioeconomic enrollment gap among top-decile test-takers. The program also reallocated students across the quality distribution of higher education: beneficiaries shifted from low-quality institutions toward high-quality—predominantly private—universities, dramatically increasing the socioeconomic diversity of historically elite campuses. Londoño-Vélez et al. (2020) further demonstrate that financial aid expanded enrollment among both eligible and ineligible students, generating net social gains. As high-ability, low-income students vacated seats in lower-tier institutions, these were filled by the next-best applicants. Greater visibility and transparent eligibility rules expanded demand, while private high-quality universities responded by modestly increasing cohort sizes.

Longer-run evaluations confirm that these shifts translated into substantial gains in educational attainment, human capital, and labor-market outcomes. Linking national administrative records through 2023, Londoño-Vélez et al. (2025) document large increases in B.A. completion at high-quality institutions (particularly in STEM fields), improved performance on standardized college exit exams, and persistent labor-market advantages: nine years after high school, recipients earn roughly 18 log points more than comparable ineligible peers and are substantially more likely to reach the upper tail of the earnings distribution. The program also narrowed socioeconomic gaps among students with comparable academic ability by equalizing access to high–value-added universities. A welfare analysis of SPP reveals high social returns: the marginal value of public funds (MVPF) ranges between 1.9 and 3.2, indicating that individuals' willingness to pay for the program substantially exceeds its net fiscal cost.

Beyond its direct beneficiaries, SPP also altered incentives upstream in the educational pipeline. Laajaj et al. (2022) show that once the program became publicly known, low-SES students in the 2015 cohort—the first aware of SPP while preparing for *Saber 11*—increased effort and achievement near the merit cutoff, leading to differential enrollment gains among the newly eligible. Related improvements in *Saber 9* scores at the school level point to broader motivational responses earlier in the schooling trajectory, consistent with higher perceived returns to education and expanded aspirational windows among disadvantaged students.

Taken together, the existing literature shows that SPP relaxed binding credit constraints, reallocated talent toward high-quality institutions, enhanced skill acquisition and labor-market outcomes, and generated sizable social returns. Yet, one dimension remains largely unexplored: how these shocks propagate *within families*. If a scholarship propels one child into a selective university, does that experience alter younger siblings' beliefs, effort, information sets, or opportunity costs? This paper examines that margin. Leveraging the eligibility criteria of SPP and rich administrative linkages across siblings, we study whether younger siblings of SPP-eligible students adjust their schooling, higher-education, labor, and behavioral trajectories. This perspective sheds light on the intergenerational and intra-household mechanisms through which large-scale merit- and need-based scholarships may amplify their benefits beyond the direct recipients.

3 Data and Descriptive Statistics

Estimating within-family spillovers requires linking family composition and following individuals within the same household across different stages of life, effectively creating a longitudinal family panel. This paper builds on a large-scale consolidation of multiple administrative datasets to construct a nationwide panel of siblings linked across education, labor, and crime outcomes between 2009 and 2023. The analysis combines administrative records with original data on criminal convictions and institutional data from a large private university that hosted a significant share of *Ser Pilo Paga* (SPP) beneficiaries.

3.1 Data Sources

We begin with the population of test-takers of the national high school exit examination, *Saber 11*, administered by the Instituto Colombiano para la Evaluación de la Educación (*ICFES*, 2009-2023). The exam measures core competencies in mathematics, critical reading, writing, and reasoning. Around 90 percent of high school seniors take the test each year, regardless of whether they plan to continue to higher education, providing an almost complete census of secondary graduates. Two exam rounds are held annually: students in private schools typically sit for the first-semester exam, while most public-school students take it in the second semester. The dataset includes basic demographic characteristics, school identifiers, and self-reported socioeconomic information.

To capture outcomes earlier in the educational trajectory, we incorporate standardized test data for grades 3, 5, and 9—*Saber 3*, *Saber 5*, and *Saber 9*. Individual-level data are available for the 2017 cohort only, covering approximately 760,000, 776,000, and 590,000 students, respectively. This cross-section allows us to estimate sibling spillover effects in earlier stages of schooling, not only at the high school graduation and college decision margin captured by *Saber 11*. In addition to test scores, the accompanying student questionnaires collect information on academic behaviors such as attendance, punctuality, and grade repetition. For example, students report how often they arrived on time, were late, or missed class in the previous month. These measures provide valuable proxies for school engagement and effort in primary and secondary education.

Socioeconomic background and program eligibility are measured using the *Sisben* (*Sistema de Identificación de Potenciales Beneficiarios de Programas Sociales*), Colombia's primary instrument for targeting social programs. Managed by the National Planning Department (DNP), *Sisben* characterizes living conditions among low-income households and assigns each family a continuous index from 0 (poorest) to 100 (least poor), based on housing, utilities, demographics, and education (Camacho and Conover, 2011). Versions I through IV of *Sisben* were implemented in 1999, 2005, 2010, and 2018, jointly covering roughly 12 million households—about 83 percent of all households in the country. The 2010–2018 version (III) defines socioeconomic eligibility for the *Ser Pilo Paga* program.

We next use the longitudinal school census (*SIMAT*, 2005–2023), which provides annual records of enrollment and grade progression for all public and private schools. These data allow us to follow students over time to measure grade advancement, dropout, and completion. Postsecondary outcomes are drawn from the National Higher Education System (*SNIES*, 2009–2023), which records enrollment and degree completion across universities and technical institutions. From *SNIES*, we identify the type of postsecondary program—vocational education and training (VET), two-year technical degrees, or four-year university programs—and the field of study, including whether it belongs to a STEM discipline. We complement these data with results from *Saber Pro*, the mandatory higher-education exit exam taken by graduating students in their final semester. Performance on *Saber Pro* serves as a proxy for college completion and allows us to assess skill acquisition at the end of tertiary education.

Labor-market outcomes are obtained from the Integrated Contribution System, *PILA* (*Planilla Integrada de Liquidación de Aportes, 2009–2021*). It consolidates employer and employee contributions to health, pension, and labor-risk insurance systems, providing monthly information on formal employment and earnings. These data enable the construction of medium-term employment and earnings trajectories after secondary and postsecondary schooling.

To capture behavioral outcomes, we complement these administrative sources with original data on criminal convictions. We web-scrapped approximately 600,000 publicly available conviction records from 17 of Colombia's 33 judicial districts. These districts encompass the country's largest and most urbanized regions, representing roughly 67 percent of the national population, 69 percent of homicides, and 83 percent of property crimes. Each record includes the type of offense, date of ruling, sentence length, and court of jurisdiction.

Finally, we complement the analysis with institutional data from a large private university that enrolled approximately 2,500 *Ser Pilo Paga* beneficiaries during the program's operation. These records include information on admissions, program enrollment, course-taking, GPA, and degree completion. Identifying older siblings within this institution provides a unique opportunity to examine the academic performance of scholarship recipients and to test potential mechanisms for sibling spillovers through exposure, role modeling, and information sharing within families.

3.2 Family Linkage and Sample Construction

We construct a family-level panel that links siblings across education, labor, and behavioral outcomes. The linkage relies on household identifiers in *Sisben*, which allow grouping individuals into family units. According to Colombia's 2018 Population and Housing Census, there are 14.24 million households nationwide, of which 11.87 million appear in *Sisben*, representing about 83 percent of all households. Among these, 9.94 million satisfy the poverty-eligibility criteria of the *Ser Pilo Paga* (SPP) program, and we restrict our analysis to this group.

We further limit the sample to households with at least two members born between 1988 and 2018—the period over which educational trajectories can be observed in our data. This restriction yields 5.05 million poverty-eligible households with at least two school-age individuals. Within these households, we define the *older sibling* as the first-born child in the family and identify them in the national school census (*SIMAT*). We find 2.03 million such first-borns in *SIMAT*, of whom 968,502 took the *Saber 11* exam between 2009 and 2019. These students constitute our cohort of "older siblings".

We then identify all *younger siblings* of these students within the same households. The 968,502 older siblings are linked to 1.69 million younger siblings, corresponding to an average of 2.75 younger siblings per older sibling. This estimate aligns closely with self-reported family size in the *Saber 11* questionnaire, where students report an average of 2.7 siblings, suggesting that our linkage captures most sibling relationships in the target population. Appendix Figure 5 provides a schematic overview of this process, illustrating how the different data sources are linked to construct the final sibling panel.

After identifying 1.69 million younger siblings, we merge them with administrative records to measure educational, employment, and behavioral outcomes. The final longitudinal panel includes families in all municipalities and follows cohorts born between 1988 and 2018, observed in schooling data from 2005 to 2023 and in higher-education and labor-market records through 2023. Appendix Figure 6 shows that about half of the older siblings in our sample have one younger sibling, with a long right tail reflecting larger families. Appendix Figure 7 displays the distribution of younger siblings' ages when the first-born takes the high school exit exam. The distribution centers around 13 years of age, providing a broad window to study sibling spillover effects both before and after key educational transitions. Together, these sources create a family-level dataset that links standardized test scores, socioeconomic characteristics, educational trajectories, employment, and criminal records. This unique linkage enables the analysis of within-family effects of higher-education access over more than two decades of administrative data.

3.3 Descriptive Statistics of Households and Siblings

Appendix Table 2 provides an overview of the households and individuals in the sibling sample. Panel A describes the socioeconomic background of families, which is shared among siblings, while Panel B reports individual characteristics of older and younger siblings. Together, these patterns indicate that the data represent a population of low-income households consistent with the program's intended target group.

Most households in our sample belong to the lowest socioeconomic strata of Colombia's classification system.[4] Eighty-eight percent of families fall within strata 1 and 2, which correspond to the poorest residential areas where households receive public-utility subsidies. Families in our sample are larger than the national average. The mean household size is 4.8, compared with 3.2 in the third wave of *Sisben*. When restricting to households in Sisben with at least two children—as in our analysis—the national average rises to 4.95, nearly identical to our estimate. Appendix Figure 8 shows that the distribution of household size in our data closely matches that of the population in the census of the poor, confirming that our linked households are representative of low-income families with children.

Parental education is low: about 45 percent of mothers and 38 percent of fathers have completed some secondary education, and only around 6 percent have any form of higher education. Consistent with this socioeconomic profile, access to durable goods and services is limited—fewer than half of households report owning a computer, 43 percent have internet access, and only 14 percent own a car. These figures align with the targeting criteria of the

[4]Strata are official socioeconomic categories assigned by municipal governments to residential properties under the *Sistema Nacional de Estratificación Socioeconómica*, as established by Law 142 of 1994 and regulated by the National Planning Department (DNP). The classification (from 1 = lowest to 6 = highest) is based on observable physical and urban characteristics of dwellings and their surroundings. It is used primarily to administer cross-subsidies in public utility tariffs: households in lower strata receive subsidies for water, electricity, and gas, financed through contributions from higher-strata users DANE (2015).

Sisben registry, which identifies the poorest households for social-program eligibility.

Turning to individual characteristics, older siblings were born on average in 1996 and took the *Saber 11* exam at age 17.7, while younger siblings were born roughly six years later, in 2002, and were about 12 years old when the first-born sat for the test. The majority of students attend public schools (91 percent), and roughly 4 percent of older siblings score above the *Ser Pilo Paga* eligibility threshold. Academic achievement is modest but comparable across siblings: both groups score about 0.1 standard deviations below the national mean on *Saber 11*. Immediate postsecondary enrollment rates are low, as expected for this population, and the distribution of fields of study is similar across siblings. Six years after high school, 40–42 percent of individuals hold formal employment, and average monthly earnings correspond to about 40–45 percent of the minimum wage.

4 Descriptive Evidence

4.1 Sibling Correlation

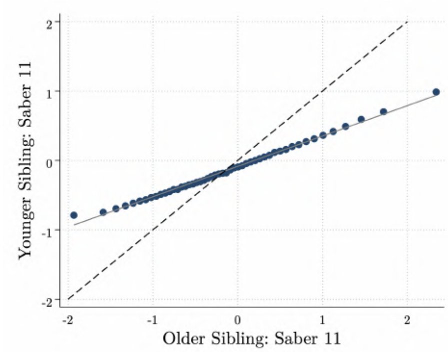
A large body of research in economics and sociology emphasizes the importance of family background in shaping long-term outcomes such as education, earnings, and social mobility. While traditional intergenerational analyses focus on links between parents and children, sibling correlations provide an alternative and complementary perspective. This approach measures the extent to which siblings resemble one another in key life outcomes, capturing shared family influences—both observed and unobserved—including common parental investments, neighborhood environments, and school quality. A high sibling correlation indicates that much of the variation in an outcome reflects differences between families rather than within them, whereas a low correlation suggests that family background plays a smaller role in generating inequality (Corcoran et al., 1976; Solon, 2018). Unlike intergenerational estimates, sibling correlations can be derived using data from a single generation, provided that siblings can be reliably identified and followed over time—a feature uniquely enabled by our linked administrative data.

Figure 1 and Appendix Table 3 present descriptive evidence on the correlation between older and younger siblings across educational, labor, and behavioral outcomes. Together, they show a consistent pattern of positive within-family associations: siblings who grow up in the same household tend to perform similarly in school, make comparable educational choices, and experience related early labor-market outcomes.

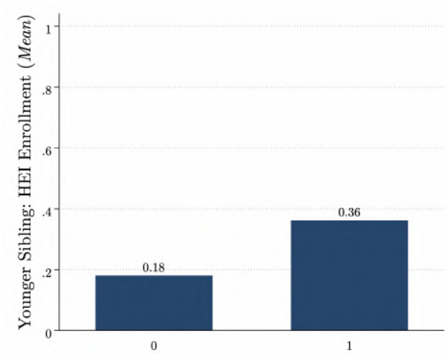
Panel A of Figure 1 plots younger siblings' scores on the national high school exit exam

Figure 1: Sibling Correlations in Education, Labor, and Crime

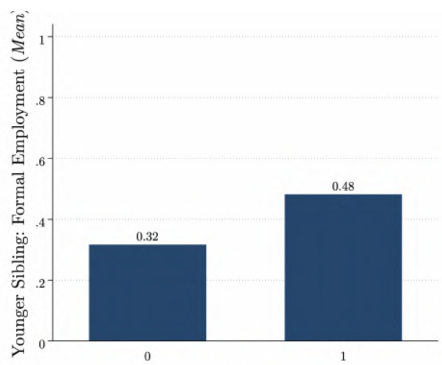
Panel A. *Saber 11*



Panel B. HEI Enrollment of Younger by Older's Enrollment

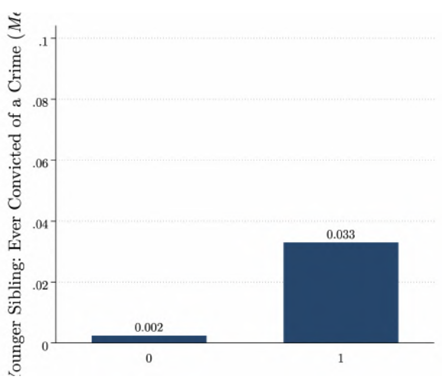


Panel C. Formal Employment (5y after HS): Younger by Older

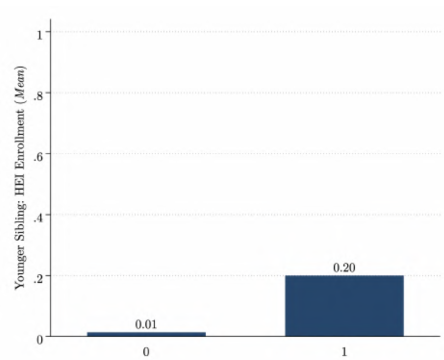


Panel D. Earnings (6y after HS): Younger by Older

Panel E. Crime Involvement: Younger by Older



Panel E. *Ser Pilo Paga* Recipient: Younger by Older



Notes: Each panel displays within-family correlations between older and younger siblings for key outcomes. Panel A plots older siblings' standardized *Saber 11* scores against those of their younger siblings; both variables are normalized to have mean zero and standard deviation one within each test cohort. Panel B shows the share of younger siblings enrolled in higher education by whether the older sibling enrolled in any higher education institution (HEI). Panel C reports the share of younger siblings formally employed six years after high school, conditional on the older sibling's formal employment status at the same horizon. Panel D presents mean earnings of younger siblings (in multiples of the minimum wage) by the older sibling's earnings six years after graduation. Panel E shows the probability that a younger sibling has any recorded criminal conviction by the older sibling's conviction status.

(*Saber 11*) against those of their older siblings. The fitted line shows a strong positive within-family correlation of 0.44, indicating that siblings' academic achievement is closely related. However, the slope is flatter than the 45-degree line, suggesting that when the older sibling performs very well, the younger sibling tends to perform slightly worse, and conversely, when the older sibling scores poorly, the younger sibling performs somewhat better. This pattern is consistent with partial convergence in performance within families rather than perfect replication of academic outcomes.

Panel B examines correlations in higher-education enrollment. Among younger siblings whose older sibling did not enroll in a higher education institution (HEI), only 18 percent enrolled themselves. In contrast, when the older sibling pursued higher education, 36 percent of younger siblings did so as well—double the rate. Consistent with this pattern, the correlation in immediate college enrollment is 0.18, and correlations across HEI types range between 0.13 and 0.19, with similar magnitudes for public and private institutions and for low- and high-quality HEIs. These correlations suggest that the link between siblings' college attendance may capture both intangible mechanisms—such as aspirations or information sharing—and tangible ones, such as shared financial capacity to fund higher education.

A similar pattern emerges in labor-market outcomes six years after high school. Panel C shows that when the older sibling is formally employed, about 50 percent of younger siblings also hold formal jobs; this share falls to 33 percent when the older sibling is not formally employed. Correspondingly, the correlation in formal employment is 0.16, and in monthly earnings—measured in multiples of the minimum wage—it is 0.17. These values are smaller than the 0.4–0.5 range typically reported for long-term sibling income correlations in the United States (e.g., Solon, 1999; Mazumder, 2008), where brothers exhibit correlations of 0.47–0.54 in earnings and wages and sisters show values around 0.34–0.45. The lower magnitudes observed in our data likely reflect both life-cycle and contextual factors: individuals in our sample are observed early in their careers—five to six years after completing high school—before earnings trajectories stabilize, and the sample is restricted to low-income, Sisben-eligible households, which exhibit limited between-family variation in resources. In addition, Colombia's labor market is characterized by high informality and mobility, which further attenuates the role of family background in explaining early labor outcomes.

Appendix Table 3 and Panel D documents sibling correlations in criminal behavior. The overall correlation in any criminal conviction is 0.034, with similar magnitudes across specific offense categories. Although these values are modest, they are meaningful given the low baseline prevalence of criminal activity in the data. The probability of ever being convicted of a crime in our sample is around 0.0025 for younger siblings—roughly half the rate observed for adult men in Brazil, where the probability of prosecution for any crime is approximately 0.0052 and even lower for specific offenses such as economic (0.0014) and violent crimes (0.0015) (Britto et al., 2022).

Differences in exposure windows also help explain the relatively low correlations and the low overall probability of conviction in our sample. Because criminal records are observed through 2023, older siblings—who are on average six years older—have had more time to appear in judicial records than their younger siblings. Appendix Figure 9 shows that sibling correlations in criminal convictions increase with the age of the younger sibling's cohort: for instance, the correlation reaches 0.15 when younger siblings were born around 1991. This pattern indicates that the smaller overall correlation is largely driven by the limited exposure of younger cohorts rather than by weaker familial clustering in criminal behavior. Because younger siblings have had fewer years of potential exposure to criminal behavior, our estimates represent lower bounds of the true long-run sibling correlations in crime. As these cohorts age and additional administrative years become available, we expect the measured correlations to rise, consistent with the pattern already visible among older cohorts.

Finally, correlations in *Ser Pilo Paga* (SPP) participation are 0.20, consistent with strong within-family similarities in high academic achievement among low-income students. Because eligibility depends on cohort-specific merit thresholds, this relatively high value suggests that families play an important role in shaping not only educational access but also the likelihood of benefiting from competitive scholarship programs.

Overall, these results reveal consistent within-family correlations across education, employment, and behavioral outcomes, even in early adulthood. Correlations are strongest in cognitive and academic outcomes (≈0.44), moderate in education and labor outcomes (≈0.17–0.19), and smaller but still positive in crime involvement (≈0.03). Relative to the U.S., where sibling correlations in education and long-term earnings often exceed 0.5–0.6, our estimates imply that family background explains a smaller fraction of inequality among poor Colombian households. This difference likely reflects a combination of (i) the early age window observed, (ii) the socioeconomic homogeneity of the Sisben sample, (iii) the volatility of early labor outcomes, and (iv) greater contextual heterogeneity and informality in the Colombian labor market. Together, these patterns provide an important benchmark for interpreting the causal sibling spillover effects.

4.2 Motivating Evidence: Spillovers and Breakdown of Local Continuity at the SPP Cutoff

A natural starting point to study how the *Ser Pilo Paga* (SPP) program affected the siblings of eligible students is to exploit its merit-based eligibility cutoff using a regression discontinuity (RD) design. The program offered student loans that converted into full scholarships upon graduation for high-achieving, low-income students whose *Saber 11* scores exceeded a national threshold. A valid RD design would require that, around this cutoff, potential outcomes for younger siblings evolve smoothly—so that those just below the threshold serve as valid counterfactuals for those just above it. Figure 2 plots younger siblings' educational outcomes against their older siblings' standardized *Saber 11* scores, normalized relative to the SPP eligibility cutoff (values ≥ 0 indicate eligibility). Each panel compares younger siblings' outcomes for pre- and post-program cohorts of older siblings (2010 and 2014) and reports local polynomial RD point estimates with robust bias-corrected confidence intervals following Calonico, Cattaneo, and Titiunik (2014).

Panel A presents results for younger siblings' *Saber 11* performance. In the pre-program period (2010), outcomes are smooth across the cutoff, and the RD estimate is small and statistically insignificant (0.048, SE = 0.069, $p = 0.48$), confirming continuity in the absence of treatment. After the introduction of SPP (2014), the fitted relationship steepens just below the cutoff, and the RD estimate becomes significantly negative (-0.11, SE = 0.048, $p = 0.022$). This negative discontinuity does not necessarily indicate a deterioration in performance. In fact, Appendix Table 4 shows that, within the RD bandwidth, younger siblings of students taking the test in 2014 (post-period) perform on average 0.10 standard deviations better than their counterparts in 2010 (pre-period), a difference that is statistically significant at the 1% level. Above the cutoff, younger siblings of post-period students also perform better than those of pre-period students, with a mean difference of 0.17 standard deviations, likewise significant at the 1% level. These results suggest that the negative RD estimate may be driven by upward improvements among younger siblings just below the threshold—consistent with indirect benefits for families with an almost-eligible student. When near-eligible households experience positive spillovers, the RD estimate can mechanically understate the true effect of the program and even reverse sign, as observed here.

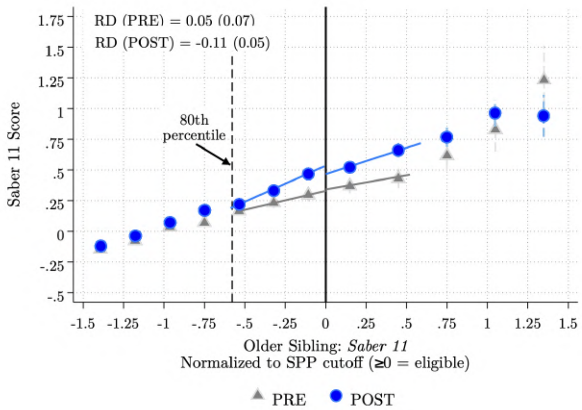
Panels B–D present analogous patterns for postsecondary enrollment outcomes. For overall higher-education enrollment (Panel B), the post-SPP RD estimate is -0.10 (SE = 0.040, $p = 0.012$), compared to an insignificant 0.013 (SE = 0.051, $p = 0.79$) before the program. These results again suggest strong local spillovers below the cutoff that bias the RD toward zero or negative values. Below the threshold, we observe a steeper slope in the probability of enrollment—implying that younger siblings' outcomes became more responsive to the older sibling's performance after SPP was introduced. This pattern is consistent with enhanced within-family transmission of educational aspirations, information, or motivation once the scholarship opportunity became available.

Furthermore, Panels C and D show that these changes are not confined to the immediate neighborhood of the cutoff. In the post-SPP period, shifts for younger siblings of eligible students emerge farther above the threshold: they are less likely to attend public HEIs and more likely to enroll in private ones. This suggests that the program's effects—and the resulting behavioral adjustments within families—extend beyond the local discontinuity in eligibility.

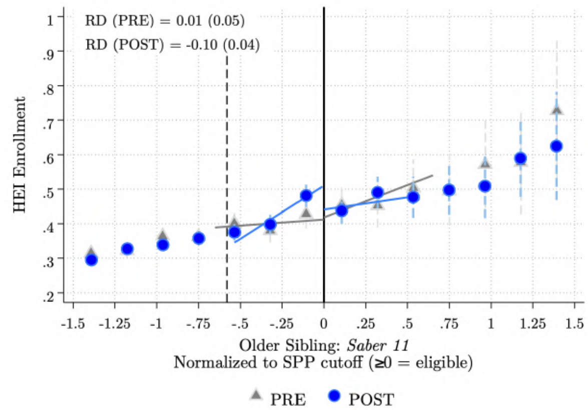
Together, these patterns indicate that the standard RD identifying assumptions do not hold in this setting. The descriptive evidence suggests that younger siblings' outcomes below the cutoff are affected by indirect exposure to treatment, and the treatment effect itself is not strictly local but spreads across a wider range of scores. For this reason, we do not rely on an RD design. Instead, the next section introduces an empirical strategy that exploits variation in older siblings' eligibility across cohorts and households—before and after the introduction of SPP—to estimate both the effect of having an older sibling almost eligible for the program and the spillover effect of having a sibling who actually qualified for SPP.

Figure 2: Younger Siblings' Education Outcomes Around the *Ser Pilo Paga* Eligiblity Cutoff

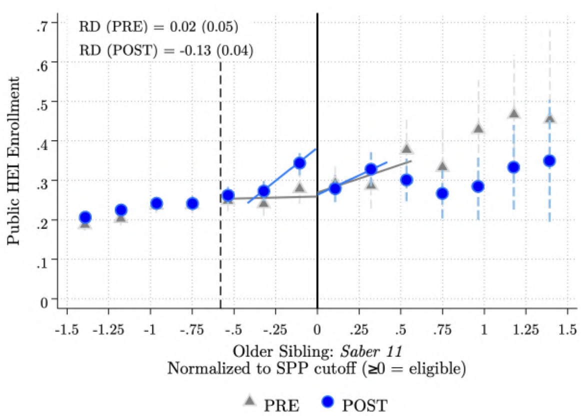
Panel A. *Saber 11* (Standardized)



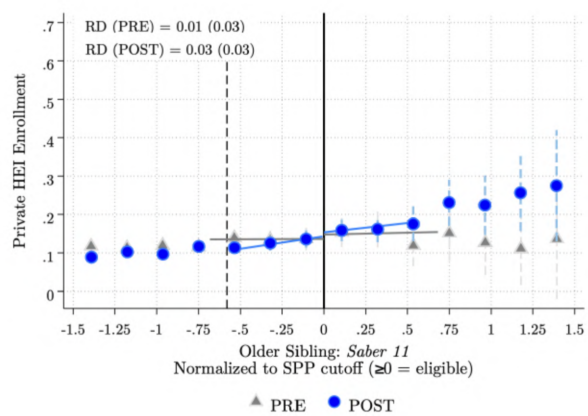
Panel B. Postsecondary Enrollment



Panel C. Postsecondary Enrollment in Public HEIs



Panel D. Postsecondary Enrollment in Private HEIs



Note: Each panel compares younger siblings' educational outcomes for pre- and post-program cohorts of older siblings (2010 and 2014). Outcomes are plotted against the older sibling's standardized *Saber 11* score, normalized relative to the *Ser Pilo Paga* (SPP) eligibility cutoff (values ≥ 0 indicate eligibility). Blue lines correspond to the post-program period (2014), and gray lines to the pre-program period (2010). The solid vertical line marks the SPP cutoff, and the dashed line at the 80th percentile highlights where slope changes in younger siblings' outcomes begin to emerge below the threshold. Each plot displays local polynomial Regression Discontinuity (RD) estimates with robust bias-corrected confidence intervals following Calonico et al. (2014). *Post-secondary enrollment* is a binary variable equal to 1 if the student enrolls in a higher education institution (HEI) in the academic year immediately following high school graduation. Pre-program outcomes evolve smoothly across the cutoff, while post-program patterns display steeper slopes below the threshold and noticeable shifts above it, suggesting sibling spillover effects that extend beyond the local discontinuity.

5 Empirical Strategy

We exploit the introduction of *Ser Pilo Paga* (SPP) in 2014 as a natural experiment to estimate spillover effects on younger siblings. The program assigned college scholarships based on two criteria: a merit cutoff on the *Saber 11* exam and a poverty threshold based on the Sisben index. We leverage the merit cutoff to construct three groups defined by the older sibling's score: children whose older sibling qualified for SPP ($T1$), children whose older sibling narrowly missed eligibility but scored above the 80th percentile ($T2$), and a control group whose older sibling scored between the 70th and 80th percentiles. Comparing how outcomes for younger siblings in these groups evolved before and after 2014 identifies the causal effect of having an older sibling eligible — or nearly eligible — for college financial aid.

We estimate the following first-stage equation:

$$\begin{aligned} SPP^{OS}_{ismt} &= \theta_s + \delta_{mt_{os}} + \alpha_1 T1_i + \alpha_2 T2_i \\ &\quad + \underbrace{\sum_{\substack{k=2010 \\ k\neq 2013}}^{2020} \boldsymbol{\beta_{1k}}(T1_i \times \delta_{t_{os}=k})}_{\text{Eligible Older Sibling}} + \underbrace{\sum_{\substack{k=2010 \\ k\neq 2013}}^{2020} \boldsymbol{\beta_{2k}}(T2_i \times \delta_{t_{os}=k})}_{\text{Almost-eligible Older Sibling}} + X_i' \times \delta_t + \epsilon_{ismt} \end{aligned} \tag{1}$$

SPP^{OS}_{ismt} is an indicator equal to one if the older sibling of child i, enrolled in school s in municipality m, received an SPP scholarship in cohort t. We also estimate the first stage using the older sibling's probability of enrolling in a higher education institution (HEI). Cohort t is defined by the year in which the older sibling sat the *Saber 11* exam. Cohorts from 2014 onward were exposed to the program; earlier cohorts serve as pre-program comparisons. $T1_i$ indicates that the older sibling scored above the SPP eligibility cutoff; $T2_i$ indicates that the older sibling scored just below the cutoff but above the 80th percentile. The control group consists of families whose older sibling scored between the 70th and 80th percentiles, well below the threshold but comparable in academic preparation and family background.

Figure 3 illustrates the construction of these groups across cohorts. For each cohort from 2009 to 2017, it plots the distribution of older siblings' *Saber 11* scores normalized by that year's cutoff, with shaded regions indicating the eligible ($T1$), almost-eligible ($T2$), and control groups. Pre-program panels (2009–2013) use the 2014 cutoff for normalization.

The terms $\delta_{t_{os}=k}$ are cohort indicators relative to 2013, the last pre-program cohort. The coefficients β_{1k} and β_{2k} trace the dynamic effects of having an eligible or almost-eligible older sibling, relative to 2013. All specifications include municipality-by-cohort and school fixed effects. The vector X_i' comprises maternal education categories and household socioeconomic stratum, each interacted with cohort fixed effects to allow outcome trajectories to vary across families with different socioeconomic backgrounds. We additionally control for the number of siblings interacted with cohort fixed effects, given the documented relationship between family size and educational investment per child (Black et al., 2005). Standard errors are clustered at the family level.

We then estimate the reduced-form equation:

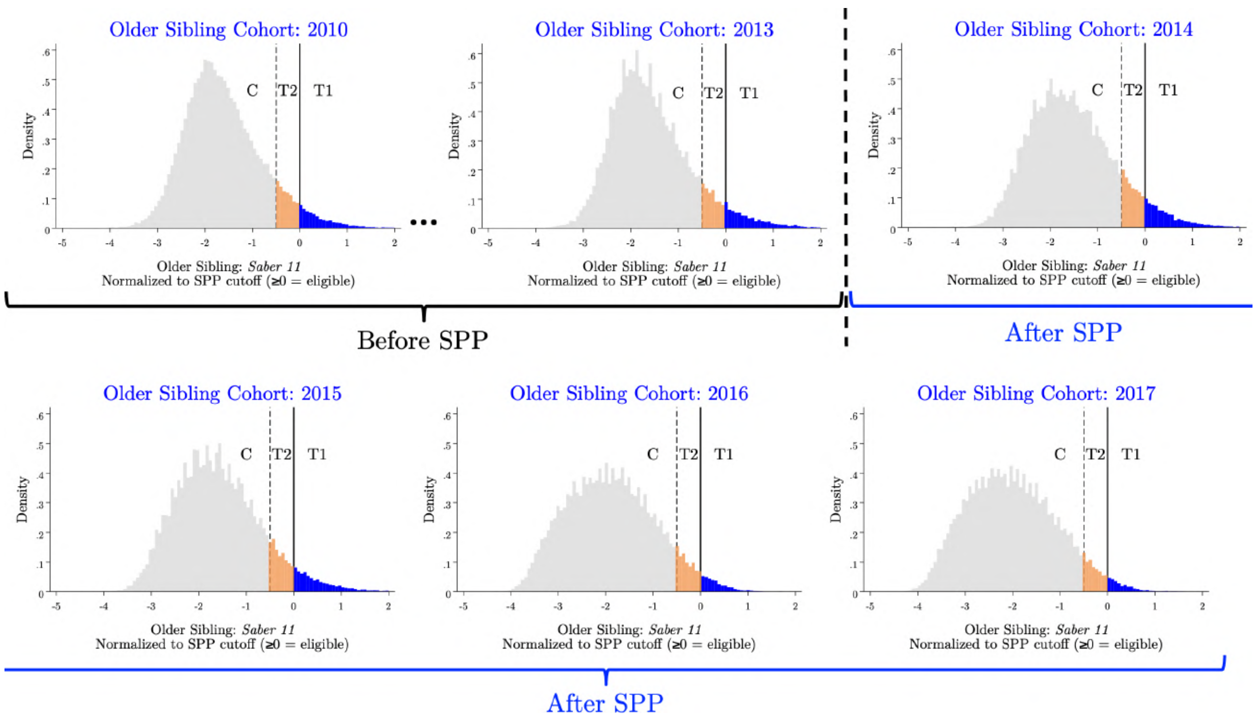
$$
\begin{aligned}
Y_{ismt} = \theta_s + \delta_{mt_{os}} + \alpha_1 T1_i + \alpha_2 T2_i \\
+ \underbrace{\sum_{\substack{k=2010 \\ k \neq 2013}}^{2020} \boldsymbol{\beta_{1k}} (T1_i \times \delta_{t_{os}=k})}_{\text{Eligible Older Sibling}} + \underbrace{\sum_{\substack{k=2010 \\ k \neq 2013}}^{2020} \boldsymbol{\beta_{2k}} (T2_i \times \delta_{t_{os}=k})}_{\text{Almost-eligible Older Sibling}} + X_i' \times \delta_t + \epsilon_{ismt}
\end{aligned}
\tag{2}
$$

where Y_{ismt} is the outcome of younger sibling i, enrolled in school s in municipality m, whose older sibling sat the *Saber 11* exam in cohort t. We study outcomes across three domains. We begin with higher education outcomes, including enrollment, institutional quality, and completion, as these are most directly linked to the treatment: a college scholarship for the older sibling may shift the younger sibling's educational aspirations, information set, or household resources in ways that affect college-going behavior. We then examine labor market outcomes, including employment and earnings from zero to six years after high school graduation. For younger siblings born after 2007, we additionally examine criminal convictions and offense type (economic, drug-related, or violent).[5]

The key identifying assumption is that, absent SPP, younger siblings' outcomes would have evolved similarly across cohorts for the eligible ($T1_i$), almost-eligible ($T2_i$), and control groups. This assumption does not require random assignment of college financial aid; it requires that the introduction of SPP was the source of differential change across groups after 2014.

[5] Appendix Section B provides detailed definitions and timing for all outcomes.

Figure 3: Illustration of the Empirical Strategy: Distribution of Older-Sibling *Saber 11* Scores by Cohort.



Note: Each panel shows the distribution of older siblings' *Saber 11* scores, where each score is linked to the younger siblings in the sample whose treatment status is defined by that score. Panels correspond to the cohort in which the older sibling took the *Saber 11* exam, and the distributions are normalized by the relevant year's cutoff. The blue, orange, and gray shaded areas indicate the eligible ($T1$), almost–eligible ($T2$), and control groups, respectively. Panels for 2010–2013 are labeled *Before SPP*, and those for 2014 onward as *After SPP*. Because no cutoff existed prior to 2014, scores for pre-SPP cohorts are normalized using the 2014 cutoff to ensure comparability across cohorts.

5.1 Validity of the Identifying Assumptions

The key identifying assumption underlying Equations (1) and (2) is that, in the absence of *Ser Pilo Paga*, the outcomes of younger siblings would have evolved similarly across the eligible ($T1$), almost-eligible ($T2$), and control groups. A potential threat to this assumption is that the composition of families near the *Saber 11* eligibility cutoff changes differentially across groups after the introduction of the program. If families respond to SPP by strategically investing in their older child's exam preparation or sorting into schools with stronger academic preparation, the eligible group may drift toward higher-educated and higher-income households relative to controls after 2014, confounding our estimates.

Table 1 investigates this concern by estimating Equation (1) with 27 pre-determined family characteristics as outcomes and reporting the coefficients on β_{1k} and β_{2k}. Six of the

27 characteristics show statistically significant differential trends at the 5 percent level for at least one of the two treatment arms. The pattern is similar across groups: both the eligible and almost-eligible groups show statistically significant increases in the shares of mothers with technical and university degrees, and a corresponding decline in the share with secondary education. This is consistent with higher-educated families near the eligibility cutoff becoming increasingly likely to have a child scoring in the relevant range after the program becomes salient. The remaining significant coefficients are concentrated in upper SES strata but are negligible in magnitude: the largest represents a shift of less than one percentage point in a stratum comprising 1 percent of the sample. The 21 remaining covariates, including family size, household assets, poverty index, school enrollment, and sibling demographics, show no evidence of differential compositional change. These results motivate our inclusion of interactions between maternal education and socioeconomic status with cohort fixed effects [$X_i' \times \delta_t$ in Equations (1) and (2)], which allow outcome trajectories to vary flexibly across families with different socioeconomic backgrounds, reducing the scope for the observed compositional shifts to confound our estimates.

The timing of the program's announcement provides additional support for the identifying assumption. Londoño-Vélez et al. (2020) document that SPP was announced in October 2014, two months after students had already sat the *Saber 11* exam. Since the exam took place before the announcement, families in the 2014 cohort had no opportunity to respond to the program's incentives by investing in exam preparation or sorting above the cutoff. To the extent that compositional sorting occurs, it is more likely concentrated in cohorts from 2015 onward. The 2014 cohort provides a clean test of the identifying assumption, free from behavioral responses to the program. We further assess the validity of the parallel trends assumption by examining whether the pre-period coefficients β_{1k} and β_{2k} for $k < 2013$ are statistically distinguishable from zero in the event-study specification.

6 Results: Sibling Spillover Effects of College Financial Aid

A merit scholarship that sends a low-income student to a selective university changes the household along multiple dimensions: it reshapes the educational trajectory the older sibling follows, changes the financial position of a family that is now supporting a college-going child, and reduces the time and resources the older sibling contributes to the household as they transition into full-time study. These changes may affect younger siblings through distinct channels: by relaxing or tightening household resource constraints, by changing the educational example the older sibling provides, or by shifting what younger siblings believe is achievable for someone like them. We first characterize the nature and magnitude of this household-level shock by documenting how SPP eligibility changes older siblings' outcomes across three domains: postsecondary enrollment and institutional quality, field of study and degree completion, and labor market trajectories. We then estimate the reduced-form effects on younger siblings' postsecondary enrollment, labor market outcomes, and criminal convictions.

6.1 First Stage: Characterizing the Older Sibling's Response

We begin by documenting the magnitude of the household-level shock.

Educational Enrollment and Institutional Quality. SPP eligibility changes older siblings' postsecondary trajectories at both the extensive and intensive margins. Table 5 reports the pooled first stage estimates. Figure 10 shows the event-study coefficients, which are small and statistically indistinguishable from zero before 2014 and rise sharply thereafter, consistent with parallel pre-trends.

The first stage is large. The probability of having an SPP recipient in the household rises by 52.0 percentage points among younger siblings with an eligible older sibling ($T_1 \times Post$). Much of this effect operates at the extensive margin: the probability of having a college-going older sibling rises by 19.2 percentage points, a 48 percent increase relative to the baseline enrollment rate of 40.2 percent, meaning that a substantial share of the gain comes from families that would not have sent the older sibling to college in the absence of the scholarship.

Beyond whether older siblings go to college, SPP changes where they go. The program redirects enrollment away from low-quality institutions and toward high-quality private universities. Low-quality enrollment falls by 10.7 percentage points, a 51 percent decline relative to baseline. Among high-quality institutions, the gain is concentrated in the private sector: high-quality private enrollment rises by 37.4 percentage points, a 570 percent increase relative to a baseline of 6.6 percent, while high-quality public enrollment falls by 7.5 percentage points. Program duration shifts in the same direction: four-year enrollment rises by 26.7 percentage points while two-year enrollment falls by 7.5 percentage points.

The almost-eligible group ($T_2 \times Post$) provides a useful contrast. These are students who scored just below the SPP eligibility threshold and did not receive the scholarship, but who may have been indirectly affected as eligible peers exited into private universities. Their overall college-going rises by only 2.4 percentage points, but the key difference from T1 households lies in the public sector. While high-quality public enrollment falls by 7.5 percentage points for T1 households, it rises by 3.3 percentage points for T2 households. High-quality private enrollment increases in both groups, but the gain is larger for T1 households, at 37.4 percentage points, than for T2 households, at 1.4 percentage points. This contrast is consistent with a displacement effect: as SPP recipients move into selective private universities, competitive public institutions become more accessible to students just below the cutoff. We cannot, however, distinguish this channel from informational or aspirational spillovers that independently lead almost-eligible students to target higher-quality public institutions.

Field of Study, Location, and Degree Completion. Table 6 reports the effects on field of study, geographic mobility, and degree completion. STEM enrollment rises by 13.9 percentage points, a 78 percent increase relative to a baseline of 17.8 percent. The probability of enrolling in a higher education institution more than 50 kilometers from the home municipality rises by 18.0 percentage points, a 203 percent increase relative to a baseline of 8.9 percent. SPP requires recipients to attend one of 33 nationally accredited institutions, most of which are located in large cities far from the low-income communities where eligible families live, consistent with geographic displacement for many SPP recipients.

The shift in enrollment translates into degree completion. Four-year program completion rises by 14.2 percentage points, a 26 percent increase relative to a baseline of 55.1 percent, while two-year program completion falls by 14.0 percentage points, a 74 percent decline relative to baseline. Any-HEI graduation rises by only 4.9 percentage points, consistent with the shift being largely compositional: eligible students move from completing shorter technical programs toward completing four-year university degrees rather than only adding new graduates at the extensive margin.

The T2 results sharpen this picture. Among almost-eligible households, STEM enrollment rises by only 1.4 percentage points and university graduation by 3.8 percentage points, consistent with the modest enrollment gains documented in Table 5. We find no significant change in geographic mobility for T2 households. Almost-eligible students access higher-quality public universities that are local to their home communities, while SPP recipients relocate to selective private institutions in major cities. That is, T1 and T2 younger siblings are exposed to different versions of the household shock, and the geographic dimension is one of the differences between them.

Field of Study, Location, and Degree Completion. Table 6 reports the effects on field of study, geographic mobility, and degree completion. STEM enrollment rises by 13.9 percentage points, a 78 percent increase relative to a baseline of 17.8 percent. The probability of enrolling in a higher education institution more than 50 kilometers from the home municipality rises by 18.0 percentage points, a 203 percent increase relative to a baseline of 8.9 percent. SPP requires recipients to attend one of 33 nationally accredited institutions, most of which are located in large cities far from the low-income communities where eligible families live, making geographic displacement a likely consequence of program participation for many recipients.

The shift in enrollment translates into degree completion. Four-year program completion rises by 14.2 percentage points, a 26 percent increase relative to a baseline of 55.1 percent, while two-year program completion falls by 14.0 percentage points, a 74 percent decline relative to baseline. Any-HEI graduation rises by 4.9 percentage points, indicating that the shift is largely but not entirely compositional: most of the gain in four-year completion comes from students who would have completed a two-year program in the absence of SPP, but a share represents new graduates who would not have completed any postsecondary degree.

Among almost-eligible households, STEM enrollment rises by only 1.4 percentage points and four-year program completion rises by 3.8 percentage points, consistent with the modest enrollment gains documented in Table 5. We find no significant change in geographic mobility for T2 households. Almost-eligible students access higher-quality public universities that are local to their home communities, while SPP recipients relocate to selective private institutions in major cities. T1 and T2 younger siblings are exposed to qualitatively different household shocks, with the geographic dimension being among one of the differences between them.

Labor Market Trajectories. Table 7 reports older siblings' formal employment, earnings, and position in the earnings distribution from zero to six years after high school. The coefficients follow a U-shaped pattern for T1 households (see Figure 12): short-run declines during the college years, followed by a recovery that exceeds the control group by year five.

Formal employment falls immediately after high school, reaching its lowest point at year two, when eligible older siblings are 5.5 percentage points less likely to be formally employed than the control group, relative to a baseline of 27.6 percent. By year four the gap closes, and by years five and six employment is significantly above the control group, at 4.0 and 2.9 percentage points respectively. Earnings follow the same trajectory. The earnings gap peaks at year three, at 7.2 percentage points below the control group relative to a baseline of 29.2 percent, before recovering to a 13.7 percentage point gain by year six, measured in multiples of the monthly minimum wage. The probability of reaching the top quartile of the earnings distribution mirrors this pattern, falling through year three and rising to 5.4 percentage points above baseline by year six. The short-run declines reflect the opportunity cost of full-time college attendance: eligible older siblings forgo formal employment during the college years in exchange for substantially higher earnings once they enter the labor market.

Among almost-eligible households, this pattern is different. Employment and earnings also fall in the short run, consistent with some college enrollment among this group. However, neither employment nor earnings recover above the control group within the six-year window. This absence of a recovery for T2 households is consistent with a smaller long-run labor market premium associated with the higher-quality public institutions that almost-eligible students attend relative to the selective private universities that SPP recipients access.

Implications for the Spillover Analysis. SPP eligibility creates two household shocks. Eligible older siblings shift into selective private universities outside their home municipalities, forgo formal employment during the college years, and enter the labor market with higher earnings. Almost-eligible older siblings follow a different path: they access higher-quality public universities within their home municipalities, with smaller and less persistent labor market gains and no significant change in geographic mobility. These shocks change the educational and labor market environment younger siblings are exposed to, both through what they observe at home and through the household's financial position. We now estimate the reduced-form effects on younger siblings' postsecondary enrollment, labor market outcomes, and criminal convictions.

6.2 Spillover Effects on Younger Siblings

We now ask whether the household shocks documented above propagate to younger siblings. We estimate spillover effects for college enrollment and institutional quality, SPP receipt, labor market outcomes, and criminal convictions.

Postsecondary Enrollment and Institutional Quality. Exposure to an SPP-eligible older sibling raises younger siblings' own probability of receiving the scholarship and redirects their postsecondary enrollment toward higher-quality institutions. Table 8 reports the reduced-form estimates. Figure 15 plots the corresponding event-study coefficients, which are small and statistically indistinguishable from zero before 2014 and rise thereafter, consistent with parallel pre-trends.

Among younger siblings in T1 households, the probability of receiving SPP rises by 2.5 percentage points, a 42 percent increase relative to a baseline take-up rate of 6.1 percent (see Figure 14). Overall postsecondary enrollment rises by 2.9 percentage points, a 9 percent increase relative to a baseline of 31.4 percent. The gains are concentrated in high-quality institutions: enrollment in high-quality HEIs rises by 3.8 percentage points, a 23 percent increase above baseline, driven entirely by the private sector (see Figure 16). The shift extends to program duration: four-year enrollment rises by 4.2 percentage points, a 22 percent increase above baseline, while two-year enrollment falls by 1.2 percentage points (see Figure 18). Younger siblings in T1 households therefore enroll more and shift toward longer and higher-quality programs.

Among almost-eligible households, the pattern differs. Overall enrollment shows no significant change, and the SPP receipt effect is not significant, consistent with almost-eligible older siblings not receiving the scholarship. The only significant enrollment effect is in high-quality public institutions, where enrollment rises by 0.8 percentage points, an 8 percent increase above baseline, mirroring the pattern documented for almost-eligible older siblings in the first stage. Four-year enrollment rises by 1.2 percentage points. The gains for T2 younger siblings are smaller and more diffuse than those for T1 younger siblings, consistent with the less dramatic change in the older sibling's educational trajectory in almost-eligible households.

Table 8, Panel B, reports placebo estimates for Sisben-ineligible families. The coefficients for high-quality enrollment, overall enrollment, and four-year programs are small and statistically indistinguishable from zero, providing no evidence that the main effects in Panel A are driven by pre-existing differences near the eligibility threshold. We do observe negative and significant effects on low-quality public enrollment and two-year programs among ineligible families. These results are consistent with a general equilibrium effect: SPP moved a large number of students out of low-quality public institutions, potentially making spots more accessible to all students near the cutoff regardless of Sisben eligibility. The significant placebo effects are confined to low-quality outcomes and do not appear for the high-quality outcomes that are the focus of our analysis.

Field of Study, Location, and Academic Choices. Table 9 reports the effects on field of study, geographic mobility, whether younger siblings attend the same institution or field as their older sibling, and degree completion.

STEM enrollment rises by 1.5 percentage points, a 12 percent increase relative to a baseline of 12.5 percent. The probability of enrolling outside the home municipality rises by 2.7 percentage points, a 33 percent increase relative to a baseline of 8.2 percent, mirroring the geographic mobility documented for older siblings in the first stage. We also examine whether younger siblings attend the same institution or study the same field as their older sibling. Younger siblings are 1.7 percentage points less likely to enroll in the same institution as their older sibling, an 18 percent decline relative to baseline, yet 1.5 percentage points more likely to enroll in the same field of study, a 16 percent increase. This combination is consistent with resource-constrained emulation: SPP granted older siblings access to Colombia's most selective universities, institutions younger siblings cannot access without financial aid, so they instead replicate their older siblings' field choices within the constraints of their own resources.

Turning to degree completion, the sample is restricted to younger siblings who take the *Saber 11* exam in 2018 or earlier. Younger siblings in later cohorts have not had sufficient time to complete a degree by the end of our observation period in 2023. The probability of graduating from any HEI shows no significant change. The probability of completing a university degree rises by 1.5 percentage points, marginally significant, while two-year program completion falls by 2.8 percentage points, a 15 percent decline relative to baseline, consistent with the earlier shift toward four-year enrollment. To the extent that some younger siblings who shifted into four-year programs are still enrolled at the end of our observation window, these estimates may understate the long-run graduation effect.

Panel B reports placebo estimates for Sisben-ineligible families. Field of study, geographic mobility, and same-field results are all statistically indistinguishable from zero, providing no evidence that these effects in Panel A reflect pre-existing differences near the eligibility threshold. We do observe a significant negative effect on same-institution enrollment (-2.7 p.p.) and a marginally significant negative effect on university graduation (-3.9 p.p.) among ineligible T1 families. However, the university graduation effect moves in the opposite direction from the main result in Panel A. We interpret the same-institution and university graduation results with more caution than the remaining outcomes, where the placebo is clean.

Labor Market Outcomes. Tables 10–12 report the effects on formal employment, wages, and the probability of reaching the top quartile of the earnings distribution, from zero to six years after high school graduation (see Figures 21–23).

For T1 younger siblings, formal employment and wages both display a U-shaped pattern: significant short-run declines during the college years, followed by a tentative recovery that is positive but not yet statistically significant within the six-year observation window. Formal employment falls by 1.4 percentage points at year one, significant at the 1 percent level relative to a baseline of 10.2 percent, and by 2.3 percentage points at year four, significant at the 5 percent level relative to a baseline of 37.9 percent. Wages follow the same pattern, falling by 1.0 percentage point at year one and by 3.0 percentage points at year four, both significant at the 5 percent level. From year five onward, both formal employment and wages turn positive, though neither reaches statistical significance within our observation window. The short-run declines concentrate in the years when younger siblings are most likely enrolled in higher education, consistent with delayed labor-market entry rather than weaker employability. The dip at year four corresponds to the final year of a four-year program, when students who shifted into longer degree programs are still enrolled and have not yet entered the labor market. Whether these educational investments translate into long-run earnings gains remains an open question. Younger siblings who shifted into four-year programs starting in 2015 would graduate no earlier than 2019, leaving limited time to accumulate labor market experience within our data window.

The T2 pattern differs. Almost-eligible younger siblings also show significant negative effects on formal employment at year zero and year one, consistent with some college enrollment among this group. However, the effects disappear from year two onward, and there is no evidence of recovery at any later horizon. The absence of a recovery for T2 households, in contrast to the positive late-period coefficients for T1 households, is consistent with the smaller and less persistent educational investments documented above for almost-eligible younger siblings.

Placebo estimates for Sisben-ineligible families are small and statistically indistinguishable from zero across all years and all three outcomes. The pattern of significant negative effects across four years in Panel A has no analog in the placebo, providing no evidence that the labor market results reflect pre-existing differences near the eligibility threshold.

Criminal Outcomes. Exposure to an SPP-eligible older sibling reduces the probability that a younger sibling is convicted of a crime. Table 13 reports the estimates and Figure 24 plots the corresponding event-study coefficients.

The probability of any conviction falls by 1.7 per 1,000 younger siblings relative to a pre-program baseline of 2.4 per 1,000, a 71 percent reduction, significant at the 1 percent level. The total number of convictions falls by 2.6 per 1,000, an 82 percent reduction, consistent with the effect operating primarily at the extensive margin: fewer younger siblings enter criminal activity rather than a reduction in repeat offending among those already convicted. The reductions are concentrated in economically motivated and drug-related offenses. Violent crime convictions show no significant change. This pattern is consistent with higher opportunity costs of criminal activity among younger siblings who invest more in education. Among almost-eligible younger siblings, the probability of any conviction falls by 0.9 per 1,000, significant at the 5 percent level. However, effects across individual crime categories and the total number of convictions are not statistically significant. Panel B reports placebo estimates for Sisben-ineligible families. All coefficients are small and statistically indistinguishable from zero for both T1 and T2 across all crime categories, providing no evidence that the crime reductions in Panel A reflect pre-existing differences near the eligibility threshold.

7 Mechanisms

The reduced-form results show that exposure to an SPP-eligible older sibling improves younger siblings' educational outcomes, delays their labor market entry, and reduces criminal convictions. These effects could operate through two broad classes of mechanisms. The first involves changes in household resources: if the scholarship relieves tuition costs previously paid out of pocket, it may free resources that benefit younger siblings through better living conditions, school inputs, or reduced pressure to work. However, because SPP primarily operates at the extensive margin, shifting older siblings from non-enrollment to enrollment at selective private universities, the direction of the resource effect is ambiguous. For many families in our sample, the relevant counterfactual is not a student who would have enrolled anyway and now pays less, but a student who would not have enrolled at all and whose college attendance now introduces new costs that the scholarship only partially offsets. The second involves non-resource mechanisms: observing an older sibling access higher education may shift younger siblings' beliefs about what is attainable for someone from their background, raising aspirations and effort regardless of any financial transfer. We present evidence that the patterns are more consistent with non-resource mechanisms than with household liquidity or income redistribution as the primary driver.

7.1 Role Model and Aspiration Effects

Four pieces of evidence support a role model and aspiration interpretation. First, younger siblings show improvements in academic performance as early as age eight or nine, before program-specific information about scholarships or financial aid could influence their behavior. Second, within a subsample of families whose older sibling attends the same higher education institution, spillover effects on younger siblings emerge only when the older sibling performs well academically, not from eligibility alone. Third, heterogeneous effects are largest where the older sibling's signal is most novel and direct: among siblings close in age, in first-generation college families, and in same-gender pairs. Fourth, almost-eligible younger siblings, whose older siblings did not qualify for the scholarship and received no financial transfer, show significant enrollment gains and reductions in criminal convictions, indicating that a financial transfer is not a necessary condition for the spillovers.

Academic Effort and Achievement. Younger siblings in T1 households show improvements in academic performance at ages when information about college admissions or financial aid is not yet relevant. Table 14 reports reduced-form estimates using the 2017 *Saber 3*, *Saber 5*, and *Saber 9* administrations, the only rounds for which individual-level records are available. Because individual-level records are restricted to a single cross-section of younger siblings observed in 2017, the sample covers approximately 8,000 younger siblings taking *Saber 3* (average age 8.8), 14,000 taking *Saber 5* (average age 10.9), and 20,000 taking *Saber 9* (average age 15.2), rather than all cohorts in the main analysis. Among younger siblings taking *Saber 3*, T1 younger siblings score 0.27 standard deviations higher in math and 0.18 standard deviations higher in Spanish. Positive effects persist among *Saber 5* takers, with score gains of 0.18 standard deviations in math and 0.20 standard deviations in Spanish, and a 7 percentage point increase in the probability of never arriving late to class, a 10 percent increase relative to baseline. Following Kraft (2020), the score gains at these ages fall within the medium-to-large range for educational interventions. Among *Saber 9* takers, the effect is significant in math at 0.13 standard deviations but not in Spanish. Children at these ages are several years away from any college decision, and the presence of significant effects as early as age 8 is consistent with a general shift in academic effort and motivation within the household.

Table 15 and Figure 25 report effects on *Saber 11* scores for younger siblings at the end of secondary education. T1 younger siblings score 0.078 standard deviations higher overall, with gains of 0.079 standard deviations in math and 0.087 standard deviations in Spanish, all significant at the 1 percent level. T2 younger siblings score 0.048 standard deviations higher overall, with similar gains across subjects. In T2 families, since their older siblings are not eligible for the scholarship, these effects cannot be attributed to direct income transfers from SPP. Placebo estimates for Sisben-ineligible families are small and statistically indistinguishable from zero across all score outcomes.

The Role of Older Sibling Academic Success. A separate question is whether the specific experience of the older sibling in college matters, or whether exposure to the eligibility threshold alone is sufficient. We answer this using administrative records from one of Colombia's leading private universities, which enrolled more than one-tenth of all SPP recipients nationwide. To test whether the older sibling's specific experience in college matters, we re-estimate Equation 2 on the subsample of need-eligible families whose older sibling enrolled in this university, and augment the specification with an interaction term between the treatment indicator and an indicator for older sibling academic success, defined as a first-semester GPA above the 75th percentile of the cohort distribution. Because all older siblings in this exercise attend the same institution, families face comparable tuition schemes, financial aid arrangements, and academic environments.

Table 16 reports the effects on younger siblings' *Saber 11* scores. The coefficient on T1 $\times$ Post is small and not statistically significant, indicating that eligibility alone does not improve younger siblings' academic outcomes within this subsample. The interaction term T1 $\times$ Post $\times$ Sibling Success captures whether spillover effects are larger when the older sibling's first-semester GPA exceeds the 75th percentile of the cohort distribution, holding institution and financial conditions constant across families. This term is positive and significant across all subjects: 0.294 standard deviations for the overall score, 0.321 standard deviations in math, and 0.255 standard deviations in Spanish. Table 17 shows the corresponding enrollment effects. The main eligibility effect is again not significant. When the older sibling performs well academically, younger siblings are 9.9 percentage points more likely to enroll in any HEI relative to a baseline of 45 percent, 10.4 percentage points more likely to enroll in a high-quality institution, and 8.2 percentage points more likely to enroll in a four-year program. The results are consistent with the older sibling's academic success functioning as a signal that raises younger siblings' own educational investment. Eligibility alone, without visible academic achievement by the older sibling, does not move younger siblings' outcomes in this sample.

Heterogeneity Consistent with Aspiration and Role Model Effects. Tables 18 and 19 report heterogeneous effects across family subgroups for enrollment and crime respectively. Three patterns stand out.

Enrollment effects are concentrated among younger siblings who are close in age to the older sibling. Among siblings with an age gap of four years or less, the T1 enrollment effect is 2.9 percentage points, significant at the 1 percent level and a 9 percent increase above baseline. Among siblings with an age gap greater than four years, the effect is 1.4 percentage points and not statistically significant (Figure 26). Enrollment effects are also concentrated in first-generation college families. Among younger siblings where neither parent has postsecondary education, the T1 enrollment effect is 3.4 percentage points, a 12 percent increase above baseline, significant at the 1 percent level. Among families where at least one parent has postsecondary education, the effect is not significant. When the older sibling is the first in the family to show that higher education is attainable for someone from their background, the spillover effect is stronger.

7.2 Ruling out the Resource Channel

Two predictions should hold if the resource channel were the primary driver of the spillovers. First, effects should be absent in households that do not receive the scholarship. Second, effects should be weaker in households where the resource disruption from the older sibling's college transition is largest. Neither prediction is supported by the data.

SPP operates through two margins. It shifts some older siblings from non-enrollment into selective private universities, and it shifts others from low-quality institutions into higher-quality ones. The direction of the resource effect depends on which margin is relevant for a given family. For families whose older sibling moves from one institution to another, the scholarship may provide genuine financial relief by covering tuition costs they were going to bear. But for families at the extensive margin, where the older sibling would not have attended college without the program, the scholarship introduces new costs rather than relieving existing ones. Even with the stipend, these families commonly bear new expenses related to housing, transportation, and relocation, and anecdotal evidence suggests the stipend is insufficient to cover rent outside the home municipality. Because a substantial share of SPP recipients come from the extensive margin, the resource channel is ambiguous. Moreover, in low-income households, adolescents frequently contribute to household income or caregiving (Edmonds and Schady, 2012; Soares et al., 2012). In fact, 16 percent of older siblings in our sample report being employed during their senior year of high school, suggesting that work was already part of older siblings' lives before the program. The older sibling's departure for college may therefore require younger siblings to compensate through increased domestic or market work (Lafortune and Lee, 2014). For many families in our sample, the net resource effect of the scholarship may be negative rather than positive in the short run.

The T2 results provide the first test of this prediction. Almost-eligible older siblings do not receive the scholarship, and to the extent that they enroll in a higher-quality public institution without financial aid, their college attendance likely introduces new costs for the family. Under a resource channel, this negative income shock should harm younger siblings' outcomes. Instead, we observe the opposite: T2 younger siblings are more likely to enroll in a four-year degree program and 38 percent less likely to have a criminal conviction, with effects concentrated among the poorest families. The resource channel fails to explain these patterns. The T2 results therefore provide direct evidence that the spillovers are driven by the older sibling's educational trajectory itself, not by financial relief.

The heterogeneity by older sibling employment provides the second test. Among families where the older sibling was employed during their senior year of high school, the T1 enrollment effect is 10.5 percentage points, a 36 percent increase above a baseline of 28.9 percent. These would be precisely the households where the resource disruption from the older sibling's transition to full-time college attendance is largest: the household loses a contributing earner while simultaneously bearing the new costs of college attendance. If resource relief were the dominant mechanism, this group should show the weakest effects. The opposite pattern weakens the hypothesis of a resource channel as the primary driver of enrollment spillovers.

Hence, we cannot fully separate informational from aspirational effects, but the evidence is more consistent with non-resource mechanisms, specifically changes in what younger siblings believe is possible and worth pursuing, than with household liquidity or income redistribution as the primary driver.

[Test 1: Extensive vs intensive margin families within T1, to do: Construct a predicted probability that the older sibling would have attended college without SPP, using pre-program characteristics as proxies. goal: under the resource channel, effects should be stronger at the intensive margin because the scholarship provides genuine financial relief for families that were already bearing college costs.]

[Test 2: Poor vs less poor families within T2. to do: interact T2 $\times$ Post $\times$ Poor indicator in Saber 3 and Saber 5 RF. Define poor using Sisben score. Goal: within T2 there is no scholarship transfer, so heterogeneity by poverty status shouldn't be explained by resources. If early academic effects are positive and concentrated among the poorest T2 families $\rightarrow$ older sibling's educational trajectory is more novel and salient as an aspiration signal for poor families. evidence for the aspiration channel because the resource channel has nothing to say about poor vs less poor families when there is no money changing hands]

8 Robustness

8.1 Alternative Control Group Definitions

This section tests how sensitive the results are to alternative control groups. Table 20 reports estimates for younger siblings' performance on the Saber 11 exam. The table includes participation, overall scores, and domain-specific results. Panel A shows the baseline results. The control group is younger siblings of students who scored between the 70th and 80th percentiles on the Saber 11 exam. These students are similar in observable characteristics to the treated groups but are far enough from the program cutoff to be unaffected by it. Panel B uses an alternative control group. Here, the control group is siblings of students who scored between the 60th and 70th percentiles. The estimated effects are slightly larger. This suggests that if some spillovers reach families near the cutoff, the baseline estimates represent a lower bound. Panel C repeats the analysis using siblings of students in the 50th to 60th percentile as the control group. Results remain similar to those in the baseline. The effects on test participation and scores are consistent across all control definitions. The results confirm that the main estimates are stable to reasonable variations in the comparison group.

We next repeat this exercise for higher education enrollment outcomes. Table 21 reports the estimated effects for the probability that younger siblings enroll in higher education and the type of institution and program they attend. Panel A presents the baseline results, where the control group consists of younger siblings of students who scored between the 70th and 80th percentiles on the Saber 11 exam. Panels B and C redefine the control group to the 60th–70th, and the 50th–60th percentiles, respectively. The results remain stable across samples. Younger siblings of eligible students are consistently more likely to enroll in higher education and to attend high-quality institutions. When the control group shifts to students in the 60th–70th percentile, the estimated effects increase slightly, suggesting that if some spillovers extend to families near the cutoff, the baseline provides a lower bound of the true effect.

When we use the 50th–60th percentile as the control group, a new pattern emerges. The probability of enrolling in low-quality institutions, particularly public ones, declines, and enrollment in two-year programs also decreases. This difference reflects who the control students are. Students in the 50th–60th percentile are weaker and more likely to select into low-quality public institutions or shorter programs. In contrast, families near the top of the distribution—those in the 70th–80th percentile—are academically stronger and already inclined toward higher-quality and longer programs. Comparing T1 and T2 households to this stronger control group compresses the difference between them, yielding smaller and sometimes statistically indistinguishable gaps. The baseline estimates are thus conservative, reflecting the proximity of the control group to the treatment threshold. Using lower-scoring controls broadens the comparison and makes the reallocation toward high-quality and four-year programs more visible. Overall, the results for higher education enrollment confirm that the estimated spillovers are robust to alternative definitions of the control group and, if anything, are understated in the baseline specification.

8.2 Machine-Learning–Based Definition of Comparison Groups

The baseline empirical strategy relies on the institutional rules of the *Ser Pilo Paga* (SPP) program. Conditional on being a member of a household eligible based on *need*, eligibility for financial aid is determined by whether the older sibling's *Saber 11* score exceeds the official threshold in a given cohort. The assignment into treatment is not modeled, predicted, or learned: all students above the cutoff are mechanically classified as *Eligible (T1)*. This categorization is taken as exogenous input to the empirical design.

The key challenge arises in defining appropriate comparison groups among *untreated* students. While program rules determine eligibility, there is no institutional analog to the "almost-eligible" or "control" groups. In the main analysis, these comparison groups are constructed using percentile bands, e.g., siblings whose older sibling scored between the 80th–*eligibility cutoff* percentile, the *Almost-Eligible* (T2) families, and the 70th–80th percentile, the *Control* group. These ranges were selected based on observed changes in the distribution of academic outcomes around the 80th percentile after the implementation of the program in 2014. Specifically, we find that younger siblings of students scoring just below the eligibility cutoff exhibit different levels and trends in key outcomes compared to their pre-SPP counterparts, suggesting a structural break in the data generating process following the introduction of the policy. Nonetheless, these choices are not dictated by the program itself and involve researcher discretion.

To evaluate the sensitivity of the results to the choice of percentile-based cutoff bands, we implement an alternative approach that leverages machine learning tools to define groups among untreated older siblings. The idea is simple: we retain the official eligible group, defined by the program's cutoff, and allow the data to guide the classification of untreated students. Specifically, we use clustering techniques to identify which untreated students most closely resemble the eligible group, thereby defining an "*Almost-Eligible (T2)*" group in a data-driven manner. The algorithm then assigns the remaining students to a broader control group composed of individuals who are not as similar to the eligible population but still represent a reasonable comparison group. This clustering-based procedure serves as a robustness check to ensure that the main findings are not driven by arbitrary definitions of percentile ranges but instead remain consistent under a technically grounded, data-driven classification strategy.

Step 1: Constructing a One-Dimensional Index via Linear Discriminant Analysis (LDA). The first step summarizes pre-treatment characteristics into a single index measuring similarity to the eligible population. To construct this index, we estimate a Linear Discriminant Analysis (LDA) model using observable characteristics of older siblings. Let $\boldsymbol{X}_i$ denote a vector of baseline covariates (test score, age, gender, parental education, socioeconomic status, school characteristics, and municipality fixed effects), and let

$$D_i = \begin{cases} 1 & \text{if older sibling is eligible for SPP} \\ 0 & \text{otherwise.} \end{cases}$$

LDA seeks a linear combination of covariates

$$\text{LDA Score}_i = \boldsymbol{w}'\boldsymbol{X}_i$$

that best separates treated from untreated observations. Formally, LDA chooses $\boldsymbol{w}$ to maximize the ratio of between-group to within-group variance:

$$\max_{\boldsymbol{w}} \quad \frac{(\boldsymbol{w}'(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0))^2}{\boldsymbol{w}'\boldsymbol{\Sigma}\boldsymbol{w}},$$

where $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_0$ are the group means and $\boldsymbol{\Sigma}$ is the pooled within-group covariance matrix. The resulting one-dimensional index assigns higher values to untreated students who resemble the eligible population along observable dimensions. Importantly, LDA does *not* modify the treatment indicator. Eligibility is pre-determined by institutional rules, LDA summarizes similarity only among untreated units.

Step 2: Clustering Untreated Units in LDA Space. With each student mapped to an LDA score, we focus exclusively on untreated students ($D_i = 0$). These observations are grouped using K-means clustering applied to their LDA scores:

$$\min_{\{C_k\}} \sum_{k=1}^{K} \sum_{i \in C_k} \|z_i - \mu_k\|^2,$$

where z_i is the LDA score for student i, and μ_k is the centroid of cluster k.

This procedure partitions untreated students into K clusters along the similarity dimension. Since the LDA score is one-dimensional, clustering is transparent and interpretable. To map clusters to empirical categories, we compare each cluster's mean LDA score to that of the eligible group. The cluster with centroid closest to the treated mean is labeled "Almost-Eligible", the next closest is labeled "Control"; remaining clusters are labeled "Other". Treated students retain their mechanically assigned category.

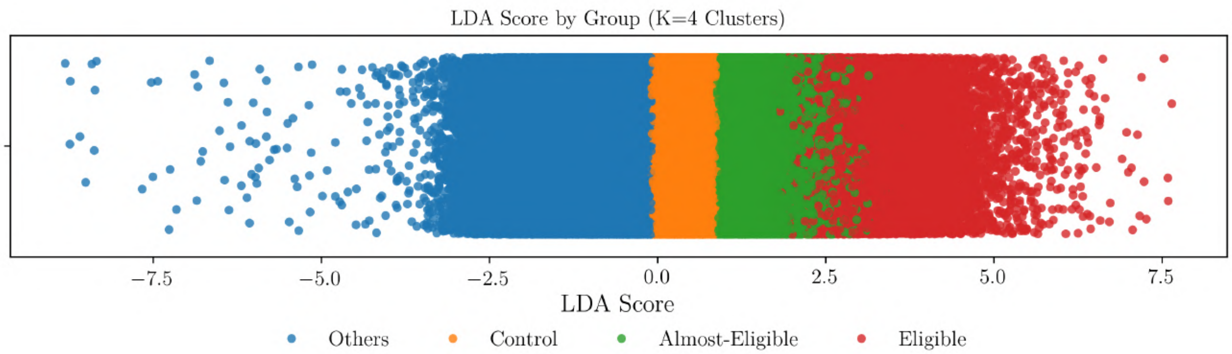
Step 3: Choosing the Number of Clusters. To determine K, we implement the *Elbow Method*: plotting the within-cluster sum of squares (WCSS) against K and looking for the largest decline before marginal improvements diminish. Formally, for $K = 1, 2, \dots, K_{\max}$,

$$\text{WCSS}(K) = \sum_{k=1}^{K} \sum_{i \in C_k} \|z_i - \mu_k\|^2.$$

As shown in Appendix Figure C.2, the curve exhibits an "elbow" at $K = 4$, which we use in the baseline robustness classification.

Step 4: Visualizing Machine-Learned Group Structure. Figure 4 plots the LDA scores of all observations, colored by their machine-assigned groups. The Eligible group (T1) occupies the upper range of the similarity distribution by construction. Immediately below lies the machine-identified "Almost-Eligible" group (T2), followed by "Control" and a large residual category. The visualization illustrates that the data-driven groups are coherent and monotonic in similarity to treated units, supporting interpretability.

Figure 4: Distribution of LDA Scores by Machine-Learning–Defined Groups (K = 4)



Note: The figure plots each observation's LDA score and colors individuals based on their group assignment using K-means clustering among untreated observations. "Eligible" students are determined mechanically based on SPP program rules (score above yearly cutoff). "Almost-Eligible" and "Control" groups are the clusters whose mean LDA score is closest and second-closest to that of eligible students, respectively; remaining clusters form the "Other" group.

Step 5: Re-estimating Spillover Effects. We re-estimate the main event-study and difference-in-differences models replacing the percentile-based comparison groups with the machine-learned definitions. The estimated spillover effects for younger siblings remain quantitatively similar. Tables C.1 and C.2 provide estimates for representative outcomes. This robustness shows that the primary results are not an artifact of how ineligible groups are defined. This exercise preserves the institutional assignment of treatment while removing researcher discretion in constructing comparison groups among untreated students. The close correspondence between main and machine-learning–based estimates reinforces the validity of the baseline empirical design.

9 Conclusion

This paper studies how expanding access to higher education affects other members of the household. Using the eligibility criteria of *Ser Pilo Paga*—a large-scale college financial aid program for high-achieving, low-income students in Colombia—we document that the benefits of financial aid extend beyond direct recipients. When an older sibling becomes eligible for the program, younger siblings experience improvements in educational outcomes, a higher likelihood of college enrollment, suggestive evidence of medium-run improvements in labor market trajectories, along with a notable decline in criminal behavior. These patterns are also visible in educational outcomes, smaller in magnitude, among younger siblings of students who narrowly miss eligibility, suggesting that exposure to opportunity alone can influence family behavior.

The analysis provides insights into how educational investments propagate within families. First, the gains are broadly shared across socioeconomic groups and family structures, indicating that they are not concentrated among households with greater financial capacity. Second, because the program operates on the extensive margin of college participation—moving older siblings from not attending college to attending, often away from home—these effects are unlikely to arise from liquidity relief or relaxed household budgets. Instead, the evidence points toward intangible channels: increased motivation, aspirations, information, and perceived returns to education triggered by observing a sibling succeed academically despite financial constraints. Additional analyses using detailed university and early-schooling data reinforce this interpretation, showing that spillovers emerge early and are stronger when the older sibling performs well in college.

Together, these findings suggest that the impact of higher-education policy can be amplified by social dynamics within families. Financial aid targeted at one student may influence the expectations, effort, and decisions of others who are not directly treated, generating broader social returns than typically captured by program evaluations. At the same time, the results highlight that such spillovers arise in a context where households face persistent resource constraints, suggesting that the motivational and informational dimensions of policy interventions may be as important as their financial components. Recognizing and measuring these indirect effects is therefore essential for understanding the full distributional consequences of higher-education policies and for designing programs that harness family networks as a channel for promoting human capital accumulation.

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10 Appendix Tables and Figures

Figure 5: Sample construction from census to younger-sibling panel

2018 Census
14.24 million households

$\downarrow$

Sisben registry
11.87 million ($\approx$ 83% of households)

$\downarrow$

Meet SPP poverty eligibility
9.94 million need-eligible households

$\downarrow$

$\geq$2 individuals born 1988–2018
5.05 million need-eligible households

$\downarrow$

First-born is observed in *School Census*
2.03 million need-eligible households

$\downarrow$

First-born took *Saber 11* between 2009–2019
968,502 individuals

$\downarrow$

Linked younger siblings
1.69 million ($\approx$ 2.4 per older)

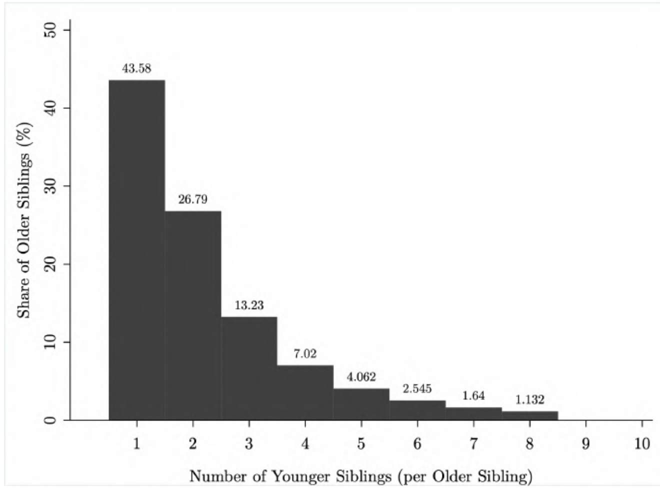
Note: the figure summarizes the sequential steps used to construct the analysis sample. The first box reports the number of households in Colombia from the 2018 Census. The sample construction begins with households listed in *Sisben*, the poverty census, which includes approximately 83 percent of Colombian households. From this registry, we identify household composition and apply the following steps to obtain the final sample of younger siblings: (i) households meeting the SPP poverty-eligibility criteria; (ii) those with at least two members (> 2) born between 1988 and 2018, ensuring that individuals are of schooling age during the years covered by our data; and (iii) those in which the first-born child is observed in the school census (*SIMAT*). The final sample of *older siblings* consists of first-borns who took the *Saber 11* exam between 2009 and 2019 in the eligible households, while *younger siblings* are defined as all other children in the household.

Table 1. Balance Check: Baseline Covariates as Outcomes

	Eligible × Post			**Almost Elig. × Post**			Pre	
	Coef.	SE	p-val.	Coef.	SE	p-val.	mean	N
Year of birth	0.013	(0.036)	0.713	0.004	(0.029)	0.886	2000.193	198,416
Male	0.013	(0.007)	0.065	0.012	(0.005)	0.029	0.477	198,416
Age at Saber 11 exam	-0.177	(0.139)	0.202	-0.068	(0.100)	0.495	17.235	198,416
Age at sibling's exam	-0.008	(0.036)	0.822	-0.007	(0.029)	0.808	12.893	198,416
Mother's education: primary	0.002	(0.006)	0.698	-0.010	(0.006)	0.065	0.260	198,416
Mother's education: secondary	-0.074	(0.008)	0.000	-0.023	(0.007)	0.000	0.492	198,416
Mother's education: technical	0.024	(0.006)	0.000	0.015	(0.004)	0.000	0.120	198,416
Mother's education: university	0.050	(0.005)	0.000	0.018	(0.004)	0.000	0.108	198,416
Mother's education: other	-0.001	(0.002)	0.682	0.001	(0.002)	0.550	0.019	198,416
Poverty index	-0.182	(0.209)	0.383	0.402	(0.166)	0.016	32.093	198,416
SES stratum 1 (poorest)	-0.020	(0.006)	0.002	-0.005	(0.005)	0.312	0.385	198,416
SES stratum 2	0.002	(0.008)	0.743	0.006	(0.006)	0.280	0.447	198,416
SES stratum 3	0.005	(0.006)	0.365	-0.004	(0.004)	0.370	0.156	198,416
SES stratum 4	0.009	(0.002)	0.000	0.001	(0.001)	0.492	0.010	198,416
SES stratum 5	0.003	(0.001)	0.003	0.002	(0.000)	0.000	0.002	198,416
SES stratum 6 (wealthiest)	0.001	(0.001)	0.391	0.000	(0.000)	0.694	0.001	198,416
High-quality HEI in mun.	0.000	(0.000)	0.995	0.000	(0.000)	0.998	0.580	198,416
Enrolled in public school	-0.001	(0.000)	0.077	-0.000	(0.000)	0.112	0.877	198,368
Urban household	0.002	(0.004)	0.627	0.002	(0.003)	0.472	0.799	198,368
Mother employed	0.013	(0.008)	0.099	0.009	(0.006)	0.164	0.443	192,992
Rooms in household	0.005	(0.015)	0.768	0.010	(0.012)	0.407	2.820	192,992
Washing machine in household	-0.000	(0.007)	0.945	-0.005	(0.006)	0.352	0.645	192,992
Computer in household	-0.007	(0.007)	0.319	-0.006	(0.006)	0.332	0.660	192,992
Internet access	0.014	(0.007)	0.051	0.000	(0.006)	0.932	0.516	192,992
Family size	0.005	(0.023)	0.840	-0.001	(0.019)	0.973	4.849	192,570
Children in household	-0.002	(0.018)	0.892	0.009	(0.015)	0.556	2.731	192,570

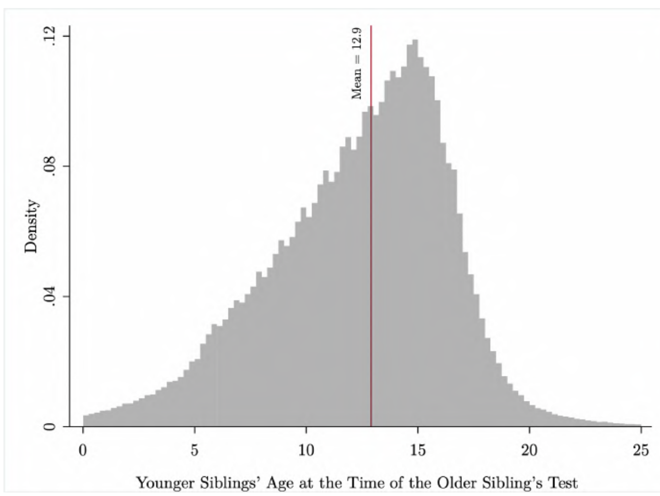
Note: Each row reports coefficients on *Eligible* × *Post* and *Almost Eligible* × *Post* from estimating Equation (1) with a pre-determined family characteristic as the outcome. All specifications absorb municipality-year and school fixed effects and cluster standard errors at the family level. The pre-period mean is computed for the 2013 cohort. The sample is identical to the main estimation sample. Null coefficients indicate that the composition of treatment and control groups does not shift differentially after the introduction of *Ser Pilo Paga*. Source: Authors' calculations based on *Sisben*, *SIMAT*, *Saber 11*, and *PILA*

Figure 6: Number of Younger Siblings Linked to Each Older Sibling



Note: The figure shows the distribution of the number of younger siblings per older sibling in the linked household data. Values on the y-axis represent the share of the 968,502 older siblings with the corresponding number of younger siblings observed in the school census (*SIMAT*)

Figure 7: Age of Younger Siblings at the Time the First-Born Takes the Test



Note: The figure shows the distribution of younger siblings' ages at the time their older sibling takes the *Saber 11* exam. The average age of younger siblings at that time is 12.9 years.

Table 2. Descriptive Statistics of the Sibling Sample

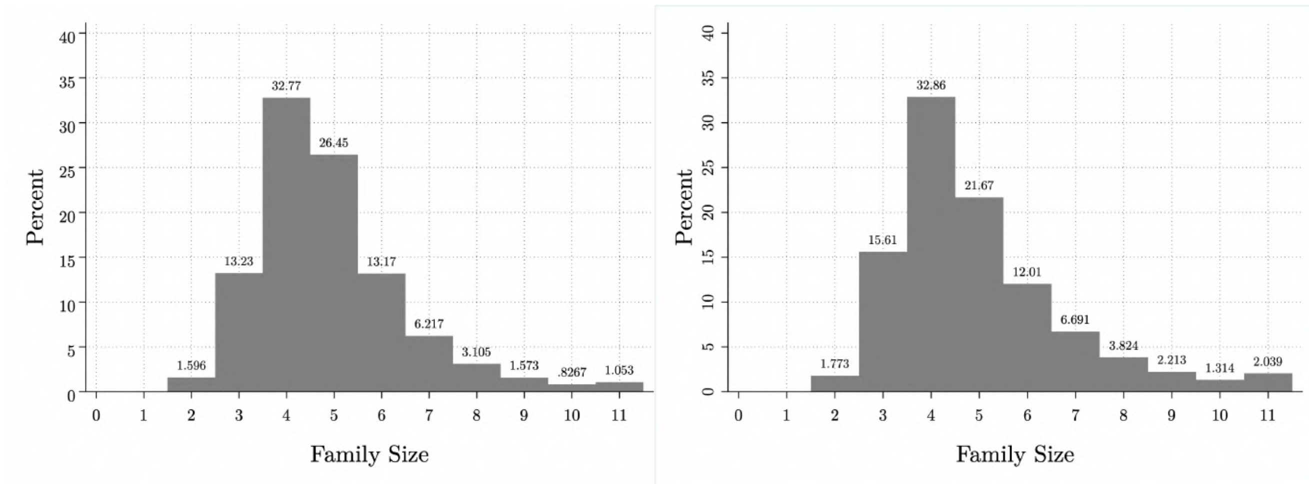
	Older Siblings		**Younger Siblings**	
Panel A. Family Background (Shared)				
	Mean	N	Mean	N
Family size	4.84	769628	4.84	769628
Children in household	2.67	769628	2.67	769628
Mother Education: primary	0.35	769628	0.35	769628
Mother Education: secondary	0.47	769628	0.47	769628
Mother Education: T&T	0.09	769628	0.09	769628
Mother Education: higher	0.07	769628	0.07	769628
Mother Education: other	0.03	769628	0.03	769628
Father Education: primary	0.39	769628	0.39	769628
Father Education: secondary	0.39	769628	0.39	769628
Father Education: T&T	0.06	769628	0.06	769628
Father Education: higher	0.06	769628	0.06	769628
Father Education: other	0.09	769628	0.09	769628
Family Poverty Index	42.88	769628	42.88	769628
High-quality HEI in mun	0.52	769628	0.52	769628
Household SES: stratum 1 (poorest)	0.50	769628	0.50	769628
Household SES: stratum 2	0.38	769628	0.38	769628
Household SES: stratum 3	0.11	769628	0.11	769628
Household SES: stratum 4	0.01	769628	0.01	769628
Household SES: stratum 5	0.00	769628	0.00	769628
Household SES: stratum 6 (wealthiest)	0.00	769628	0.00	769628
Mother stays home	0.57	769628	0.57	769628
Rooms in household	2.74	769628	2.74	769628
Has washing machine	0.61	769628	0.61	769628
Has computer	0.49	769628	0.49	769628
Has internet	0.40	769628	0.40	769628
Has car	0.14	769628	0.14	769628
Panel B. Individual Characteristics of Older and Younger Siblings				
Year of birth	1996.37	968519	2001.69	1694090
Age when older sibling takes test	.	0	12.23	1694090
Average age gap (years)	.	0	5.73	1694090
Age at Saber 11 test	17.72	968519	17.48	989167
SPP-eligible (=1)	0.04	968519	0.03	989167
Public school (=1)	0.91	967166	0.92	1693065
Saber 11 Score (SD)	-0.12	968153	-0.14	986577
Immediate post-secondary enrollment	0.22	966011	0.22	910298
HEI enrollment away from home	0.06	966011	0.06	910298
Two-year HEI degree	0.10	966011	0.10	910298
Four-year HEI degree	0.11	966011	0.11	910298
HEI Field of study: STEM	0.08	966011	0.08	910298
Earnings 6 years after HS	0.44	560245	0.42	296867
Formal employment 6 years after HS	0.42	560245	0.40	296867

Notes: This table reports summary statistics for the linked sibling sample. Columns (1) and (2) display means for older and younger siblings, respectively. Family background variables in Panel A are shared within households, while Panel B reports individual-level characteristics. The stratum system is a government classification of residential areas into six socioeconomic levels, from 1 (poorest) to 6 (wealthiest). The designation is based on the physical characteristics of a neighborhood, such as the quality of housing and surrounding infrastructure, rather than the residents' individual income. Source: Authors' calculations based on *Sisben*, *SIMAT*, *Saber 11*, and *PILA*

Figure 8: Distribution of Household Size in the Sample and the Census of the Poor

Panel A. Linked Sibling Sample (N = 968,519)

Panel B. Census of the Poor (*Sisben*)



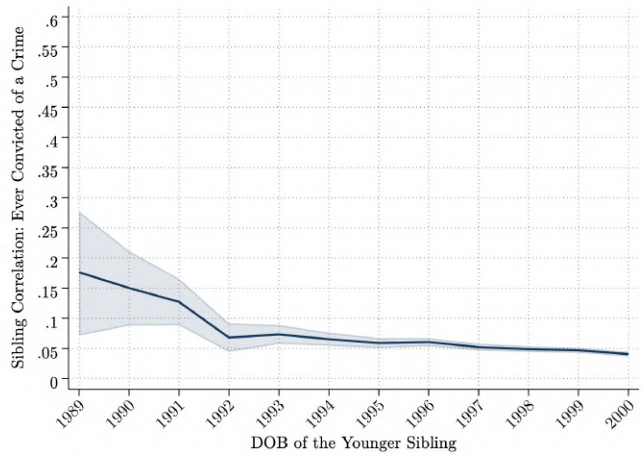
Note: The figure compares the household-size distribution in the linked sample with that in the 2018 *Sisben* census of the poor. The close overlap between the two distributions indicates that the sibling sample is representative of low-income families with children nationwide.

Table 3. Sibling Correlations: Education, College, and Labor Outcomes

Outcome Variable	Sibling Correlation	N
Saber 11: Performance in High School Exit Exam	0.435	986,154
College Enrollment	0.179	908,473
Enrollment in Public HEI	0.138	908,473
Enrollment in Private HEI	0.194	908,473
Enrollment in Low-Quality (Non-Accredited) HEI	0.130	908,473
Enrollment in Public, Low-Quality HEI	0.127	908,473
Enrollment in Private, Low-Quality HEI	0.128	908,473
Enrollment in High-Quality (Accredited) HEI	0.193	908,473
Enrollment in Public, High-Quality HEI	0.149	908,473
Enrollment in Private, High-Quality HEI	0.185	908,473
Enrollment in College Outside Home Municipality	0.176	908,473
Formal Employment (5 Years After High School)	0.166	316,131
Wage (in Monthly MW, 5 Years After High School)	0.172	316,131
Ser Pilo Paga Recipient	0.201	98,636
Any Criminal Conviction	0.034	546,342
Economic or Property Crime	0.018	546,342
Drug-Related Crime	0.028	546,342
Violent Crime	0.031	546,342
Other Crime	0.030	546,342

Notes: The table shows the value of the correlation between outcomes of older and younger siblings.

Figure 9: Sibling Correlation in Crime by Birth-Cohort of the Younger Sibling



Note: Each point represents the correlation coefficient between the crime outcomes of older and younger siblings, computed for all pairs in which the younger sibling was born before the indicated cutoff year. The x-axis therefore shows cumulative cohorts: for example, the 1988 point includes all sibling pairs where the younger sibling was born before December 31, 1988. The shaded area corresponds to 95% confidence intervals. The upward pattern reflects differences in exposure time: older cohorts have had more years to be observed in crime records, while younger cohorts—still at earlier ages—have lower probabilities of conviction and less variation in outcomes, mechanically leading to lower observed sibling correlations.

Table 4. Younger Siblings' Outcomes Below and Above the *Ser Pilo Paga* Cutoff, Pre- and Post-Program

Younger Siblings' Outcomes	Mean (Pre)	Mean (Post)	Diff	SE	p-value
Below Cutoff					
Saber 11 Score	0.237	0.334	0.097	0.018	0.0000
HEI Enrollment	0.399	0.416	0.017	0.013	0.2051
Public HEI Enrollment	0.255	0.303	0.048	0.014	0.0006
Private HEI Enrollment	0.136	0.123	-0.013	0.009	0.1519
Above Cutoff					
Saber 11 Score	0.392	0.568	0.177	0.030	0.0000
HEI Enrollment	0.466	0.456	-0.010	0.023	0.6620
Public HEI Enrollment	0.302	0.299	-0.003	0.022	0.8912
Private HEI Enrollment	0.151	0.164	0.014	0.017	0.4049

Note: This table reports mean outcomes for younger siblings before (2010) and after (2014) the introduction of the *Ser Pilo Paga* program, separately for observations below and above the eligibility cutoff of the older sibling's standardized *Saber 11* score. Means are computed within the optimal bandwidth on each side of the threshold, as determined by the robust local polynomial estimation of Calonico et al. (2014). Columns show the pre- and post-program means, their difference, standard error, and corresponding p-value. Differences capture descriptive changes across periods and do not represent causal effects.

10.1 Appendix: First Stage

Table 5. First Stage: Older Sibling's College Enrollment and Institutional Quality

			Low-quality			High-quality			Program Duration	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	SPP	Any	Any	Public	Private	Any	Public	Private	Two-years	Four-years
Differences-in-Differences Model:										
T1 × Post	0.520***	0.192***	-0.107***	-0.063***	-0.044***	0.299***	-0.075***	0.374***	-0.075***	0.267***
	(0.005)	(0.008)	(0.006)	(0.004)	(0.004)	(0.007)	(0.006)	(0.006)	(0.005)	(0.007)
T2 × Post	0.000	0.024***	-0.023***	-0.011***	-0.011***	0.047***	0.033***	0.014***	-0.016***	0.041***
	(0.001)	(0.006)	(0.005)	(0.004)	(0.003)	(0.005)	(0.004)	(0.003)	(0.005)	(0.005)
Observations	198416	198416	198416	198416	198416	198416	198416	198416	198416	198416
Mean	-	0.4018	0.2079	0.1428	0.0651	0.1939	0.1283	0.0656	0.1730	0.2282
Effect rel. to mean (T1)	-	48%	-51%	-44%	-68%	154%	-58%	570%	-43%	117%
Effect rel. to mean (T2)	-	6%	-11%	-8%	-17%	24%	25%	21%	-9%	18%

Note: Each column reports difference-in-differences estimates from Equation 1. The dependent variables are defined as follows. Column (1) indicates whether the older sibling received an SPP scholarship. Column (2) indicates enrollment in any higher education institution (HEI). Columns (3)–(8) disaggregate by institution type and quality, distinguishing between low- and high-quality institutions, each broken down by public and private sector. Columns (9) and (10) capture the duration of the older sibling's program, distinguishing between two-year and four-year degrees. Low-quality and high-quality institutions are defined according to the *High-Quality Accreditation* granted by the Ministry of Education, as described in Section 2. All specifications include school and municipality-by-cohort fixed effects, and the following family characteristics interacted with cohort fixed effects: number of siblings, maternal education categories, and household socioeconomic stratum. The sample consists of younger siblings in need-eligible households whose older sibling took the *Saber 11* exam between 2009 and 2017. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6. First Stage: Older Sibling's Field of Study, Location, and Degree Completion

	Field of Study				Migration	Graduation Outcomes		
	STEM	STEM Plus	Arts	S.S.H.	HEI Away from Home	Any HEI	University	TVET
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Differences-in-Differences:								
T1 × Post	0.139***	0.151***	0.004**	0.037***	0.180***	0.049***	0.142***	-0.140***
	(0.007)	(0.008)	(0.002)	(0.004)	(0.005)	(0.007)	(0.008)	(0.005)
T2 × Post	0.014***	0.016***	-0.001	0.007***	0.003	0.012*	0.038***	-0.029***
	(0.005)	(0.006)	(0.001)	(0.002)	(0.003)	(0.006)	(0.006)	(0.005)
Observations	198416	198416	198416	198416	198416	198416	198416	198416
Mean	0.1783	0.3552	0.0098	0.0361	0.0885	0.6623	0.5507	0.1882
Effect rel. to mean (T1)	78%	43%	45%	104%	203%	7%	26%	-74%
Effect rel. to mean (T2)	8%	5%	n.s.	18%	n.s.	2%	7%	-15%

Note: Each column reports difference-in-differences estimates from Equation 1. The dependent variables are defined as follows. Columns (1)–(4) capture the field of study of the older sibling, distinguishing between STEM, STEM Plus, Arts, and Social Sciences and Humanities. Column (5) indicates whether the older sibling enrolled in a higher education institution outside their home municipality. Columns (6)–(8) capture graduation outcomes, distinguishing between any HEI, university, and TVET completion. All specifications include school and municipality-by-cohort fixed effects, and the following family characteristics interacted with cohort fixed effects: number of siblings, maternal education categories, and household socioeconomic stratum. The sample consists of younger siblings in need-eligible households whose older sibling took the *Saber 11* exam between 2009 and 2017. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

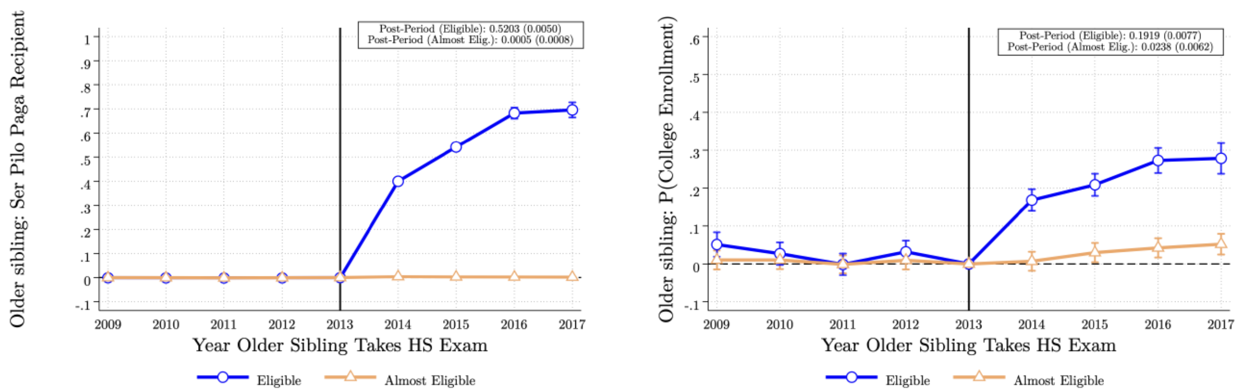
Table 7. First Stage: Older Sibling's Formal Employment, Earnings, and Earnings Distribution

	Formal Employment Ψ years after High School						
	Zero	One	Two	Three	Four	Five	Six
Differences-in-Differences Model:							
Eligible × Post	-0.006***	-0.032***	-0.055***	-0.043***	-0.004	0.040***	0.029***
	(0.002)	(0.005)	(0.007)	(0.008)	(0.008)	(0.009)	(0.010)
Almost Eligible × Post	-0.004**	-0.013***	-0.026***	-0.017***	-0.008	-0.005	0.000
	(0.002)	(0.004)	(0.006)	(0.006)	(0.007)	(0.007)	(0.008)
Observations	178585	178585	178585	178585	178585	165797	147679
Mean	0.0132	0.0984	0.2758	0.3408	0.3934	0.4617	0.5044
	Earnings Ψ years after High School						
	Zero	One	Two	Three	Four	Five	Six
Differences-in-Differences Model:							
Eligible × Post	-0.008**	-0.029***	-0.057***	-0.072***	-0.054***	0.016	0.137***
	(0.003)	(0.006)	(0.008)	(0.009)	(0.010)	(0.013)	(0.017)
Almost Eligible × Post	-0.005**	-0.011**	-0.018***	-0.016**	-0.008	-0.014	0.001
	(0.002)	(0.005)	(0.007)	(0.008)	(0.009)	(0.010)	(0.013)
Observations	178585	178585	178585	178585	178585	165797	147679
Mean	0.0183	0.0980	0.2437	0.2921	0.3479	0.4166	0.5178
	Top 25% of the earnings distribution Ψ years after High School						
	Zero	One	Two	Three	Four	Five	Six
Differences-in-Differences Model:							
Eligible × Post	-0.002	-0.005***	-0.010***	-0.018***	-0.014***	0.006	0.054***
	(0.001)	(0.002)	(0.003)	(0.004)	(0.005)	(0.006)	(0.007)
Almost Eligible × Post	-0.000	0.000	-0.001	0.000	-0.002	-0.002	0.002
	(0.001)	(0.002)	(0.003)	(0.004)	(0.004)	(0.005)	(0.006)
Observations	178585	178585	178585	178585	178585	165797	147679
Mean	0.0036	0.0147	0.0431	0.0790	0.1087	0.1430	0.1833

Note: Each panel reports difference-in-differences estimates from Equation 1, where each column corresponds to a separate regression by years since high school completion. Panel A presents the probability that the older sibling is formally employed zero to six years after high school. Panel B reports the older sibling's total earnings, expressed in multiples of the monthly minimum wage. Panel C reports the probability that the older sibling is in the top quartile of the earnings distribution. All specifications include school and municipality-by-cohort fixed effects, and the following family characteristics interacted with cohort fixed effects: number of siblings, maternal education categories, and household socioeconomic stratum. The sample consists of younger siblings in need-eligible households whose older sibling took the *Saber 11* exam between 2010 and 2017; the analysis begins with the 2010 cohort because formal employment records from *PILA* are available from that year onward. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

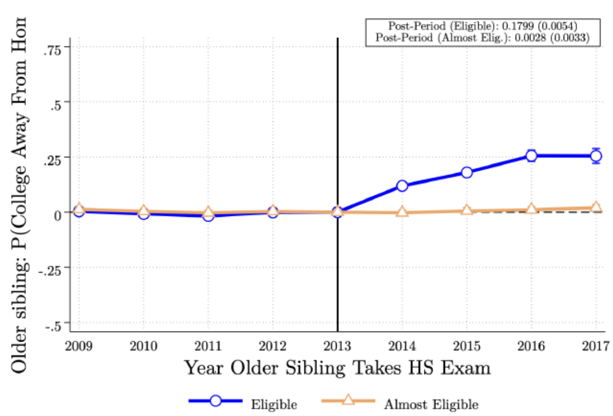
Figure 10: First Stage: Older Sibling's Program Take-Up and Postsecondary Enrollment

(a) Program Take-Up **(b) Enrollment in Any HEI**



Note: Each panel plots event-study coefficients estimated from Equation 1, where the dependent variable is an indicator for the older sibling's educational outcome. Panel A shows the probability that the older sibling received an SPP scholarship. Panel B shows the probability that the older sibling enrolled in any higher education institution (HEI). Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes in each panel display the corresponding pooled difference-in-differences estimate. The sample consists of younger siblings in need-eligible households whose older sibling took the *Saber 11* exam between 2009 and 2017. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

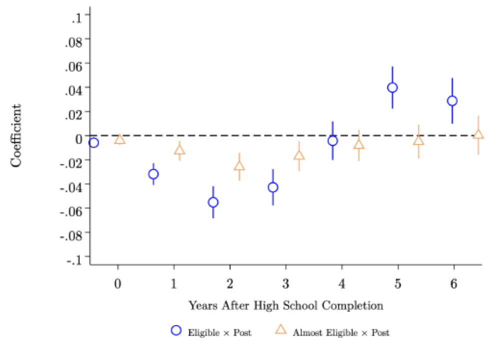
Figure 11: First Stage: Older Sibling's Enrollment in a Higher Education Institution Outside the Home Municipality



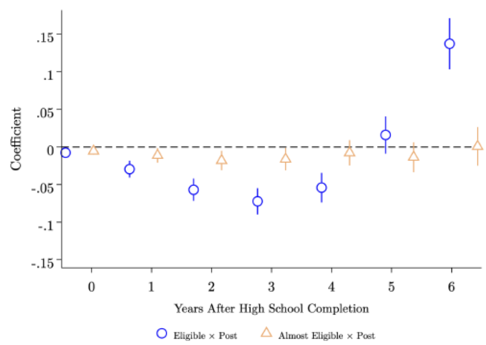
Note: The figure plots event-study coefficients estimated from Equation 1, where the dependent variable is an indicator for whether the older sibling enrolled in a higher education institution outside their home municipality. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The box displays the corresponding pooled difference-in-differences estimate. The sample consists of younger siblings in need-eligible households whose older sibling took the *Saber 11* exam between 2009 and 2017. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Figure 12: First Stage: Older Sibling's Formal Employment, Earnings, and Earnings Distribution

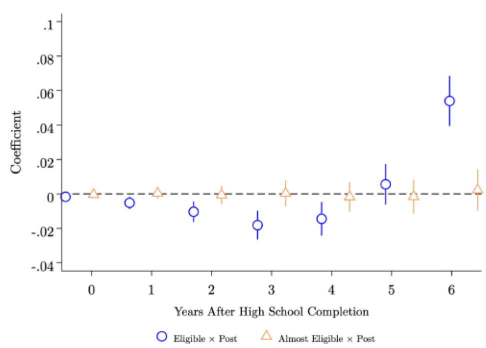
(a) Formal Employment



(b) Earnings (multiples of the monthly minimum wage)



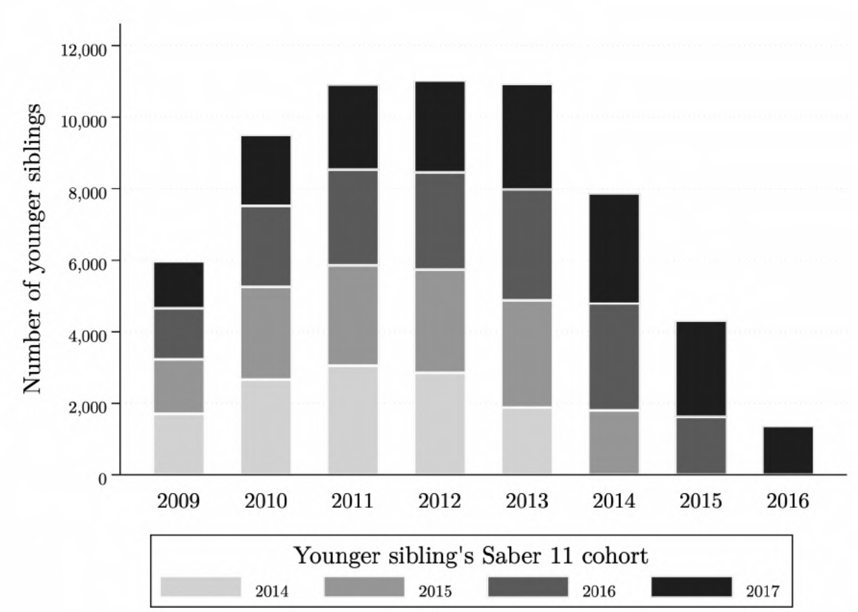
(c) Earnings: Top Quartile of the Distribution



Note: Each panel reports difference-in-differences estimates from Equation 1. Each point on the x-axis corresponds to a separate regression where the dependent variable is the older sibling's outcome measured at that number of years after high school graduation. For example, the point at zero corresponds to formal employment in the year of high school graduation, the point at one to formal employment one year later, and so on. For each regression, two coefficients are plotted: the circle markers correspond to $T_1 \times Post$ and the triangle markers to $T_2 \times Post$. Panel (a) shows the probability of formal employment, Panel (b) reports total earnings expressed in multiples of the monthly minimum wage, and Panel (c) shows the probability of being in the top quartile of the earnings distribution. The coefficients follow a U-shaped pattern, declining during the college years and recovering in later periods, consistent with the temporary displacement of labor market activity during enrollment. The sample consists of younger siblings in need-eligible households whose older sibling took the *Saber 11* exam between 2010 and 2017. Formal employment records from *PILA* are available from 2010 onward. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

10.2 Younger siblings

Figure 13: Distribution of Younger Siblings' *Saber 11* Cohort by Older Sibling's Cohort

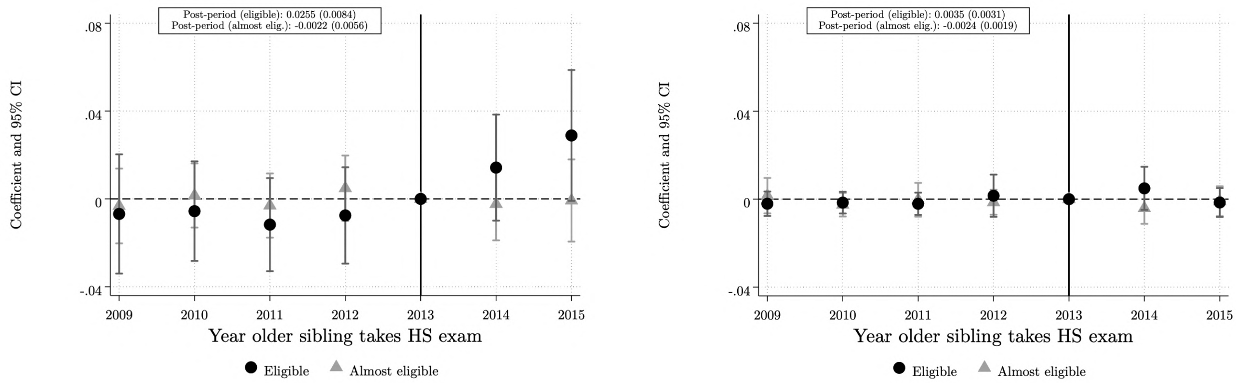


Note: Each bar corresponds to an older sibling cohort, defined by the year in which the older sibling took the *Saber 11* exam. The stacked segments show the distribution of the year in which the younger sibling took the *Saber 11* exam, restricted to the SPP window (2014–2017). The sample consists of younger siblings in need-eligible households. Older sibling cohorts from 2017 onward are excluded because younger siblings of those cohorts have no remaining years within the SPP window to take the exam and potentially receive the scholarship themselves.

Figure 14: Sibling Spillover Effects on SPP Take-Up: Event Study

Panel A: Main Results (Sisben-eligible families)

Panel B: Placebo (Sisben-ineligible families)

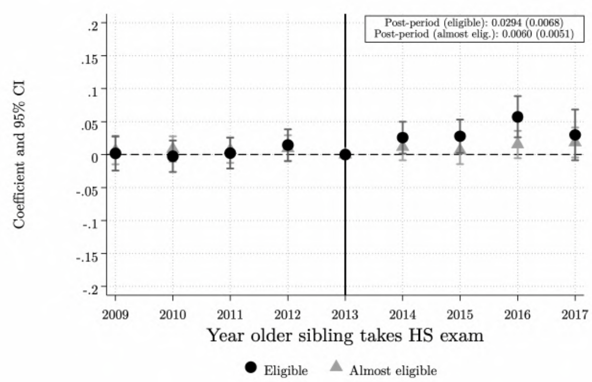


Note: Each panel plots event-study coefficients estimated from Equation 2, where the dependent variable is an indicator for whether the younger sibling received an SPP scholarship. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes display the corresponding pooled difference-in-differences estimates. The sample consists of younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2015. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Figure 15: Sibling Spillover Effects on Immediate Postsecondary Enrollment: Event Study

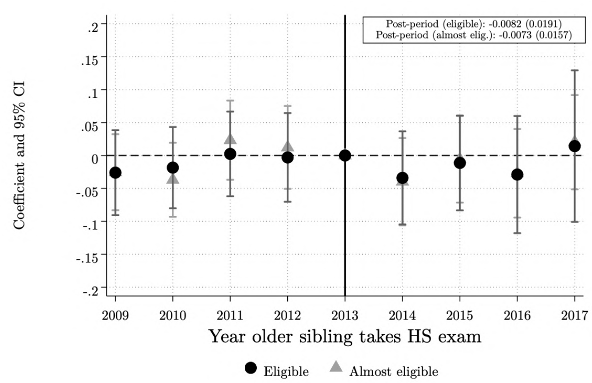
Panel A. Main Results

Any HEI Enrollment



Panel B. Placebo

Any HEI Enrollment

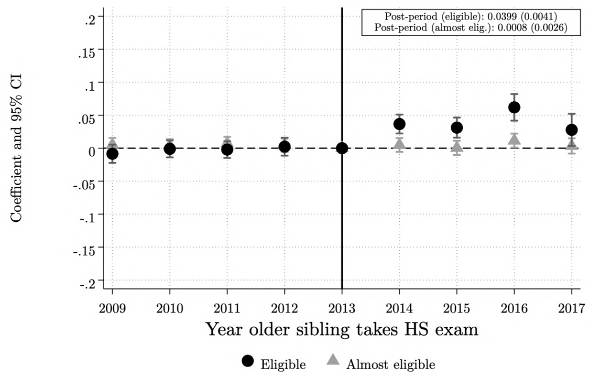


Note: Each panel plots event-study coefficients estimated from Equation 2, where the dependent variable is an indicator for enrollment in any higher education institution. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes display the corresponding pooled difference-in-differences estimates. The sample consists of younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2017. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

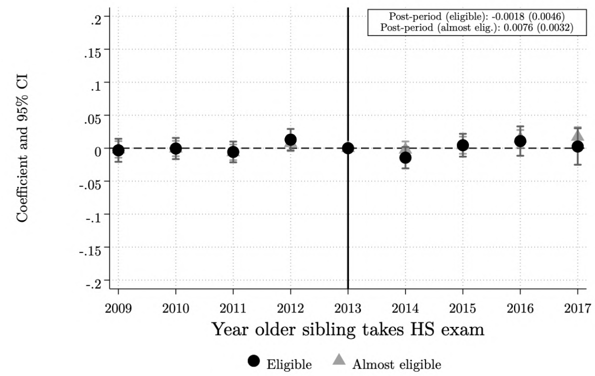
Figure 16: Sibling Spillover Effects on Enrollment in High-Quality Institutions by Sector: Event Study

Panel A. Main Results

Private High-Quality HEI

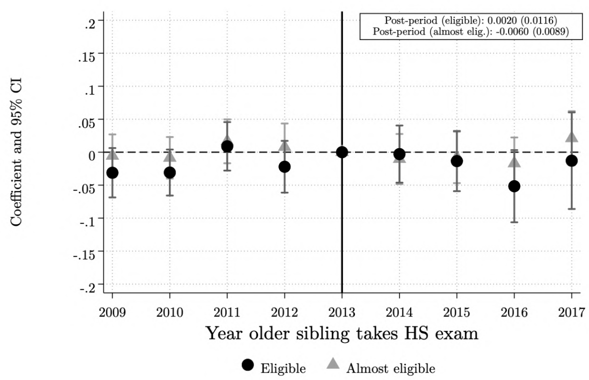


Public High-Quality HEI

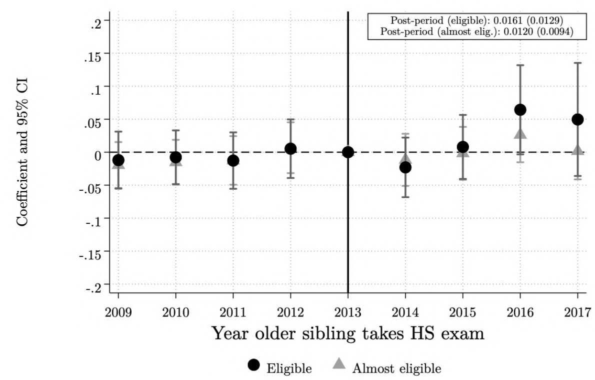


Panel B. Placebo

Private High-Quality HEI



Public High-Quality HEI

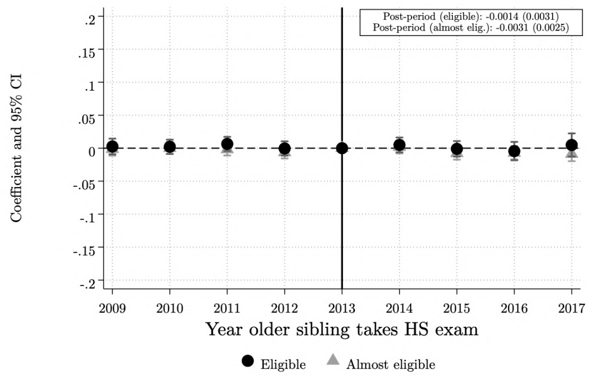


Note: Each panel plots event-study coefficients estimated from Equation 2, where the dependent variable is an indicator for enrollment in a high-quality higher education institution, distinguished by sector. High-quality institutions are defined as those holding the *High-Quality Accreditation* granted by the Ministry of Education, as described in Section 2. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes display the corresponding pooled difference-in-differences estimates. The sample consists of younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2017. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

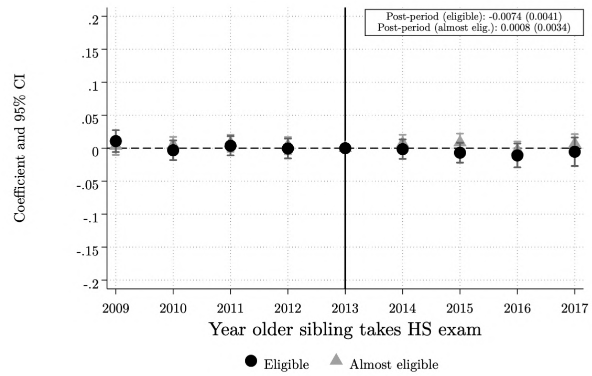
Figure 17: Sibling Spillover Effects on Enrollment in Low-Quality Institutions by Sector: Event Study

Panel A. Main Results

Private Low-Quality HEI

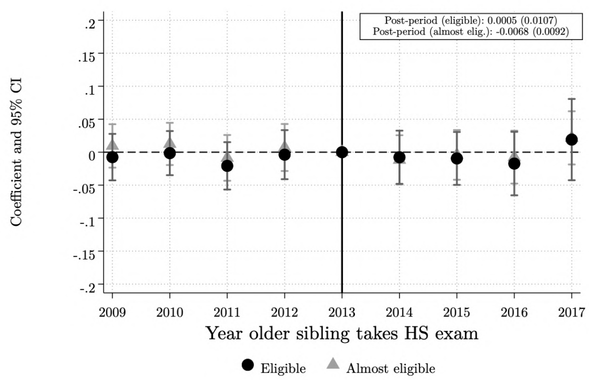


Public Low-Quality HEI

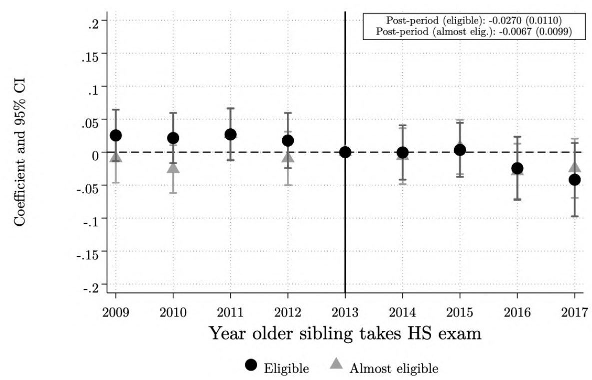


Panel B. Placebo

Private Low-Quality HEI



Public Low-Quality HEI

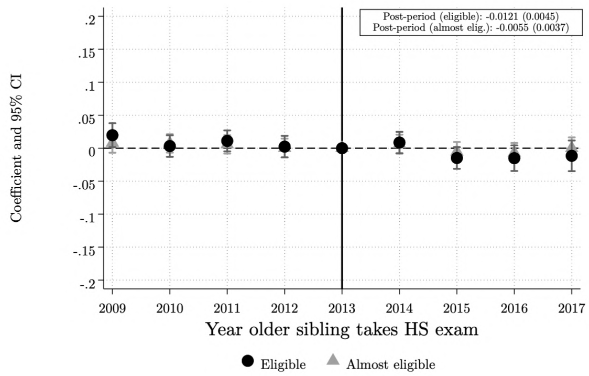


Note: Each panel plots event-study coefficients estimated from Equation 2, where the dependent variable is an indicator for enrollment in a low-quality higher education institution, distinguished by sector. Low-quality institutions are defined as those without the *High-Quality Accreditation* granted by the Ministry of Education, as described in Section 2. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes display the corresponding pooled difference-in-differences estimates. The sample consists of younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2017. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

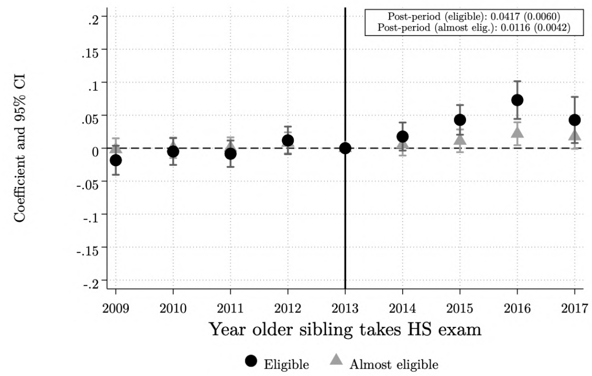
Figure 18: Sibling Spillover Effects on Enrollment by Program Duration: Event Study

Panel A. Main Results

Two-Year Program

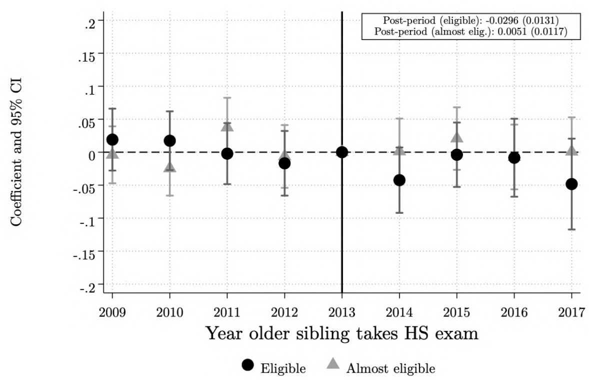


Four-Year Program

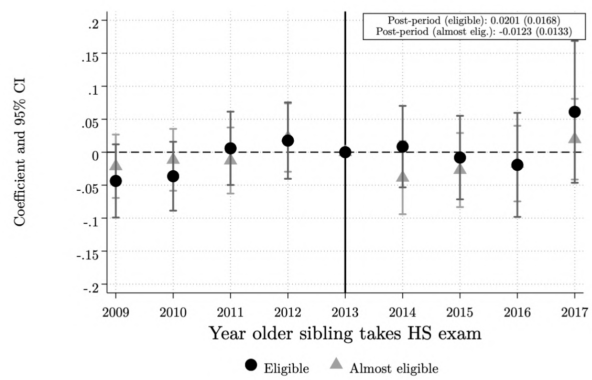


Panel B. Placebo

Two-Year Program



Four-Year Program



Note: Each panel plots event-study coefficients estimated from Equation 2, where the dependent variable is an indicator for enrollment in a two-year or four-year higher education program. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes display the corresponding pooled difference-in-differences estimates. The sample consists of younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2017. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Table 8. Sibling Spillover Effects on Immediate Postsecondary Enrollment by Institution Type, Quality and Program Duration

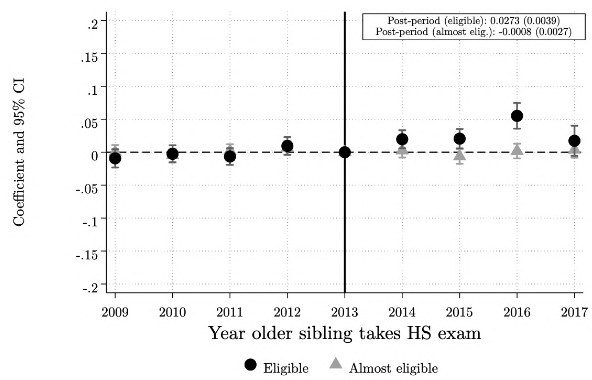
			High-quality			Low-quality			Program Duration	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	SPP	Any	Any	Public	Private	Any	Public	Private	Two-years	Four-years
Panel A: Main Results (Sisben-eligible families)										
T1 × Post	0.025***	0.029***	0.038***	-0.002	0.040***	-0.009*	-0.007*	-0.001	-0.012***	0.042***
	(0.008)	(0.007)	(0.006)	(0.005)	(0.004)	(0.005)	(0.004)	(0.003)	(0.004)	(0.006)
T2 × Post	-0.002	0.006	0.008**	0.008**	0.001	-0.002	0.001	-0.003	-0.005	0.012***
	(0.006)	(0.005)	(0.004)	(0.003)	(0.003)	(0.004)	(0.003)	(0.002)	(0.004)	(0.004)
Observations	60461	198416	198416	198416	198416	198416	198416	198416	198416	198416
Mean	0.0611	0.3144	0.1682	0.0991	0.0691	0.1463	0.0973	0.0490	0.1198	0.1936
Effect rel. to mean (T1)	42%	9%	23%	n.s.	58%	-6%	-8%	n.s.	-10%	22%
Effect rel. to mean (T2)	n.s.	n.s.	5%	8%	n.s.	n.s.	n.s.	n.s.	n.s.	6%
Panel B: Placebo (Sisben-ineligible families)										
T1 × Post	0.004	-0.008	0.018	0.016	0.002	-0.026*	-0.027**	0.000	-0.030**	0.020
	(0.003)	(0.019)	(0.016)	(0.013)	(0.012)	(0.015)	(0.011)	(0.011)	(0.013)	(0.017)
T2 × Post	-0.002	-0.007	0.006	0.012	-0.006	-0.013	-0.007	-0.007	0.005	-0.012
	(0.002)	(0.016)	(0.012)	(0.009)	(0.009)	(0.013)	(0.010)	(0.009)	(0.012)	(0.013)
Observations	8583	30251	30251	30251	30251	30251	30251	30251	30251	30251
Mean	0.0012	0.4023	0.2351	0.1234	0.1117	0.1672	0.0857	0.0815	0.1286	0.2731
Effect rel. to mean (T1)	n.s.	n.s.	n.s.	n.s.	n.s.	-16%	-32%	n.s.	-23%	n.s.
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.

Note: Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Column (1) reports the probability of being an SPP recipient. Column (2) shows enrollment in any HEI. Columns (3)–(5) and (6)–(8) distinguish between low- and high-quality HEIs by sector. Columns (9) and (10) report enrollment in two-year and four-year programs respectively. All specifications include school and municipality-by-cohort fixed effects, and the following family characteristics interacted with cohort fixed effects: number of siblings, maternal education categories, and household socioeconomic stratum. The sample consists of younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2017. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 19: Sibling Spillover Effects on Enrollment Away from Home Municipality: Event Study

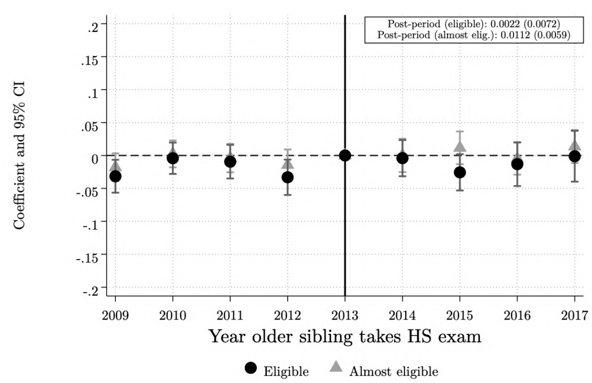
Panel A. Main Results

HEI Enrollment Away from Home Municipality



Panel B. Placebo

HEI Enrollment Away from Home Municipality

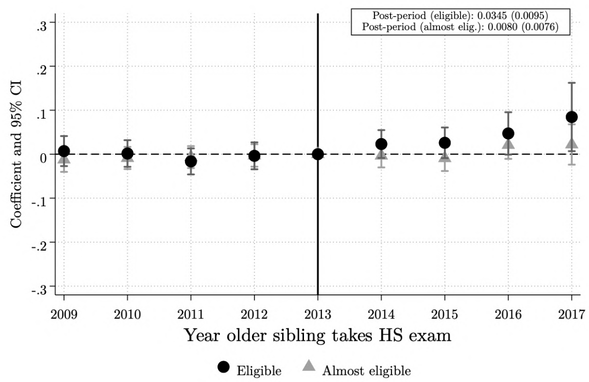


Note: Each panel plots event-study coefficients estimated from Equation 2, where the dependent variable is an indicator for whether the younger sibling enrolled in a higher education institution outside their home municipality. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes display the corresponding pooled difference-in-differences estimates. The sample consists of younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2017. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

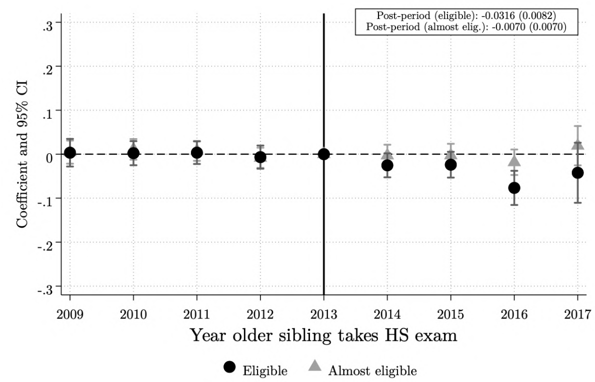
Figure 20: Sibling Spillover Effects on Graduation: Event Study

Panel A. Main Results

University Graduation (Saber PRO)

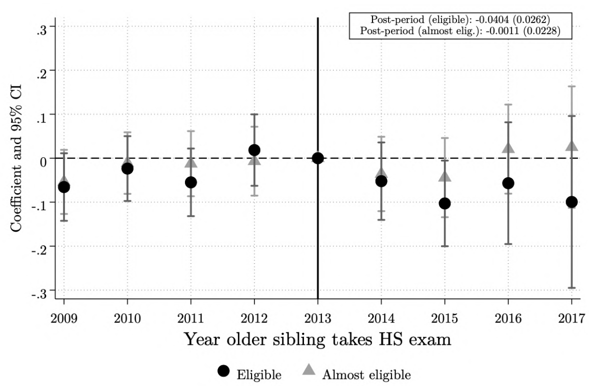


TVET Graduation (Saber TyT)

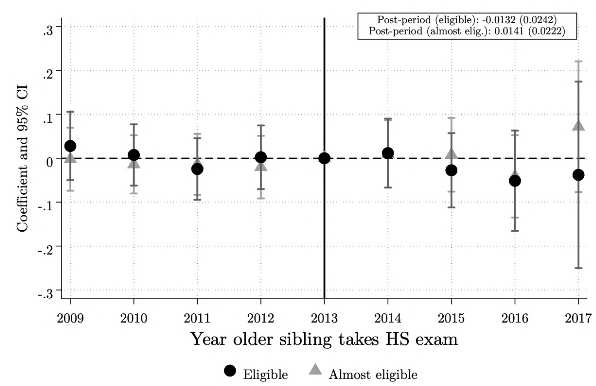


Panel B. Placebo

University Graduation (Saber PRO)



TVET Graduation (Saber TyT)



Note: Each panel plots event-study coefficients estimated from Equation 2. University graduation is measured using *Saber PRO* records, available from 2012 onward. TVET graduation is measured using *Saber TyT* records, available from 2016 onward. Both outcomes are defined over younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2019, subject to sufficient time for graduation to be observed. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes display the corresponding pooled difference-in-differences estimates. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Table 9. Sibling Spillover Effects on Field of Study, Migration, Mimicking, and Graduation

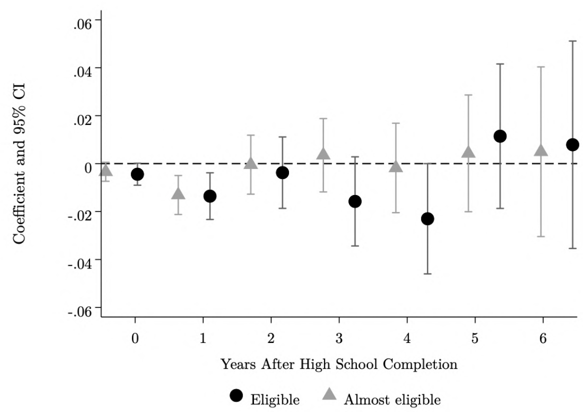
	Field of Study				Migration	Mimicking		Graduation		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	STEM	STEM Plus	Arts	S.S.H.	HEI Away	Same HEI	Same Field	Any HEI	Four-years	Two-years
Panel A: Main Results (Sisben-eligible families)										
T1 × Post	0.015***	0.023***	0.003*	0.002	0.027***	-0.017***	0.015***	-0.002	0.015*	-0.028***
	(0.005)	(0.007)	(0.002)	(0.003)	(0.004)	(0.004)	(0.005)	(0.009)	(0.008)	(0.007)
T2 × Post	0.003	0.005	-0.000	-0.001	-0.001	0.002	0.003	0.003	0.003	-0.004
	(0.004)	(0.005)	(0.001)	(0.002)	(0.003)	(0.003)	(0.003)	(0.007)	(0.006)	(0.006)
Observations	198416	198416	198416	198416	198416	198416	198416	143680	143680	143680
Mean	0.1253	0.2634	0.0127	0.0340	0.0821	0.0939	0.0960	0.4658	0.3235	0.1869
Effect rel. to mean (T1)	12%	9%	22%	n.s.	33%	-18%	16%	n.s.	5%	-15%
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
Panel B: Placebo (Sisben-ineligible families)										
T1 × Post	-0.001	0.000	-0.005	-0.006	0.002	-0.027**	0.009	-0.038	-0.039*	-0.015
	(0.014)	(0.018)	(0.006)	(0.009)	(0.007)	(0.012)	(0.013)	(0.024)	(0.023)	(0.019)
T2 × Post	-0.015	0.001	0.002	-0.014**	0.011*	-0.007	0.000	0.010	-0.001	0.011
	(0.011)	(0.015)	(0.005)	(0.007)	(0.006)	(0.010)	(0.010)	(0.020)	(0.019)	(0.017)
Observations	30251	30251	30251	30251	30251	30251	30251	23373	23373	23373
Mean	0.1614	0.3294	0.0205	0.0474	0.0563	0.1109	0.1403	0.5696	0.4460	0.1895
Effect rel. to mean (T1)	n.s.	n.s.	n.s.	n.s.	n.s.	-24%	n.s.	n.s.	-9%	n.s.
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	-29%	20%	n.s.	n.s.	n.s.	n.s.	n.s.

Note: Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Columns (1)–(4) report enrollment by field of study: STEM, STEM Plus, Arts, and Social Sciences and Humanities. Column (5) reports enrollment in an HEI outside the home municipality. Columns (6) and (7) report whether the younger sibling enrolls in the same HEI and same field as the older sibling, respectively. Columns (8)–(10) report graduation from any HEI, a four-year program (Saber PRO), and a two-year program (Saber TyT), respectively. All specifications include school and municipality-by-cohort fixed effects, and the following family characteristics interacted with cohort fixed effects: number of siblings, maternal education categories, and household socioeconomic stratum. The sample consists of younger siblings whose older sibling took the *Saber 11* exam between 2009 and 2017. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 21: Sibling Spillover Effects on Formal Employment: Event Study

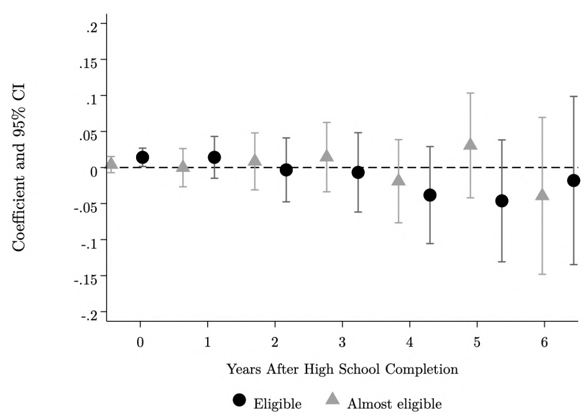
Panel A. Main Results

Formal Employment



Panel B. Placebo

Formal Employment

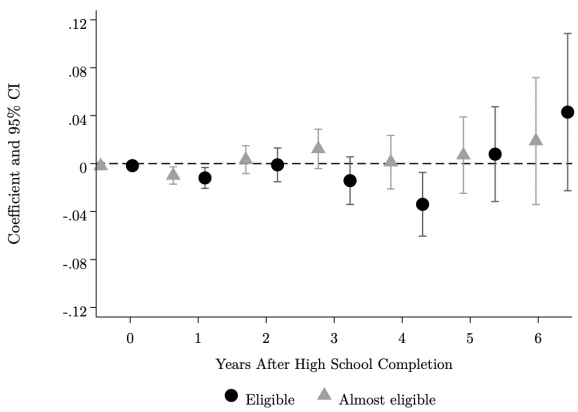


Note: Each panel reports difference-in-differences estimates from Equation 2. Each point on the x-axis corresponds to a separate regression where the dependent variable is an indicator for formal employment of the younger sibling at that number of years after high school graduation. Circle markers correspond to $T_1 \times Post$ (eligible) and triangle markers to $T_2 \times Post$ (almost eligible). Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. The sample shrinks for higher values of k as only earlier cohorts are observed far enough post-graduation. Formal employment records from *PILA* are available from 2010 onward. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Figure 22: Sibling Spillover Effects on Wages: Event Study

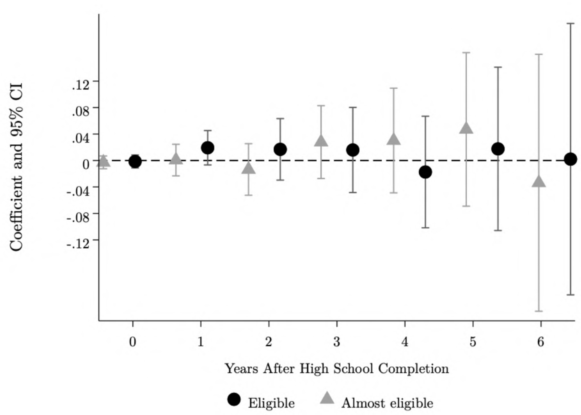
Panel A. Main Results

Wages (Minimum Wage Units)



Panel B. Placebo

Wages (Minimum Wage Units)

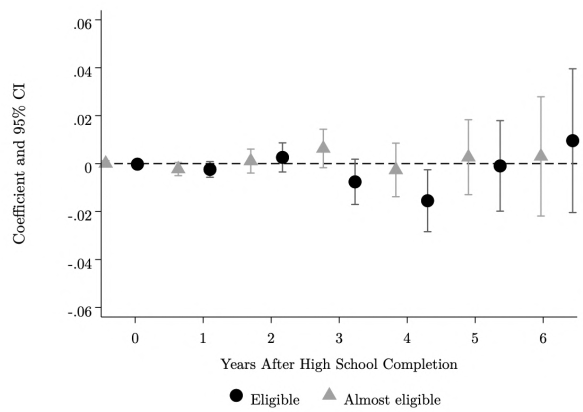


Note: Each panel reports difference-in-differences estimates from Equation 2. Each point on the x-axis corresponds to a separate regression where the dependent variable is the younger sibling's earnings expressed in multiples of the monthly minimum wage at that number of years after high school graduation. Circle markers correspond to $T_1 \times Post$ (eligible) and triangle markers to $T_2 \times Post$ (almost eligible). Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. The sample shrinks for higher values of k as only earlier cohorts are observed far enough post-graduation. Earnings records from *PILA* are available from 2010 onward. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Figure 23: Sibling Spillover Effects on Top Quartile of Wage Distribution: Event Study

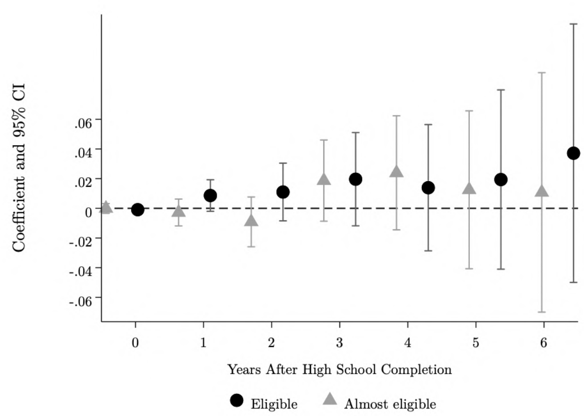
Panel A. Main Results

Top 25% of Wage Distribution



Panel B. Placebo

Top 25% of Wage Distribution



Note: Each panel reports difference-in-differences estimates from Equation 2. Each point on the x-axis corresponds to a separate regression where the dependent variable is an indicator for whether the younger sibling is in the top quartile of the earnings distribution at that number of years after high school graduation. Circle markers correspond to $T_1 \times Post$ (eligible) and triangle markers to $T_2 \times Post$ (almost eligible). Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. The sample shrinks for higher values of k as only earlier cohorts are observed far enough post-graduation. Earnings records from *PILA* are available from 2010 onward. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Table 10. Sibling Spillover Effects on Formal Employment

	Years since High School Graduation						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	k=0	k=1	k=2	k=3	k=4	k=5	k=6
Panel A: Main Results (Sisben-eligible families)							
Eligible × Post	-0.004*	-0.014***	-0.004	-0.016*	-0.023**	0.011	0.008
	(0.002)	(0.005)	(0.008)	(0.009)	(0.012)	(0.015)	(0.022)
Almost Eligible × Post	-0.003*	-0.013***	-0.000	0.004	-0.002	0.004	0.005
	(0.002)	(0.004)	(0.006)	(0.008)	(0.010)	(0.012)	(0.018)
Observations	162559	145043	127013	107154	87462	68295	49720
Mean	0.0267	0.1023	0.2534	0.3348	0.3788	0.4359	0.4667
Panel B: Placebo (Sisben-ineligible families)							
Eligible × Post	0.012*	0.006	-0.009	-0.004	-0.027	-0.044	-0.000
	(0.006)	(0.014)	(0.022)	(0.027)	(0.033)	(0.041)	(0.056)
Almost Eligible × Post	0.000	-0.005	-0.002	0.019	-0.012	0.025	0.002
	(0.005)	(0.013)	(0.019)	(0.023)	(0.028)	(0.035)	(0.053)
Observations	28061	25809	23373	20524	17704	14870	12108
Mean	0.0226	0.1252	0.3067	0.3820	0.4427	0.5050	0.5505

Note: Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. The dependent variable is an indicator for formal employment (contributing to PILA) at k years after high school graduation, for k = 0 to 6. Each column corresponds to a separate regression. The sample shrinks for higher values of k as only earlier cohorts are observed far enough post-graduation. Formal employment records from *PILA* are available from 2010 onward. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11. Sibling Spillover Effects on Wages

	Years since High School Graduation						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	k=0	k=1	k=2	k=3	k=4	k=5	k=6
Panel A: Main Results (Sisben-eligible families)							
Eligible × Post	-0.002	-0.010**	-0.002	-0.013	-0.030**	0.013	0.048
	(0.002)	(0.004)	(0.007)	(0.010)	(0.013)	(0.020)	(0.033)
Almost Eligible × Post	-0.003*	-0.009**	0.004	0.009	-0.001	0.013	0.018
	(0.002)	(0.004)	(0.006)	(0.008)	(0.011)	(0.016)	(0.026)
Observations	180841	162580	143680	122777	101774	81192	61047
Mean	0.0131	0.0746	0.1902	0.2731	0.3266	0.3904	0.4878
Panel B: Placebo (Sisben-ineligible families)							
Eligible × Post	-0.001	0.019	0.017	0.016	-0.017	0.018	0.002
	(0.005)	(0.013)	(0.024)	(0.033)	(0.043)	(0.063)	(0.105)
Almost Eligible × Post	-0.003	0.001	-0.014	0.028	0.030	0.047	-0.034
	(0.005)	(0.012)	(0.020)	(0.028)	(0.040)	(0.059)	(0.099)
Observations	28061	25809	23373	20524	17704	14870	12108
Mean	0.0156	0.0933	0.2404	0.3311	0.3965	0.4912	0.6116

Note: Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. The dependent variable is the lower bound of monthly wages measured in minimum wage units at k years after high school graduation, for k = 0 to 6. Each column corresponds to a separate regression. The sample shrinks for higher values of k as only earlier cohorts are observed far enough post-graduation. Earnings records from *PILA* are available from 2010 onward. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 12. Sibling Spillover Effects on Top Quartile of Wage Distribution

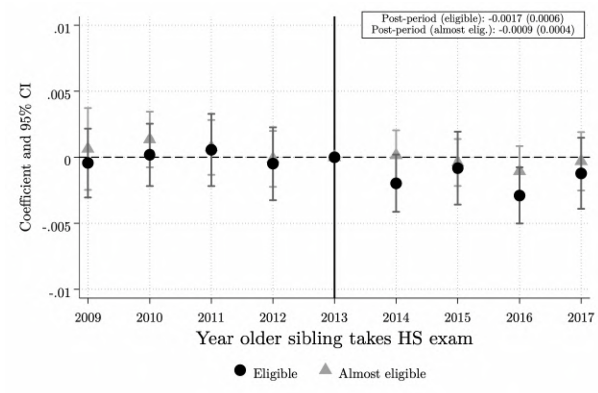
	Years since High School Graduation						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	k=0	k=1	k=2	k=3	k=4	k=5	k=6
Panel A: Main Results (Sisben-eligible families)							
Eligible × Post	-0.000	-0.002	0.003	-0.005	-0.012*	0.006	0.015
	(0.001)	(0.002)	(0.003)	(0.005)	(0.006)	(0.009)	(0.015)
Almost Eligible × Post	-0.000	-0.002*	0.001	0.006	-0.002	0.008	0.006
	(0.001)	(0.001)	(0.002)	(0.004)	(0.006)	(0.008)	(0.012)
Observations	180841	162580	143680	122777	101774	81192	61047
Mean	0.0023	0.0099	0.0286	0.0597	0.0851	0.1105	0.1370
Panel B: Placebo (Sisben-ineligible families)							
Eligible × Post	-0.001	0.009	0.011	0.020	0.014	0.019	0.037
	(0.002)	(0.005)	(0.010)	(0.016)	(0.022)	(0.031)	(0.044)
Almost Eligible × Post	-0.000	-0.003	-0.009	0.019	0.024	0.012	0.011
	(0.002)	(0.005)	(0.009)	(0.014)	(0.020)	(0.027)	(0.041)
Observations	28061	25809	23373	20524	17704	14870	12108
Mean	0.0024	0.0148	0.0424	0.0901	0.1248	0.1574	0.1886

Note: Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. The dependent variable is an indicator for whether the younger sibling is in the top quartile of the earnings distribution at k years after high school graduation, for k = 0 to 6. Each column corresponds to a separate regression. The sample shrinks for higher values of k as only earlier cohorts are observed far enough post-graduation. Earnings records from *PILA* are available from 2010 onward. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 24: Younger sibling: probability of conviction

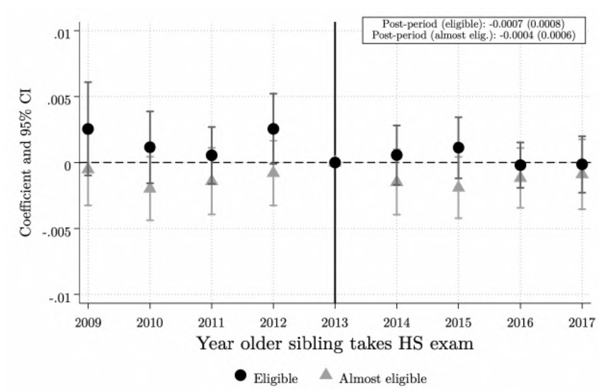
Panel A. Main Results

Any Crime



Panel B. Placebo

Any Crime



Note: Each panel plots event-study coefficients from Equation 2, capturing the effect of having an older sibling eligible for college financial aid on the probability that a younger sibling is convicted of any crime. Coefficients are normalized relative to the 2013 cohort, the last cohort unexposed to the program. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Ninety-five-percent confidence intervals are shown, with standard errors clustered at the household level.

Table 13. Younger sibling: probability of conviction

		Type of offense				
	(1)	(2)	(3)	(4)	(5)	(6)
	Any	Economic	Drugs	Violent	Others	*No.* convictions
Panel A: Main Results (Sisben-eligible families)						
T1 × Post	-0.0017***	-0.0008**	-0.0006*	0.0001	-0.0001	-0.0026***
	(0.0006)	(0.0004)	(0.0003)	(0.0001)	(0.0003)	(0.0008)
T2 × Post	-0.0009**	-0.0001	-0.0003	-0.0001	-0.0003	-0.0009
	(0.0004)	(0.0003)	(0.0002)	(0.0001)	(0.0003)	(0.0008)
Observations	189033	189033	189033	189033	189033	189033
Mean	0.0024	0.0011	0.0005	0.0003	0.0005	0.0031
Effect rel. to mean (T1)	-71%	-72%	-117%	n.s.	n.s.	-82%
Effect rel. to mean (T2)	-38%	n.s.	n.s.	n.s.	n.s.	n.s.
Panel B: Placebo (Sisben-ineligible families)						
T1 × Post	-0.0007	-0.0001	0.0001	-0.0001	-0.0003	-0.0005
	(0.0008)	(0.0005)	(0.0003)	(0.0002)	(0.0003)	(0.0011)
T2 × Post	-0.0004	0.0004	-0.0001	-0.0003	-0.0002	-0.0007
	(0.0006)	(0.0004)	(0.0003)	(0.0002)	(0.0003)	(0.0010)
Observations	104725	104725	104725	104725	104725	104725
Mean	0.0022	0.0011	0.0006	0.0001	0.0004	0.0028
Effect rel. to mean (T1)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.

Note: Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Each column reports coefficients from a Difference-in-Differences specification estimated on younger siblings' criminal outcomes. The dependent variables indicate whether the younger sibling was ever convicted of (1) any crime, (2) an economic offense, (3) a drug-related offense, (4) a violent offense, or (5) another type of crime; column (6) reports the total number of convictions. The mean row reports the pre-program probability of conviction for each outcome. The baseline probability of committing any crime is 0.24 percent (2.4 per 1,000 individuals), and the pre-program average number of convictions is 0.0031, indicating that most convicted individuals have only one conviction. Standard errors are clustered at the household level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 14. Younger Siblings: *Saber 3*, *Saber 5*, and *Saber 9* Outcomes (2017 ICFES Data)

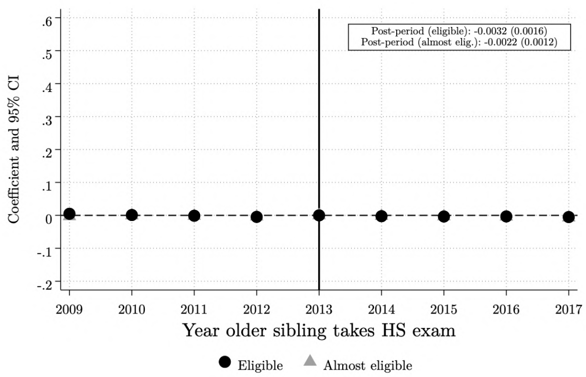
	Saber 3		Saber 5				Saber 9			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Math	Spanish	Math	Spanish	Perfect Attendance	Never late to class	Math Math	Spanish Spanish	Perfect Attendance	Never late to class
Differences-in-Differences Model:										
T1 × Post	0.27***	0.18*	0.18***	0.20***	0.04	0.07**	0.13***	0.02	-0.02	0.01
	(0.10)	(0.10)	(0.07)	(0.07)	(0.04)	(0.03)	(0.05)	(0.05)	(0.03)	(0.03)
T2 × Post	-0.09	-0.02	0.03	0.09*	-0.01	0.04	0.05	-0.05	0.01	0.01
	(0.07)	(0.08)	(0.05)	(0.05)	(0.03)	(0.03)	(0.04)	(0.03)	(0.02)	(0.02)
Observations	8103	8103	14185	14185	14185	14185	20131	20131	20131	20131
Mean	0.18	0.25	0.30	0.31	0.60	0.64	0.25	0.25	0.50	0.48
Effect rel. to mean (T1)					n.s.	10%			n.s.	n.s.
Effect rel. to mean (T2)					n.s.	n.s.			n.s.	n.s.
Average age	8.8		10.9				15.2			

Note: Each column reports coefficients from a Difference-in-Differences specification estimated using the 2017 *Saber 3*, *Saber 5*, and *Saber 9* test scores and self-reported behavioral indicators. *Perfect Attendance* equals 1 if the student reports not missing any school days in the previous month; *Never Late* equals 1 if the student reports never arriving late. All models include school and cohort-by-municipality fixed effects and baseline family characteristics interacted with cohort dummies. Standard errors are clustered at the household level.

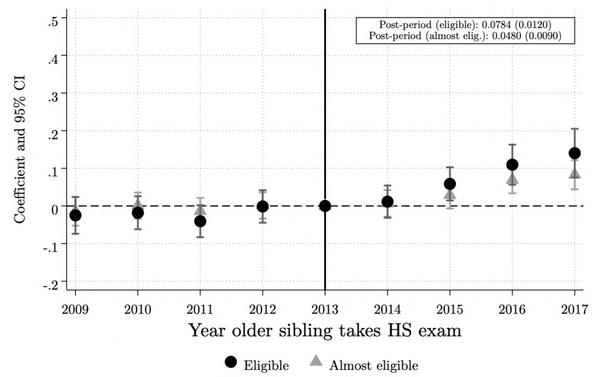
Figure 25: Sibling Spillover Effects on High School Graduation and Saber 11 Performance: Event Study

Panel A. Main Results

HS Graduation (Extensive Margin)

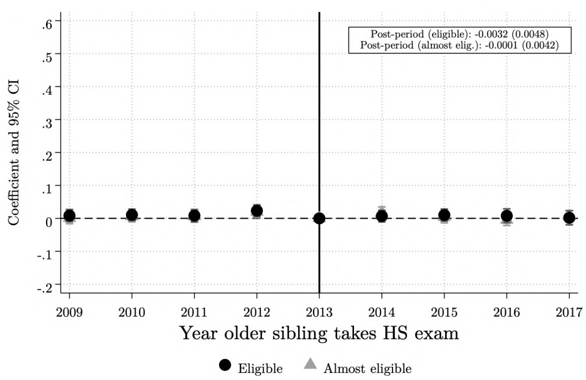


Saber 11 Score (Intensive Margin)

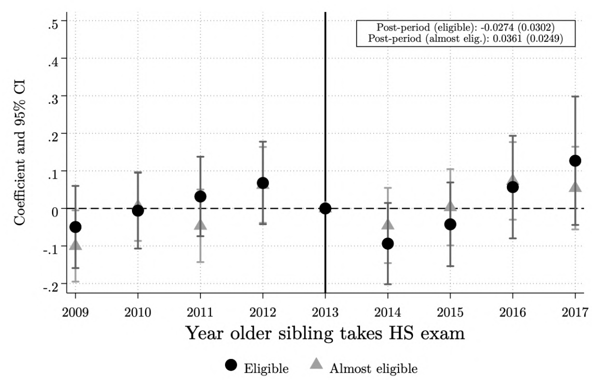


Panel B. Placebo

HS Graduation (Extensive Margin)



Saber 11 Score (Intensive Margin)



Note: Each panel plots event-study coefficients estimated from Equation 2. The left graph shows the probability that the younger sibling graduates high school, measured as an indicator for taking the *Saber 11* exam. The right graph shows standardized scores on the overall *Saber 11* exam. Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Coefficients represent differences relative to the 2013 cohort, the last cohort before the introduction of SPP. The boxes display the corresponding pooled difference-in-differences estimates. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Table 15. Sibling Spillover Effects on High School Graduation and Saber 11 Performance

	Extensive Margin	Intensive Margin		
	(1)	(2)	(3)	(4)
	HS Graduation	Overall	Math	Spanish
Panel A: Main Results (Sisben-eligible families)				
T1 × Post	-0.003**	0.078***	0.079***	0.087***
	(0.002)	(0.012)	(0.012)	(0.012)
T2 × Post	-0.002*	0.048***	0.045***	0.048***
	(0.001)	(0.009)	(0.009)	(0.009)
Observations	276452	212502	212502	212502
Mean	0.7799	0.3979	0.3804	0.3564
Panel B: Placebo (Sisben-ineligible families)				
T1 × Post	-0.003	-0.027	-0.001	0.002
	(0.005)	(0.030)	(0.031)	(0.031)
T2 × Post	-0.000	0.036	0.035	0.034
	(0.004)	(0.025)	(0.025)	(0.026)
Observations	36505	31567	31567	31567
Mean	0.8889	0.7522	0.6703	0.6456

Note: Panel A reports estimates for younger siblings in Sisben-eligible families. Panel B reports the placebo estimates for younger siblings in Sisben-ineligible families, using the same score cutoffs. Column (1) reports the probability of high school graduation, measured as an indicator for taking the *Saber 11* exam. Columns (2)–(4) report standardized scores on the overall, math, and Spanish *Saber 11* exam respectively, standardized by year to have mean zero and standard deviation one. All specifications include school and municipality-by-cohort fixed effects, and the following family characteristics interacted with cohort fixed effects: number of siblings, maternal education categories, and household socioeconomic stratum. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 16. Younger Sibling: *Saber 11* Outcomes by Older Sibling Success in a Top Private University

	Takes *Saber 11*	Overall Score	Math Score	Spanish Score
T1 × Post	−0.041 (0.133)	−0.244 (0.223)	−0.229 (0.192)	0.262 (0.348)
T1 × Post × **Sibling Success**	0.020 (0.030)	0.294*** (0.076)	0.321*** (0.066)	0.255*** (0.090)
Observations	1,234	994	994	994
Mean	0.84	1.16	1.07	0.96

Note: Each column reports coefficients from a Difference-in-Differences specification estimated on the subsample of families where the older sibling enrolled in the same private university. The interaction term $T1 \times Post \times Sibling\ Success$ captures whether spillovers are stronger when the older sibling's GPA in the first semester is above the 75th percentile of the cohort. Standard errors are clustered at the household level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 17. Younger Sibling: Postsecondary Enrollment by Older Sibling Success in a Top Private University

Panel A: Any Enrollment and Low-Quality Institutions

	(1)	(2)	(3)	(4)
	Any	Low-Quality		
		Any	Public	Private
T1 × Post	0.026	0.019	−0.001	0.020
	(0.178)	(0.159)	(0.108)	(0.078)
T1 × Post × **Sibling Success**	0.099**	−0.004	0.003	−0.008
	(0.042)	(0.034)	(0.027)	(0.019)
Observations	915	915	915	915
Mean	0.45	0.09	0.06	0.04

Panel B: High-Quality Institutions

	(5)	(6)	(7)
	High-Quality		
	Any	Public	Private
T1 × Post	0.007	−0.092*	0.099*
	(0.076)	(0.050)	(0.053)
T1 × Post × **Sibling Success**	0.104*	0.040	0.064**
	(0.053)	(0.043)	(0.029)
Observations	915	915	915
Mean	0.36	0.12	0.23

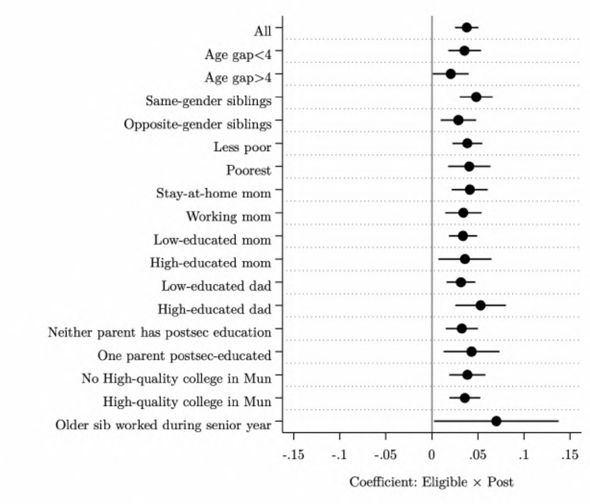
Panel C: Duration

	(8)	(9)
	Duration	
	Two-Year	Four-Year
T1 × Post	0.069	−0.038
	(0.132)	(0.100)
T1 × Post × **Sibling Success**	0.016	0.082**
	(0.029)	(0.038)
Observations	915	915
Mean	0.07	0.38

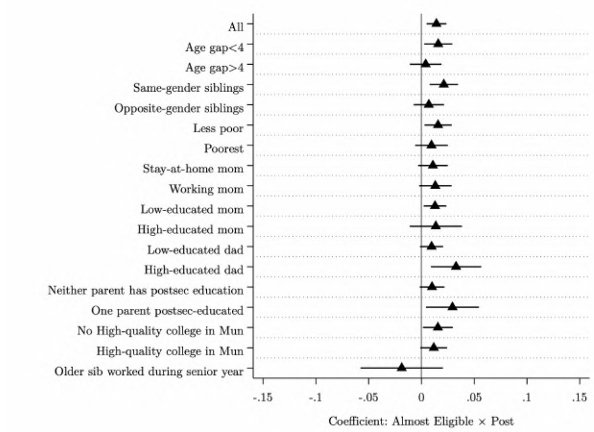
Note: Each column reports Difference-in-Differences coefficients for younger siblings' postsecondary enrollment outcomes, by institution type, quality, and degree length. The interaction $T1 \times Post \times Sibling\ Success$ indicates whether spillovers are amplified when the older sibling's GPA in the first semester exceeds the 75th percentile. Standard errors are clustered at the household level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 26: Heterogeneity Analysis: Immediate Postsecondary Enrollment

Panel A. Eligible × Post



Panel B. Almost Eligible × Post



Note: Each panel shows subgroup-specific estimates of the interaction term ($T1 \times Post$ or $T2 \times Post$) for the probability that the younger sibling enrolls in any higher-education institution. Subgroups are defined by age gap between siblings, gender pairing, household poverty index, parental education, maternal employment, local HEI availability, and older sibling employment during senior year. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Table 18. Heterogeneity Analysis – Postsecondary Enrollment

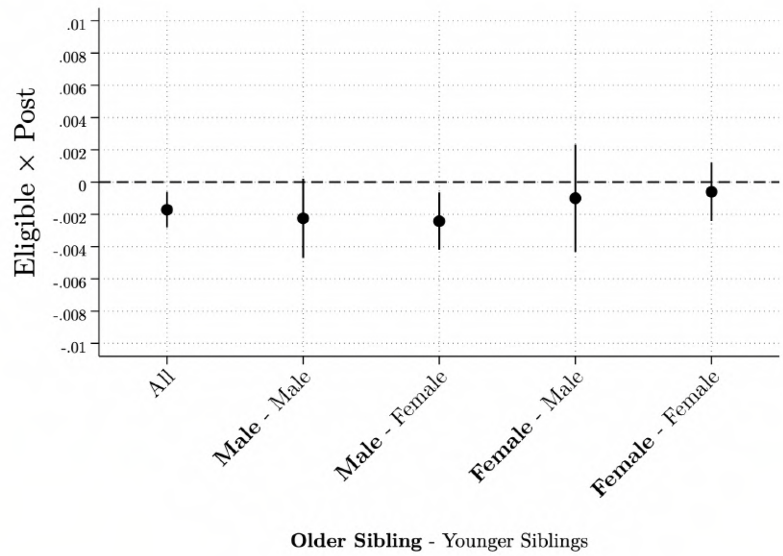
Differences-in-Differences Model:

	(1) All	(2) Age gap ≤ 4 years	(3) Age gap > 4 years	(4) Same-Gender Siblings	(5) Opposite-Gender Siblings	(6) SES: Less Poor
T1 × Post	0.0294***	0.0294***	0.0137	0.0281***	0.0283***	0.0266***
	(0.0068)	(0.0090)	(0.0115)	(0.0098)	(0.0103)	(0.0102)
T2 × Post	0.0060	0.0085	0.0002	0.0016	0.0106	-0.0033
	(0.0051)	(0.0068)	(0.0090)	(0.0074)	(0.0079)	(0.0080)
Observations	198416	112398	81358	102645	90977	96781
Mean	0.3144	0.3275	0.3007	0.3224	0.3065	0.3390
Effect rel. to mean (T1)	9%	9%	n.s.	9%	9%	8%
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
	(7) SES: Poorest	(8) Stay-at-home Mom	(9) Working Mom	(10) Low-educated Mom	(11) High-educated Mom	(12) Low-educated Dad
T1 × Post	0.0290***	0.0424***	0.0228**	0.0356***	0.0140	0.0330***
	(0.0101)	(0.0107)	(0.0111)	(0.0083)	(0.0154)	(0.0083)
T2 × Post	0.0146*	0.0133*	0.0002	0.0146**	-0.0195	0.0073
	(0.0075)	(0.0080)	(0.0090)	(0.0059)	(0.0138)	(0.0060)
Observations	97393	87678	72976	149766	42221	146390
Mean	0.2920	0.2957	0.3394	0.2891	0.3950	0.2937
Effect rel. to mean (T1)	10%	14%	7%	12%	n.s.	11%
Effect rel. to mean (T2)	5%	5%	n.s.	5%	n.s.	n.s.
	(13) High-educated Dad	(14) Neither parent has postsecondary education	(15) One parent has postsecondary education	(16) No High-Quality College in Mun	(17) High-Quality College in Mun in Mun	(18) Older sibling worked in Senior Year
T1 × Post	0.0100	0.0344***	0.0142	0.0252**	0.0304***	0.1050***
	(0.0149)	(0.0093)	(0.0165)	(0.0110)	(0.0088)	(0.0390)
T2 × Post	-0.0043	0.0089	0.0084	0.0000	0.0079	-0.0074
	(0.0130)	(0.0064)	(0.0136)	(0.0081)	(0.0068)	(0.0252)
Observations	45610	126603	38825	81899	114368	13056
Mean	0.3736	0.2822	0.3463	0.2838	0.3364	0.2886
Effect rel. to mean (T1)	n.s.	12%	n.s.	9%	9%	36%
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.

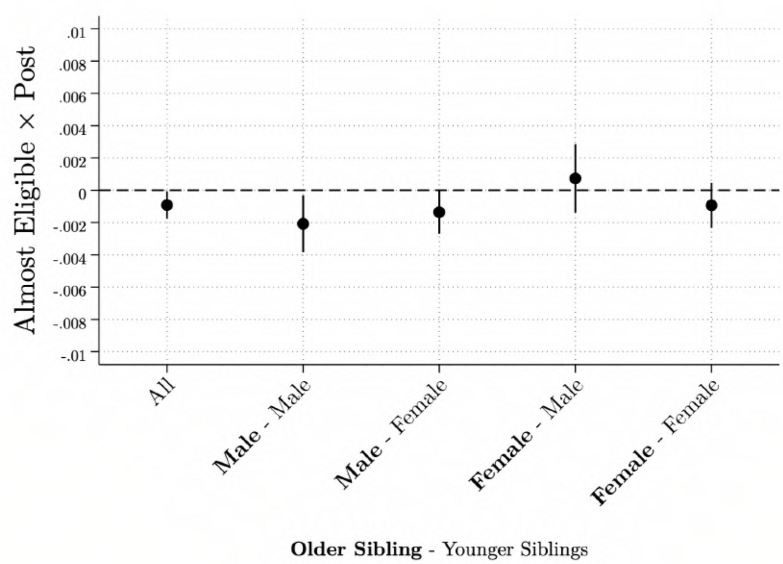
Notes: Each column reports separate estimates of Equation (2) for the indicated subsample. Column (1) includes all younger siblings. Columns (2) and (3) restrict the sample to families where the age gap with the older sibling is less than or equal to four years, and greater than four years, respectively. Columns (4) and (5) include same-gender and opposite-gender sibling pairs. Columns (6) and (7) split the sample by socioeconomic status using the poverty index: less poor corresponds to the upper half and poorest to the lower half of the distribution. Columns (8) and (9) distinguish households with a stay-at-home versus working mother. Columns (10)–(13) split the sample by parental education: mother or father with low versus high education, and whether neither or at least one parent has postsecondary education. Columns (14) and (15) separate municipalities without and with at least one accredited high-quality HEI. Column (18) restricts the sample to households where the older sibling was employed during the last year of high school. Standard errors are clustered at the household level. All specifications include the same set of controls and fixed effects as in the baseline model.

Figure 27: Younger Sibling: Probability of Conviction by Sibling Gender Pair

Panel A. Eligible × Post



Panel B. Almost Eligible × Post



Note: The figure plots Difference-in-Differences estimates of the effect of having an older sibling eligible for college financial aid on the probability that a younger sibling is convicted of any crime, by sibling gender pair. Panel A reports the coefficient on Eligible × Post (T_1) and Panel B reports the coefficient on Almost Eligible × Post (T_2). Each point corresponds to a different older sibling–younger sibling gender combination, where the label indicates older sibling gender followed by younger sibling gender. Ninety-five percent confidence intervals are shown. Standard errors are clustered at the household level.

Table 19. Heterogeneity Analysis – Younger Sibling: Ever Being Convicted of a Crime

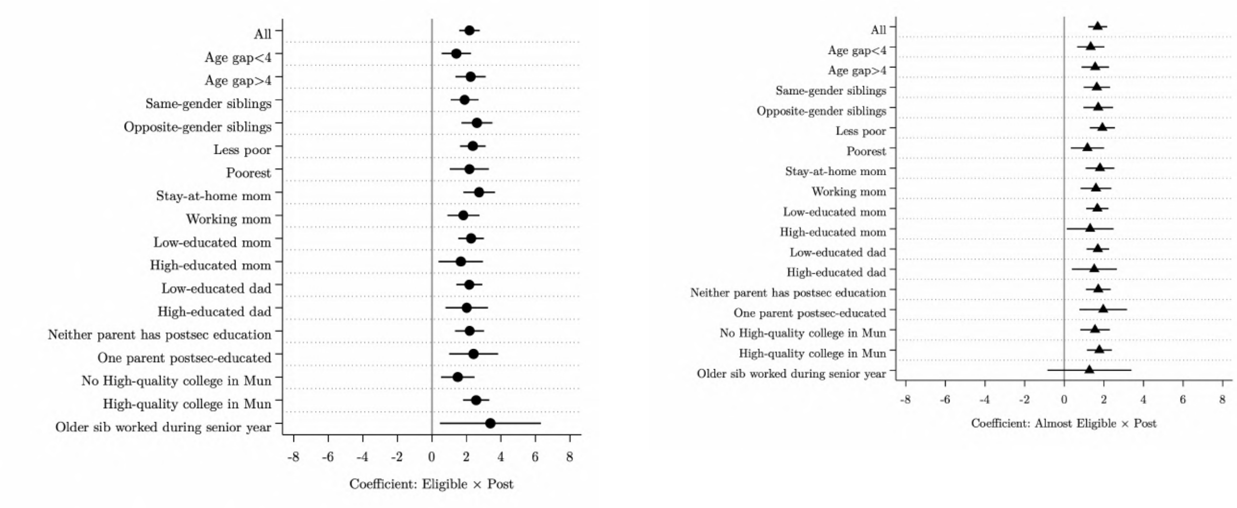
Differences-in-Differences Model:

	(1) All	(2) Age gap $\leq$ 4 years	(3) Age gap $>$ 4 years	(4) Same-Gender Siblings	(5) Opposite-Gender Siblings	(6) SES: Less Poor
T1 $\times$ Post	-0.0017***	-0.0017**	-0.0017**	-0.0018**	-0.0015	-0.0014
	(0.0006)	(0.0009)	(0.0007)	(0.0008)	(0.0010)	(0.0009)
T2 $\times$ Post	-0.0009**	-0.0019**	0.0002	-0.0016**	-0.0002	-0.0008
	(0.0004)	(0.0008)	(0.0005)	(0.0006)	(0.0006)	(0.0006)
Observations	189033	96460	92467	98829	90080	103062
Mean	0.0024	0.0032	0.0017	0.0021	0.0028	0.0021
Effect rel. to mean (T1)	-71%	-54%	-98%	-86%	n.s.	n.s.
Effect rel. to mean (T2)	-38%	-61%	n.s.	-76%	n.s.	n.s.
	(7) SES: Poorest	(8) Stay-at-home Mom	(9) Working Mom	(10) Low-educated Mom	(11) High-educated Mom	(12) Low-educated Dad
T1 $\times$ Post	-0.0018**	-0.0021**	-0.0011	-0.0019**	-0.0019*	-0.0020***
	(0.0008)	(0.0008)	(0.0010)	(0.0007)	(0.0011)	(0.0008)
T2 $\times$ Post	-0.0011	-0.0005	-0.0008	-0.0006	-0.0022***	-0.0013**
	(0.0008)	(0.0007)	(0.0007)	(0.0006)	(0.0009)	(0.0006)
Observations	85885	85112	73240	122671	33906	118806
Mean	0.0028	0.0024	0.0021	0.0024	0.0010	0.0020
Effect rel. to mean (T1)	-63%	-85%	n.s.	-80%	-196%	-96%
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	n.s.	-235%	-66%
	(13) High-educated Dad	(14) Neither parent has postsecondary education	(15) One parent has postsecondary education	(16) No High-Quality College in Mun	(17) High-Quality College in Mun in Mun	(18) Older sibling worked in Senior Year
T1 $\times$ Post	0.0006	-0.0023***	0.0001	-0.0006	-0.0023***	0.0013
	(0.0015)	(0.0009)	(0.0012)	(0.0011)	(0.0008)	(0.0046)
T2 $\times$ Post	0.0004	-0.0010	-0.0006	-0.0012*	-0.0008	-0.0042
	(0.0012)	(0.0007)	(0.0011)	(0.0007)	(0.0005)	(0.0027)
Observations	28956	103188	34972	68609	120421	9879
Mean	0.0022	0.0022	0.0023	0.0025	0.0024	0.0027
Effect rel. to mean (T1)	n.s.	-106%	n.s.	n.s.	-96%	n.s.
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	-49%	n.s.	n.s.

Notes: Each column reports separate estimates of Equation (2) for the indicated subsample. Column (1) includes all younger siblings. Columns (2) and (3) restrict the sample to families where the age gap with the older sibling is less than or equal to four years, and greater than four years, respectively. Columns (4) and (5) include same-gender and opposite-gender sibling pairs. Columns (6) and (7) split the sample by socioeconomic status using the poverty index: less poor corresponds to the upper half and poorest to the lower half of the distribution. Columns (8) and (9) distinguish households with a stay-at-home versus working mother. Columns (10)–(13) split the sample by parental education: mother or father with low versus high education, and whether neither or at least one parent has postsecondary education. Columns (14) and (15) separate municipalities without and with at least one accredited high-quality HEI. Column (18) restricts the sample to households where the older sibling was employed during the last year of high school. Standard errors are clustered at the household level. All specifications include the same set of controls and fixed effects as in the baseline model.

Figure 28: Heterogeneity Analysis: *Saber 11* Test Scores

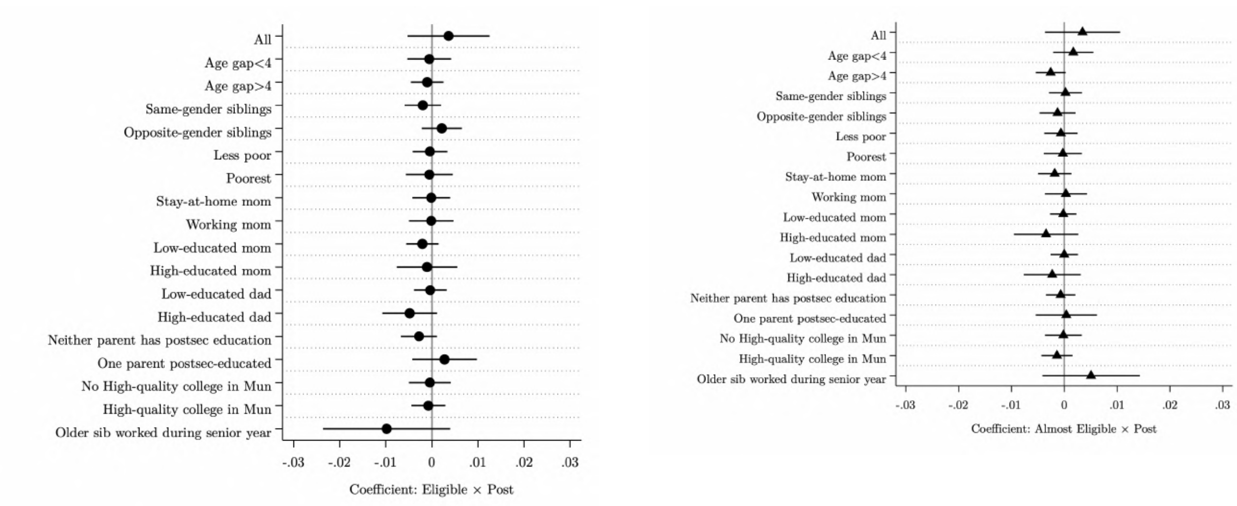
Panel A. Eligible × Post | **Panel B. Almost Eligible × Post**



Note: The figures plot subgroup-specific estimates of the treatment interaction terms ($T1 \times Post$ and $T2 \times Post$) from Equation (2), where the outcome is the standardized *Saber 11* test score. Subgroups include sibling characteristics (age gap, gender), household poverty index, parental education, and maternal employment. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Figure 29: Heterogeneity Analysis: Probability of Taking the *Saber 11* Exam

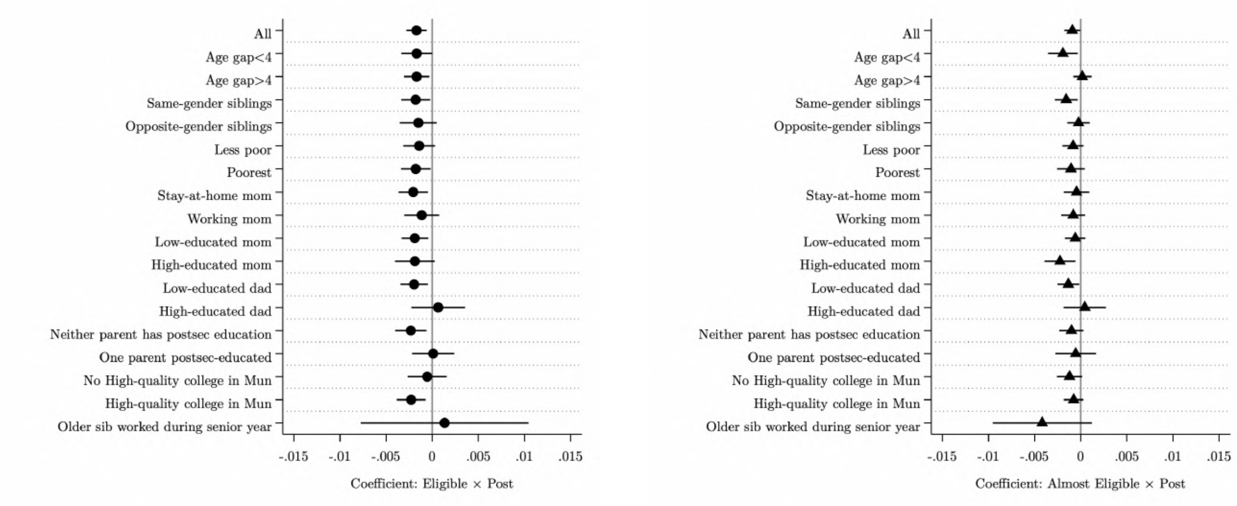
Panel A. Eligible × Post | **Panel B. Almost Eligible × Post**



Note: Each panel displays subgroup-specific estimates of the post-treatment interaction term ($T1 \times Post$ or $T2 \times Post$) from Equation (2). The dependent variable equals one if the younger sibling takes the national high-school exit exam (*Saber 11*). Vertical lines denote 95 percent confidence intervals. Standard errors are clustered at the household level.

Figure 30: Heterogeneity Analysis: Ever Being Convicted of a Crime

Panel A. Eligible × Post **Panel B. Almost Eligible × Post**



Note: Each panel shows subgroup-specific estimates of the interaction term ($T1 \times Post$ or $T2 \times Post$) for the probability that the younger sibling is ever convicted of a crime. Subgroups are defined by household poverty index, parental education, and the presence of a high-quality HEI in the municipality. Vertical bars represent 95 percent confidence intervals. Standard errors are clustered at the household level.

Table 20. Robustness Check – Younger Sibling: Performance in National Standardized Test

	Extensive Margin	Intensive Margin		
	(1)	(2)	(3)	(4)
	HS Graduation	Overall	Math	Spanish
Differences-in-Differences Model:				
Panel A: Baseline				
T1 × Post	-0.003**	0.079***	0.078***	0.087***
	(0.002)	(0.012)	(0.012)	(0.012)
T2 × Post	-0.002*	0.048***	0.045***	0.048***
	(0.001)	(0.009)	(0.009)	(0.009)
Observations	276802	212764	212764	212764
Mean	0.7799	0.3979	0.3804	0.3564
Panel B: Control: 60th – 70th percentile				
T1 × Post	-0.001	0.093***	0.097***	0.099***
	(0.002)	(0.012)	(0.012)	(0.012)
T2 × Post	-0.000	0.060***	0.061***	0.056***
	(0.001)	(0.009)	(0.009)	(0.009)
Observations	287505	217932	217932	217932
Mean	0.7706	0.3330	0.3217	0.2985
Panel C: Control: 50th – 60th percentile				
T1 × Post	-0.002	0.099***	0.110***	0.103***
	(0.002)	(0.012)	(0.012)	(0.012)
T2 × Post	-0.000	0.066***	0.071***	0.062***
	(0.001)	(0.009)	(0.009)	(0.009)
Observations	296217	221698	221698	221698
Mean	0.7583	0.2822	0.2765	0.2524

Note: The table reports Difference-in-Differences estimates from Equation (2) for younger siblings' educational outcomes. Each column corresponds to a separate regression. The coefficients on $T1 \times Post$ and $T2 \times Post$ capture the change in outcomes for younger siblings of eligible and almost-eligible older siblings, respectively, relative to the control group of families whose older sibling scored well below the SPP cutoff. High-school graduation is measured as an indicator equal to one if the younger sibling takes the Saber 11 exam. *Saber 11* outcomes are standardized by year to have mean zero and standard deviation one. All regressions include school and municipality-by-cohort fixed effects and control for baseline family characteristics interacted with cohort dummies. Standard errors are clustered at the family level.

Table 21. Sibling Spillover Effects on Postsecondary Enrollment by Institution Type, Quality and Program Duration

Differences-in-Differences Model:			Low-quality			High-quality			Program Duration	
	(1) SPP	(2) Any	(3) Any	(4) Public	(5) Private	(6) Any	(7) Public	(8) Private	(9) Two-years	(10) Four-years
Panel A: Baseline										
T1 × Post	0.013***	0.029***	-0.009*	-0.007*	-0.001	0.038***	-0.002	0.040***	-0.012***	0.042***
	(0.003)	(0.007)	(0.005)	(0.004)	(0.003)	(0.006)	(0.005)	(0.004)	(0.004)	(0.006)
T2 × Post	-0.002	0.006	-0.002	0.001	-0.003	0.008**	0.008**	0.001	-0.005	0.012***
	(0.002)	(0.005)	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)
Observations	151741	198416	198416	198416	198416	198416	198416	198416	198416	198416
Mean	0.0262	0.3144	0.1463	0.0973	0.0490	0.1682	0.0991	0.0691	0.1198	0.1936
Effect rel. to mean (T1)	50%	9%	-6%	-8%	n.s.	23%	n.s.	58%	-10%	22%
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	n.s.	n.s.	5%	8%	n.s.	n.s.	6%
Panel B: Control: 60th – 70th percentile										
T1 × Post	0.013***	0.038***	-0.007	-0.012***	0.004	0.046***	0.001	0.044***	-0.015***	0.053***
	(0.003)	(0.007)	(0.005)	(0.004)	(0.003)	(0.006)	(0.005)	(0.004)	(0.005)	(0.006)
T2 × Post	-0.002	0.014***	-0.000	-0.004	0.003	0.014***	0.011***	0.004	-0.008**	0.022***
	(0.002)	(0.005)	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)
Observations	155196	182994	182994	182994	182994	182994	182994	182994	182994	182994
Mean	0.0241	0.3014	0.1447	0.0967	0.0480	0.1567	0.0932	0.0636	0.1188	0.1818
Effect rel. to mean (T1)	53%	13%	n.s.	-12%	n.s.	29%	n.s.	70%	-12%	29%
Effect rel. to mean (T2)	n.s.	5%	n.s.	n.s.	n.s.	9%	12%	n.s.	-7%	12%
Panel C: Control: 50th – 60th percentile										
T1 × Post	0.012***	0.031***	-0.015***	-0.016***	0.001	0.045***	0.004	0.042***	-0.017***	0.048***
	(0.003)	(0.007)	(0.005)	(0.004)	(0.003)	(0.006)	(0.005)	(0.004)	(0.005)	(0.006)
T2 × Post	-0.003**	0.004	-0.010**	-0.009***	-0.000	0.014***	0.013***	0.001	-0.013***	0.017***
	(0.002)	(0.005)	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)	(0.002)	(0.004)	(0.004)
Observations	157125	186057	186057	186057	186057	186057	186057	186057	186057	186057
Mean	0.0237	0.2945	0.1428	0.0962	0.0466	0.1517	0.0905	0.0612	0.1165	0.1771
Effect rel. to mean (T1)	50%	10%	-10%	-17%	n.s.	30%	n.s.	68%	-14%	27%
Effect rel. to mean (T2)	-14%	n.s.	-7%	-10%	n.s.	9%	14%	n.s.	-11%	10%

Note: Each column reports separate estimates of Equation (2) for different higher education enrollment outcomes. Column (1) reports the probability that the younger sibling receives a *Ser Pilo Paga* (SPP) scholarship. Column (2) reports the probability of enrolling in any higher education institution (HEI). Columns (3)–(5) and (6)–(8) separate low- and high-quality HEIs by sector type (public or private). Columns (9) and (10) report enrollment in two-year and four-year programs. Panel A uses as a control group younger siblings of students scoring in the 70th–80th percentile of the Saber 11 exam, Panel B uses the 60th–70th percentile, and Panel C uses the 50th–60th percentile. All specifications include the same controls and fixed effects as in the baseline model, and standard errors are clustered at the household level.

Table 22. Younger Sibling: Dropouts and Absenteeism

	Grade of the Younger Sibling										
	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Differences-in-Differences Model:											
	Younger Sibling Drops out										
Eligible × Post	-0.001	0.000	0.002*	-0.000	0.001	-0.001	0.001	0.000	-0.001	-0.001	-0.002*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Almost Eligible × Post	-0.000	0.001	0.000	0.001	0.001	0.001	0.001	0.000	-0.001	-0.001	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Observations	247119	271296	290262	300848	304404	306228	287153	261734	232018	215521	202319
Mean	0.0112	0.0097	0.0096	0.0081	0.0134	0.0331	0.0312	0.0340	0.0352	0.0315	0.0045
Effect relative to the mean (Elig.)	n.s.	n.s.	17%	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	-35%
Effect relative to the mean (Almost Elig.)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
	Younger Sibling is absent for a year										
Eligible × Post	-0.001	0.003	0.000	-0.003	-0.002	-0.004	-0.002	0.002	-0.002	-0.002	-0.001
	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.001)
Almost Eligible × Post	0.000	0.001	0.001	-0.000	0.002	0.002	-0.001	0.000	-0.002	-0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.001)
Observations	247961	272688	292738	304506	310379	314757	298660	276801	249233	217544	184559
Mean	0.0532	0.0428	0.0428	0.0368	0.0441	0.0631	0.0626	0.0693	0.0787	0.0796	0.0060
Effect relative to the mean (Elig.)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
Effect relative to the mean (Almost Elig.)	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.

A Appendix: Replication

A.1 Differences in Immediate Enrollment: Comparison with SPP Estimates by Londoño-Vélez et al. (2020)

We replicate the effect of the Ser Pilo Paga program estimated by Londoño-Vélez et al. (2020) for the first cohort, launched in 2014. Our results are presented in Figure A.1. We find that the probability of immediate enrollment in a higher education institution increases by 26.6 percentage points. This estimate is lower than the 32.8 percentage point increase reported by the authors. The difference can be explained by the data sources used in each analysis. Our estimates are based on SNIES data, which includes enrollment in vocational training institutions. In contrast, Londoño-Vélez et al. (2020) uses SPADIES data, which does not capture this type of enrollment. This distinction is particularly relevant for students just below the eligibility cutoff, who are more likely to attend institutions such as SENA.

Figure A.1: Immediate Enrollment in Postsecondary Education: Older Siblings and All the test-takers in the Fall 2014

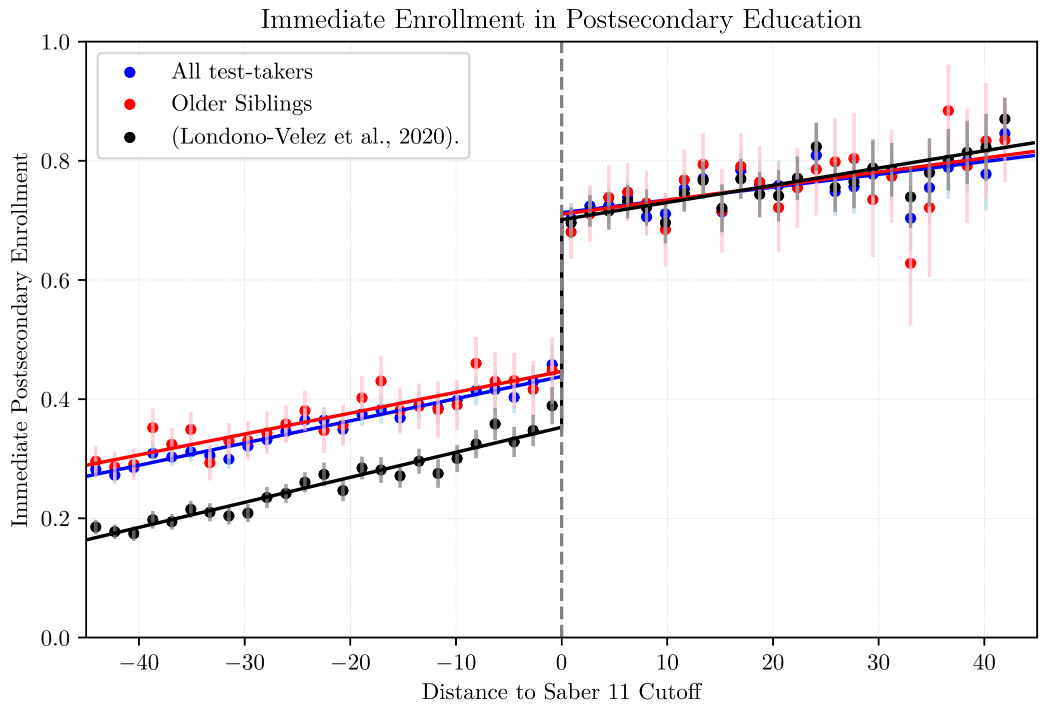


Table A.1. Immediate Enrollment in Postsecondary Education, by Type of Institution

		High quality			Low quality		
	Any	Any	Private	Public	Any	Private	Public
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Overall Effects							
RF	0.266	0.361	0.433	-0.075	-0.093	-0.037	-0.056
	(0.026)	(0.024)	(0.013)	(0.011)	(0.009)	(0.005)	(0.007)
Mean Control	0.439	0.276	0.076	0.203	0.161	0.06	0.099
Observations	266402	266402	266402	266402	266402	266402	266402
BW loc. poly.	32.346	28.979	30.87	22.493	23.328	33.577	23.502
Effect obs. control	34253	28953	31834	19894	20864	36279	21358
Effect obs. treat	11331	10758	11104	9332	9551	11524	9646
Panel B. Older Siblings							
RF	0.246	0.365	0.428	-0.062	-0.116	-0.044	-0.071
	(0.027)	(0.025)	(0.022)	(0.017)	(0.016)	(0.01)	(0.013)
Mean Control	0.44	0.257	0.074	0.184	0.183	0.07	0.112
Observations	80955	80955	80955	80955	80955	80955	80955
BW loc. poly.	20.451	22.658	22.411	30.686	27.034	29.155	27.449
Effect obs. control	5708	6504	6492	10015	8503	9341	8667
Effect obs. treat	2960	3117	3117	3635	3432	3541	3451
Panel C. Overall Effects (Londoño-Vélez et al., 2020)							
RF	0.328	0.484	0.484	-0.002	-0.16	-0.069	-0.088
	(0.013)	(0.012)	(0.011)	(0.007)	(0.012)	(0.007)	(0.009)
Mean Control	0.372	0.104	0.03	0.075	0.27	0.109	0.159
Observations	266402	266402	266402	266402	266402	266402	266402
BW loc. poly.	26.48	28.297	25.787	27.09	22.016	25.776	24.551
Effect obs. control	25050	27856	24542	26174	19439	24542	22425
Effect obs. treat	10230	10614	10159	10393	9231	10071	9806

Note: The dependent variables are the likelihood of enrolling in a higher education institution, with distinctions in columns (2) to (4) and (5) to (7) between high-quality and low-quality higher education institutions by sector type. The sample are older siblings whose families meet the Sisben eligibility criteria, i.e., those in need.

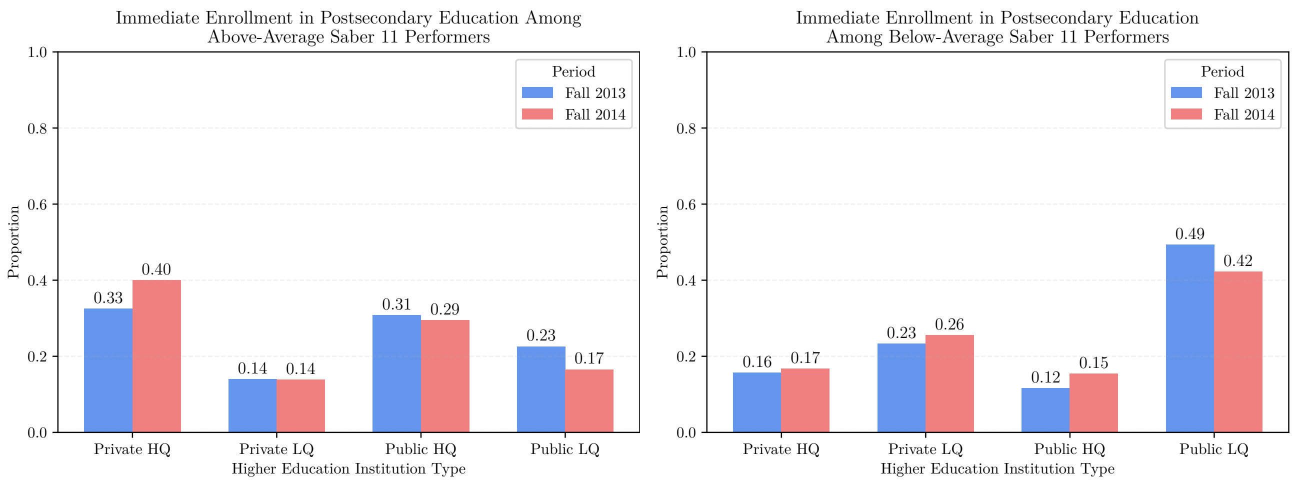
B Appendix: HEI Supply Analysis

B.1 Do SPP-Eligible Students Free Up Seats in Public Universities?

In the results, we show that the increase in enrollment rates is driven by eligible students gaining greater access to private, high-quality higher education institutions. We also observe that almost-eligible students experience increases in enrollment, driven primarily by higher attendance at high-quality public institutions, which tend to be more competitive.

One hypothesis is that students eligible for SPP enroll in high-quality private universities, thereby freeing up seats in competitive public institutions for students just below the eligibility threshold. Figure B.1 illustrates this pattern: on average, students performing above the cutoff in the standardized high school exit exam between 2013 and 2014 increase their attendance at private high-quality institutions and reduce their attendance at public institutions, particularly those of lower quality. In contrast, Panel B of Figure B.1 shows that students below the cutoff decrease their attendance at low-quality public institutions and increase enrollment in private low-quality and public high-quality universities.

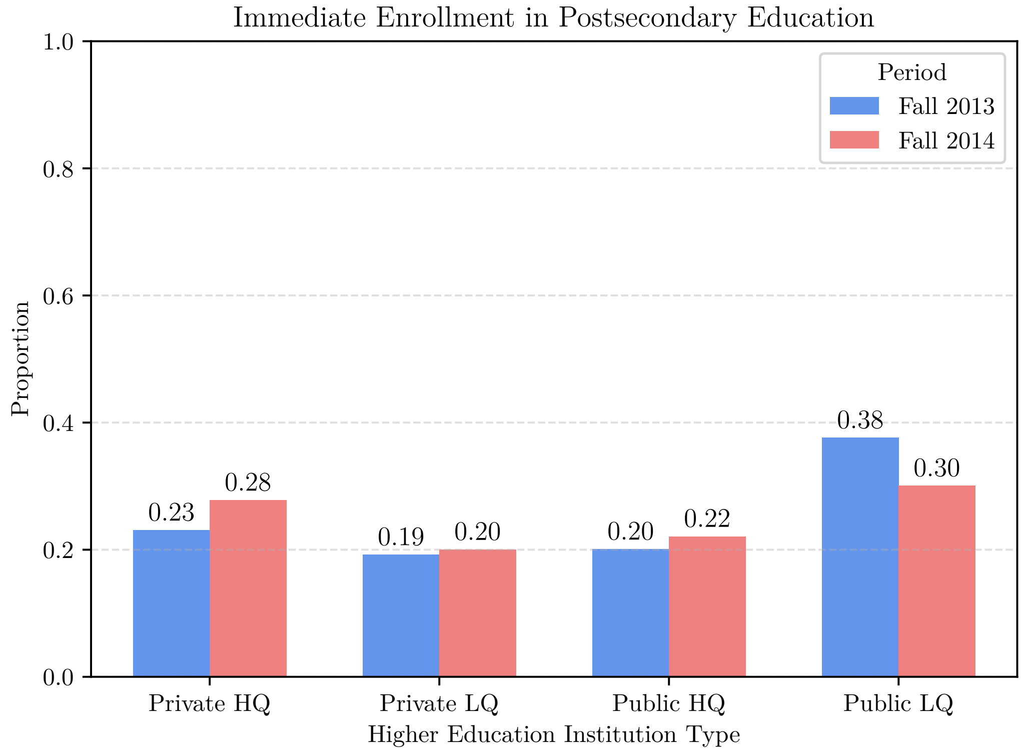
Figure B.1: Immediate Enrollment in Postsecondary Education, by Type of Institution



(b) Above-Average Saber 11 **(d) Below-Average Saber 11**

Note: Authors' calculations.

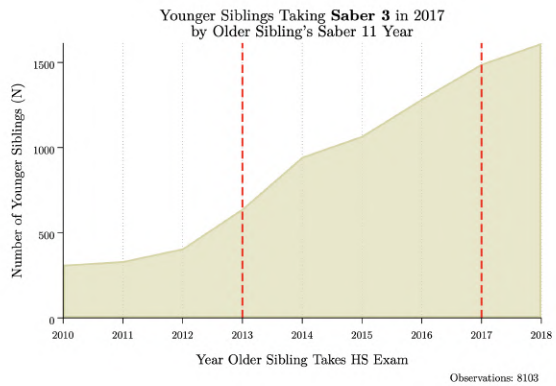
Figure B.2: Immediate Enrollment in Postsecondary Education: Older Siblings and All the test-takers in the Fall 2014



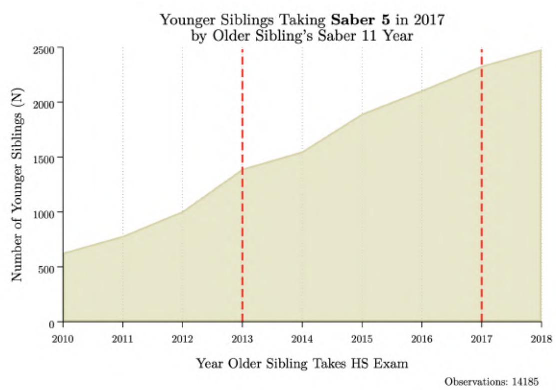
C Appendix: Additional Tables and Figures

Figure C.1: Number of Younger Siblings Taking Saber 3,5, 9 in 2017

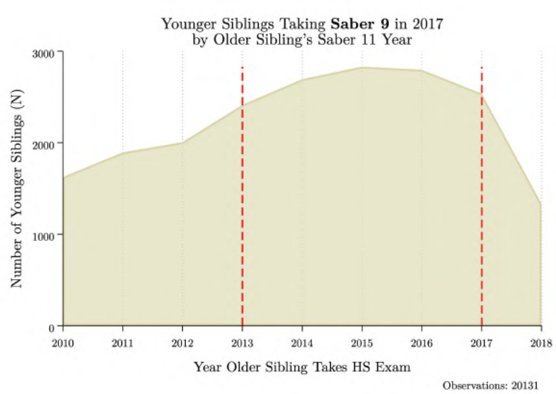
(a) Saber 3



(b) Saber 5



(c) Saber 9



Note: The figure plots the distribution of younger siblings in our sample who took the *Saber 3*, *Saber 5*, and *Saber 9* exams in 2017, by the year in which their older sibling sat for the high-school exam (*Saber 11*). The x-axis represents the cohort of the older sibling (2010–2019), and the y-axis shows the corresponding number of younger siblings observed in each cohort. Panel (a) corresponds to younger siblings taking *Saber 3*, Panel (b) to those taking *Saber 5*, and Panel (c) to those taking *Saber 9*.

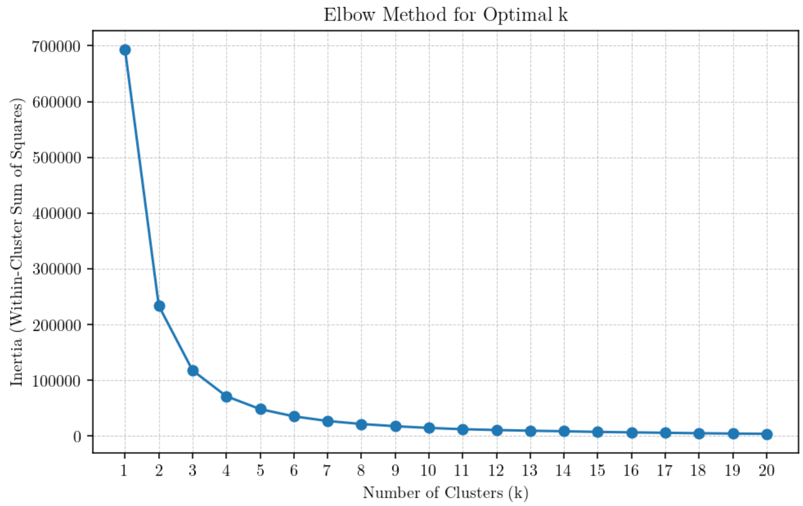


Figure C.2: Elbow Method for Determining Optimal Number of Clusters

Table C.1. Younger Sibling: Performance in National Standardized Test

	Extensive Margin	Intensive Margin		
	(1)	(2)	(3)	(4)
	HS Graduation	Overall	Math	Spanish
Differences-in-Differences Model:				
Panel A. Baseline				
T1 × Post	-0.003**	0.079***	0.078***	0.087***
	(0.002)	(0.012)	(0.012)	(0.012)
T2 × Post	-0.002*	0.048***	0.045***	0.048***
	(0.001)	(0.009)	(0.009)	(0.009)
Observations	276802	212764	212764	212764
Mean	0.7799	0.3979	0.3804	0.3564
Panel B. Machine-Learning–Defined Groups				
T1 × Post	-0.001	0.069***	0.078***	0.072***
	(0.001)	(0.011)	(0.011)	(0.011)
T2 × Post	0.000	0.050***	0.055***	0.042***
	(0.001)	(0.006)	(0.007)	(0.007)
Observations	510208	373570	373570	373570
Mean	0.7441	0.1794	0.1836	0.1696

Note: The table reports Difference-in-Differences estimates from Equation (2) for younger siblings' educational outcomes. Each column corresponds to a separate regression. The coefficients on $T1 \times Post$ and $T2 \times Post$ capture the change in outcomes for younger siblings of eligible and almost-eligible older siblings, respectively, relative to the control group of families whose older sibling scored well below the SPP cutoff. High-school graduation is measured as an indicator equal to one if the younger sibling takes the Saber 11 exam. *Saber 11* outcomes are standardized by year to have mean zero and standard deviation one. All regressions include school and municipality-by-cohort fixed effects and control for baseline family characteristics interacted with cohort dummies. Standard errors are clustered at the family level.

Table C.2. Sibling Spillover Effects on Postsecondary Enrollment by Institution Type, Quality and Program Duration

			Low-quality			High-quality			Program Duration	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	SPP	Any	Any	Public	Private	Any	Public	Private	Two-years	Four-years
Differences-in-Differences Model:										
Panel A. Baseline										
T1 × Post	0.013***	0.029***	-0.009*	-0.007*	-0.001	0.038***	-0.002	0.040***	-0.012***	0.042***
	(0.003)	(0.007)	(0.005)	(0.004)	(0.003)	(0.006)	(0.005)	(0.004)	(0.004)	(0.006)
T2 × Post	-0.002	0.006	-0.002	0.001	-0.003	0.008**	0.008**	0.001	-0.005	0.012***
	(0.002)	(0.005)	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)
Observations	151741	198416	198416	198416	198416	198416	198416	198416	198416	198416
Mean	0.0262	0.3144	0.1463	0.0973	0.0490	0.1682	0.0991	0.0691	0.1198	0.1936
Effect rel. to mean (T1)	50%	9%	-6%	-8%	n.s.	23%	n.s.	58%	-10%	22%
Effect rel. to mean (T2)	n.s.	n.s.	n.s.	n.s.	n.s.	5%	8%	n.s.	n.s.	6%
Panel B. Machine-Learning–Defined Groups										
T1 × Post	0.013***	0.031***	-0.013***	-0.014***	0.002	0.044***	0.001	0.042***	-0.015***	0.046***
	(0.003)	(0.006)	(0.004)	(0.004)	(0.003)	(0.005)	(0.004)	(0.004)	(0.004)	(0.005)
T2 × Post	-0.001	0.008**	-0.003	-0.004	0.001	0.011***	0.009***	0.002	-0.005**	0.014***
	(0.001)	(0.004)	(0.003)	(0.002)	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)
Observations	297618	349338	349338	349338	349338	349338	349338	349338	349338	349338
Mean	0.0177	0.2782	0.1431	0.0964	0.0467	0.1351	0.0795	0.0556	0.1164	0.1609
Effect rel. to mean (T1)	76%	11%	-9%	-15%	n.s.	32%	n.s.	76%	-13%	29%
Effect rel. to mean (T2)	n.s.	3%	n.s.	n.s.	n.s.	9%	11%	n.s.	-5%	9%

Note: The dependent variables are defined as follows. Column (1) reports the probability of being a *Ser Pilo Paga* (SPP) scholarship recipient. Column (2) shows the enrolling in a higher education institution (HEI). Columns (3)–(5) and (6)–(8) distinguish between low- and high-quality HEIs by sector type. Columns (9) and (10) report the probability of enrolling in a two-year and four-year HEI program, respectively.

Figure C.3: Age of Children in the Family in 2021